

Manifolds

Joe Orii

Contents

0	Preliminaries	1
0.1	Plan for the Course	1
0.2	Introduction and Motivations	1
1	Introduction to Smooth Manifolds	3
1.1	Topological Manifolds	3
1.2	Smooth Structures	9
2	Differentiation on Maps between Manifolds	19
2.1	Smooth Maps	19
2.2	Tangent Space	28
2.3	More on Smooth Maps	44
2.4	Exercises from [1]	62
3	Vector Bundles	65
3.1	Vector Bundles	65
3.2	Differential Forms	78
4	Integration	97
4.1	Orientation	97
4.2	Integration of Forms	100
4.3	The Stokes' Theorem	106
	Bibliography	110

0 Preliminaries

§0.1 Plan for the Course

Here is an indicative list of topics to be covered.

1. Structure of Smooth Manifolds
2. Functions/ Maps between Manifolds
3. Tangent Maps
4. Differential Forms
5. Integration

The main reference for this book is [1].

§0.2 Introduction and Motivations

Recall that $\mathbb{R}^n$ is the set of n -tuples with entries in $\mathbb{R}$, and this has *nice* analytic (differentiability, smoothness), algebraic (addition, multiplication, linear algebraic), and topological (open sets) properties. There are two notions for topological spaces: Hausdorffness and second-countability. We will define these shortly and elaborate on them in the next chapter.

1 Introduction to Smooth Manifolds

§1.1 Topological Manifolds

Definition 1.1.1. Suppose X is a topological space. We say that X is a **topological manifold of dimension n** or a **topological n -manifold** if it satisfies the following:

- X is a **Hausdorff space**: for every pair of points $p, q \in X$, there are disjoint open subsets $U, V \subseteq X$ such that $p \in U$ and $q \in V$.
- X is **second-countable**: there exists a countable basis for the topology of X .
- X is **locally Euclidean** of dimension n : each point of X has an open neighborhood that is homeomorphic to an open subset of $\mathbb{R}^n$.

Remark 1.1.2 — To elaborate a bit on the third property, this means that for each $p \in M$, we can find an open subset $U \subseteq M$ which contains p , an open subset $\hat{U} \subset \mathbb{R}^n$, and a homeomorphism $\varphi: U \rightarrow \hat{U}$.

Exercise 1.1.3. Show that equivalent definitions of manifolds are obtained if instead of allowing U to be homeomorphic to any open subset of $\mathbb{R}^n$, we require it to be homeomorphic to an open ball in $\mathbb{R}^n$, or to $\mathbb{R}^n$ itself.

Solution: The real change is just the locally Euclidean-ness. First, if we assume that U must be homeomorphic to an open ball, it is clear that U must be homeomorphic to an open subset of $\mathbb{R}^n$. Now, we show how our definition implies this. Suppose that U is homeomorphic to an open subset V of $\mathbb{R}^n$ under φ . Since V is open, there exists a small $r > 0$ such that $B_r(x) \subseteq V$, where $x = \varphi(p)$. Then, let $U' = \varphi^{-1}(B_r(x))$. Then, $\varphi|_{U'}: U' \rightarrow B_r(x)$ is a homeomorphism. $\square$

Notation 1.1.4. If M is a topological manifold, we abbreviate the dimension of M as $\dim M$.

Abuse of Notation 1.1.5. We may say ‘let M^n be a manifolds’ to mean ‘let M be a manifolds of dimension n ’.

Remark 1.1.6 — We note that every topological manifold has, by definition, a specific, well-defined dimension. Therefore, we don’t consider spaces of mixed dimension, such as the disjoint union of a plane and a line, to be manifolds at all.

Example 1.1.7

The empty set satisfies the definition of a topological n -manifolds for every n .

Remark 1.1.8 — We will often ignore the case, but because it is sometimes useful to include the empty manifold, we will not exclude it from the definition.

We require manifolds to share these properties to help ensure that manifolds behave in the ways we expect from that of Euclidean spaces. For instance, we can verify that in a Hausdorff space, finite subsets are closed and limits of convergent sequences are unique (prove this!). The motivation for second-countability is a bit less evident, but it will have important consequences throughout the course, mostly based on the existence of partitions of unity. We note that in practice, both the Hausdorff and second-countability properties are usually easy to check, especially for spaces that are *built* from other manifolds, since these properties are inherited by subspaces and finite products. Particularly, it follows that every open subset of a topological n -manifold is itself a topological n -manifold with the subspace topology.

Example 1.1.9

It is clear that if a topological space X is not Hausdorff or not second-countable, then it’s not a topological manifold.

Exercise 1.1.10. Find examples which satisfy the above.

Solution: Consider the space

$$\ell := (\{(x, 1) \in \mathbb{R}^2 : x \in \mathbb{R}\} \cup \{(x, 2) \in \mathbb{R}^2 : x \in \mathbb{R}\}) / \sim,$$

where $(x, 1) \sim (x, 2)$ except when $x = 0$. This is not Hausdorff. $\square$

Exercise 1.1.11. A topological manifold is locally connected, locally path-connected, and locally compact.

Solution: First note that a topological manifold has a countable basis of open sets which are homeomorphic to open balls, by the equivalent characterization of a manifold. Then, it is clear that a topological manifold is locally connected and locally path connected by this then. As for locally compactness, we take some closed ball B contained in the open ball. Then, since the chart map was a homeomorphism, we can restrict its inverse (which is continuous) to this compact set. This provides a compact set around the initial point. $\square$

Exercise 1.1.12. Show that $\mathbb{S}^1$ is not homeomorphic to $\mathbb{R}$.

Solution: If they were, then $\mathbb{R}$ would be compact. This is not true. $\square$

Definition 1.1.13. Let X be a topological n -manifold. A **coordinate chart** or simply **chart** on X is a pair (U, φ) , where U is an open subset of X and $\varphi: U \rightarrow \widehat{U}$ is a homeomorphism from U to an open subset $\widehat{U} = \varphi(U) \subseteq \mathbb{R}^n$.

Remark 1.1.14 — We note that by the definition of a topological manifold, each point $p \in X$ is contained in the domain of some chart (U, φ) . Thus, X is covered by these charts. If $\varphi(p) = 0$, we say that the chart is **centered at p** . We also note that if (U, φ) is any chart whose domain contains p , it’s easy to obtain a new chart centered at p by subtracting the constant vector $\varphi(p)$.

Definition 1.1.15. Given a chart (U, φ) , we call the set U a **coordinate domain** or a **neighborhood** of each of its points. If, further, $\varphi(U)$ is an open ball in $\mathbb{R}^n$, then we call U a **coordinate ball**. If $\varphi(U)$ is an open cube, U is a **coordinate cube**.

Definition 1.1.16. Given a chart (U, φ) we call the map φ a **(local) coordinate map**, and the component functions $(x_1, \dots, x_n)$ of φ , defined by $\varphi(p) = (x_1(p), \dots, x_n(p))$, are called the **local coordinates** on U .

Remark 1.1.17 — We may sometimes write ‘ (U, φ) is a chart containing p ’ as shorthand for ‘ (U, φ) is a chart whose domain U contains p ’. If we wish to emphasize the coordinate functions $(x_1, \dots, x_n)$ instead of the coordinate map φ , we sometimes denote the chart by $(U, (x_1, \dots, x_n))$ or $(U, (x_i))$.

§1.1.i Examples of Topological Manifolds

Example 1.1.18 ($\mathbb{R}^n$)

$\mathbb{R}^n$ is a topological manifold. Let us verify the three conditions. Hausdorff-ness and second-countability are clear; in particular, we can prove second-countability by considering the rationals. As for the locally Euclidean property, we can take φ to be the identity map.

Example 1.1.19 (The Circle $\mathbb{S}^1$)

$\mathbb{S}^1 = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 = 1\}$, with respect to the subspace topology, is a topological manifold. Recall that open sets of this are intersections of open sets of $\mathbb{R}^2$ (with the usual topology) with $\mathbb{S}^1$. Hausdorff-ness and second-countability are clear, since $\mathbb{S}^1$ is a subset of $\mathbb{R}^2$. Thus, we prove the locally-Euclidean property. For this, choose a $p \in \mathbb{S}^1$ and an arc U in the circle which contains p . This is open in $\mathbb{S}^1$. We claim that U is homeomorphic to an open set in $\mathbb{R}^1$. For this, we may use the projection map and verify that this is a continuous bijection with continuous inverse (for small intervals and appropriate points). We may also use an ‘angle function’ (with appropriate modulus) to get an appropriate map. The point here is that *there exists* a neighborhood—not all.

Example 1.1.20 (Graphs of Continuous Functions)

Let $U \subseteq \mathbb{R}^n$ be an open subset, and let $f: U \rightarrow \mathbb{R}^k$ be a continuous function. The **graph** of f is the subset of $\mathbb{R}^n \times \mathbb{R}^k$ defined by

$$\Gamma(f) = \{(x, y) \in \mathbb{R}^n \times \mathbb{R}^k : x \in U \text{ and } y = f(x)\},$$

endowed with the subspace topology. Let $\pi_1: \mathbb{R}^n \times \mathbb{R}^k \rightarrow \mathbb{R}^n$ denote the projection onto the first factor, and let $\varphi: \Gamma(f) \rightarrow U$ be the restriction of π_1 to $\Gamma(f)$:

$$\varphi(x, y) = x, \quad (x, y) \in \Gamma(f).$$

Since φ is the restriction of a continuous map, it is continuous itself. Further, it’s a homeomorphism since the inverse, given by $\varphi^{-1}(x) = (x, f(x))$, is continuous. Therefore, $\Gamma(f)$ is a topological n -manifold. In fact, $\Gamma(f)$ is homeomorphic to U itself, and $(\Gamma(f), \varphi)$ is a global coordinate chart, called **graph coordinates**. We can apply the same observation to any subset of $\mathbb{R}^{n+k}$ defined by setting any k of the coordinates (not necessarily the last k) equal to some continuous function of the other n , which are restricted to lie in an open subset of $\mathbb{R}^n$.

Example 1.1.21 (A Non-Example)

Consider $X = \{(x, y) \in \mathbb{R}^2 : xy = 0\}$ with the subspace topology from $\mathbb{R}^2$. Hausdorffness and second-countability are clear, since X is a subset of $\mathbb{R}^2$. The claim now is that the locally-Euclidean property fails at the origin. Everywhere that is not the origin looks like an interval in $\mathbb{R}^1$. The neighborhood of the origin looks like an intersection of two branches. This is not homeomorphic to an open set in $\mathbb{R}^n$, for any n . Anything above 2 is not possible, so we start from $n = 2$. This is definitely not possible, so we next move to $n = 1$; if we remove the origin from the picture, we get four disjoint components (as opposed to two disjoint components for open sets in $\mathbb{R}^1$). Thus, this is not a topological manifold.

Example 1.1.22 (A Non-Example)

Consider the closed interval $[a, b] \subseteq \mathbb{R}$. Similar to before, Hausdorff and second countable are true for subspaces, so they hold for $[a, b]$. However, what about locally Euclidean property? For any interior point p , we see that this is locally Euclidean. However, this does not hold at the endpoints. A neighborhood of a in $[a, b]$ looks like $[a, a + \epsilon)$. Indeed, this is not homeomorphic to an open subset of $\mathbb{R}$, since removing the point $\{a\}$ still gives a connected set—a contradiction.

Remark 1.1.23 (Manifolds With Boundary) — We see that $[a, b]$ is not a topological manifold; however, it is a manifold *with boundary*. We will not talk about such manifolds, and in this course we shall only talk about manifolds *without* boundary.

We can actually go one step further and generalize this to manifolds *with boundary and corners*. In vague terms, this applies to examples to spaces which have *singularities* or *sharp points* (Such as stars, squares, etc. in the case of subsets of $\mathbb{R}^2$). Again, these are not the focus of this course.

Definition 1.1.24. The n -dimensional real projective space, denoted $\mathbb{RP}^n$, is defined to be the set of 1-dimensional linear subspaces of $\mathbb{R}^{n+1}$, equipped with the quotient topology determined by the natural map $\pi: \mathbb{R}^{n+1} \setminus \{0\} \rightarrow \mathbb{RP}^n$ sending each $x \in \mathbb{R}^{n+1} \setminus \{0\}$ to the subspace spanned by x .

Remark 1.1.25 — We may also realize $\mathbb{RP}^n = \mathbb{R}^{n+1} \setminus \{0\} / x \sim \lambda x$, for all real $\lambda \neq 0$.

Notation 1.1.26. For any point $x \in \mathbb{R}^{n+1} \setminus \{0\}$, let $[x] = \pi(x) \in \mathbb{RP}^n$ denote the line spanned by x .

Definition 1.1.27. $\mathbb{RP}^2$ is called the **projective plane**.

Example 1.1.28 (Real Projective Space)

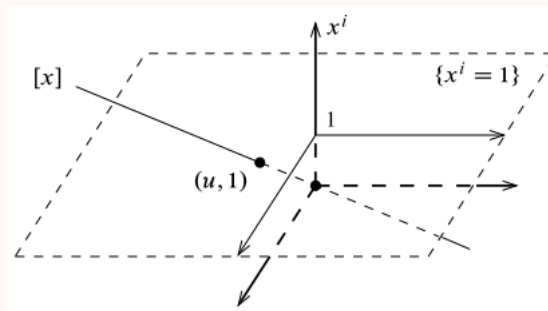
Let π be the natural map $\pi: \mathbb{R}^{n+1} \setminus \{0\} \rightarrow \mathbb{RP}^n$ sending each $x \in \mathbb{R}^{n+1} \setminus \{0\}$ to the subspace spanned by x . For each $i = 1, \dots, n+1$, let $\widetilde{U}_i \subseteq \mathbb{R}^{n+1} \setminus \{0\}$ be the set where $x_i \neq 0$, and let $U_i = \pi(\widetilde{U}_i) \subseteq \mathbb{RP}^n$. Since $\widetilde{U}_i$ is a saturated open subset, U_i is open and $\pi|_{\widetilde{U}_i}: \widetilde{U}_i \rightarrow U_i$ is a quotient map. Now define a map $\varphi_i: U_i \rightarrow \mathbb{R}^n$ by

$$\varphi[x_1, \dots, x_{n+1}] = \left(\frac{x_1}{x_i}, \dots, \frac{x_{i-1}}{x_i}, \frac{x_{i+1}}{x_i}, \dots, \frac{x_{n+1}}{x_i} \right).$$

This map is well defined because its value is unchanged by multiplying x by a nonzero constant. Because $\varphi_i \circ \pi$ is continuous, φ_i is continuous by the characteristic property of quotient maps. In fact φ_i is a homeomorphism because it has a continuous inverse given by

$$\varphi_i^{-1}(u^1, \dots, u^n) = [u^1, \dots, u^{i-1}, 1, u^i, \dots, u^n].$$

Geometrically, $\varphi([x]) = u$ means $(u, 1)$ is the point in $\mathbb{R}^{n+1}$ where the line $[x]$ intersects the affine hyperplane where $x_i = 1$. Because the sets $U_1, \dots, U_{n+1}$ cover $\mathbb{RP}^n$, this shows that $\mathbb{RP}^n$ is locally Euclidean of dimension n . The remaining properties are an exercise. The following is a chart for $\mathbb{RP}^n$.



Exercise 1.1.29. Show that $\mathbb{RP}^n$ is Hausdorff, second countable, and compact. *Hint: for compactness, show that the restriction of π to $\mathbb{S}^n$ is surjective.*

Solution: $\mathbb{RP}^n$ is second countable because it is simply a quotient of $\mathbb{R}^{n+1}$, which is second countable.

Next is Hausdorff-ness. This is more involved...

Compactness follows from viewing $\mathbb{RP}^n \cong \mathbb{S}^n / x \sim -x$. Then, $\mathbb{S}^n$ is clearly compact by the Heine Borel theorem. Then, since $\mathbb{RP}^n$ is just the image of $\mathbb{S}^n$ under the continuous map (just the quotient), $\mathbb{RP}^n$ is compact. $\square$

Example 1.1.30 (Open Subsets of Manifolds are Manifolds)

Let X be a topological manifold, and M be an open subset of X . Then, M is a topological manifold. Indeed, Hausdorff-ness and second countability are inherited from X via the subspace topology. As for the locally Euclidean property, we can take $p \in M \subset X$. Take the chart (U, φ) about p in X . Then, we can take $(U \cap M, \varphi|_{U \cap M})$ to be a chart around p in M .

Example 1.1.31 (Product Manifolds)

Suppose $M_1, \dots, M_k$ are topological manifolds of dimensions $n_1, \dots, n_k$, respectively. We claim that the product space $M_1 \times \dots \times M_k$ is a topological manifold of dimension $n_1 + \dots + n_k$.

This is Hausdorff and second-countable, so we only check the locally Euclidean property. Given a point $(p_1, \dots, p_k) \in M_1 \times \dots \times M_k$, we choose a coordinate chart (U_i, φ_i) for each M_i with $p_i \in U_i$. The product map

$$\varphi_1 \times \dots \times \varphi_k: U_1 \times \dots \times U_k \rightarrow \mathbb{R}^{n_1 + \dots + n_k}$$

is a homeomorphism onto its image, which is a product open subset of $\mathbb{R}^{n_1 + \dots + n_k}$. Therefore, $M_1 \times \dots \times M_k$ is indeed a topological manifold of dimension $n_1 + \dots + n_k$, with charts of the form $(U_1 \times \dots \times U_k, \varphi_1 \times \dots \times \varphi_k)$.

Lemma 1.1.32 (Smooth Manifold Chart Lemma)

Let M be a set, and suppose we are given a collection $\{U_\alpha\}$ of subsets of M together with maps $\varphi_\alpha: U_\alpha \rightarrow \mathbb{R}^n$, such that the following properties are satisfied:

- (i) For each α , φ_α is a bijection between U_α and an open subset $\varphi_\alpha(U_\alpha) \subseteq \mathbb{R}^n$.
- (ii) For each α and β , the sets $\varphi_\alpha(U_\alpha \cap U_\beta)$ and $\varphi_\beta(U_\alpha \cap U_\beta)$ are open in $\mathbb{R}^n$.
- (iii) Whenever $U_\alpha \cap U_\beta \neq \emptyset$, the map

$$\varphi_\beta \circ \varphi_\alpha^{-1}: \varphi_\alpha(U_\alpha \cap U_\beta) \rightarrow \varphi_\beta(U_\alpha \cap U_\beta)$$

is smooth.

- (iv) Countably many of the sets U_α cover M .
- (v) Whenever p, q are distinct points in M , either there exists some U_α containing both p and q , or there exist disjoint sets U_α, U_β with $p \in U_\alpha$ and $q \in U_\beta$.

Then M has a unique smooth manifold structure such that each $\{(U, \varphi)\}$ is a smooth chart.

Proof. We define the topology by taking all sets of the form $\varphi_\alpha^{-1}(V)$, with V an open subset of $\mathbb{R}^n$, as a basis. To prove that this is a basis for a topology, we need to show that for any point p in the intersection of two basis sets $\varphi_\alpha^{-1}(V)$ and $\varphi_\beta^{-1}(W)$, there is a third basis set containing p and contained in the intersection. It suffices to show that $\varphi_\alpha^{-1}(V) \cap \varphi_\beta^{-1}(W)$ is itself a basis set. To see this, observe that (iii) implies that

$$(\varphi_\beta \circ \varphi_\alpha^{-1})^{-1}(W)$$

is an open subset of $\varphi_\alpha(U_\alpha \cap U_\beta)$, and (ii) implies that this set is also open in $\mathbb{R}^n$. It follows that

$$\varphi_\alpha^{-1}(V) \cap \varphi_\beta^{-1}(W) = \varphi_\alpha^{-1}(V \cap (\varphi_\beta \circ \varphi_\alpha^{-1})^{-1}(W))$$

is also a basis set, as claimed.

Each map φ_α is then a homeomorphism onto its image (essentially by definition), so M is locally Euclidean of dimension n . The Hausdorff property follows easily from (v), and second-countability follows from (iv) and the result of Exercise 1.1.33, because each U_α is second-countable. Finally, (iii) guarantees that the collection $\{(U_\alpha, \varphi_\alpha)\}$ is a smooth atlas. It is clear that this topology and smooth structure are the unique ones satisfying the conclusions of the lemma. $\square$

Exercise 1.1.33. Let X be a topological space, and suppose X admits a countable open cover $\{U_i\}$ such that each set U_i is second-countable in the subspace topology. Show that X is second-countable.

Solution: Let $\mathcal{B}_i$ be a countable basis for U_i . Since the members in $\mathcal{B}_i$ are open in U_i , it follows that they are open in X . Therefore, the union $\mathcal{B} := \cup_i \mathcal{B}_i$ is a countable family of open sets in X . Let $x \in X$ and suppose that U is open in X and contains x . Since the U_i cover X , it must be that $x \in U_i$ for some i . By definition of a base, it follows that there is some $B \in \mathcal{B}$ such that $x \in B \subseteq U \cap U_i$. This B is in the union $\mathcal{B}$ and $x \in BU$, so $\mathcal{B}$ is a countable basis for X . $\square$

§1.2 Smooth Structures

Recall that:

Definition 1.2.1. A map $f: \mathbb{R}^n \rightarrow \mathbb{R}^m$ is **smooth** if it has derivatives of all orders.

Definition 1.2.2. A smooth map $f: U \rightarrow V$ is a **diffeomorphism** if the following hold:

1. f is bijective
2. f^{-1} is also smooth.

Here is a useful lemma.

Lemma 1.2.3

Let $U \subseteq \mathbb{R}^n$ and $V \subseteq \mathbb{R}^m$ be open subsets. If $f: U \rightarrow V$ is a diffeomorphism, then $n = m$.

Proof. Consider the Jacobian matrix of f at $p \in U$. If we denote f by $f(x_1, \dots, x_n) = (f_1(x_1, \dots, x_n), \dots, f_m(x_1, \dots, x_n))$, then the Jacobian matrix is the following.

$$J(f) = \begin{pmatrix} \frac{\partial f_1}{\partial x_1} & \cdots & \frac{\partial f_1}{\partial x_n} \\ \vdots & & \vdots \\ \frac{\partial f_m}{\partial x_1} & \cdots & \frac{\partial f_m}{\partial x_n} \end{pmatrix}$$

To prove that f^{-1} is smooth, we note that the Jacobian of f^{-1} is the inverse of $J(f)$. Since f is bijective, $J(f)$ is invertible, so $n = m$. $\square$

Remark 1.2.4 — Whenever we have smoothness, the Jacobian is a useful tool.

Definition 1.2.5. Let X be a topological manifold. We say that two charts (U, φ) and (V, ψ) are **smoothly compatible** if: either $U \cap V = \emptyset$ or $\varphi \circ \psi^{-1}: \psi(U \cap V) \rightarrow \varphi(U \cap V)$ is a diffeomorphism.

Remark 1.2.6 — The lemma above tells us that the images of the U and V under φ and ψ , respectively, live inside the same dimension of Euclidean space.

Definition 1.2.7. A **smooth atlas** of a topological manifold X is a collection of charts $\{(U_i, \varphi_i)\}$ such that

1. $\cup_i U_i = X$.
2. any two charts are smoothly compatibly.

Definition 1.2.8. A smooth atlas $\mathcal{A}$ is said to be **maximal** if any chart that is smoothly compatible with every chart in $\mathcal{A}$ is already in $\mathcal{A}$.

Remark 1.2.9 — To unravel this: Let $\mathcal{A} = \{(U_i, \varphi_i)\}$. Choose a chart (U, ψ) be a chart on X . If (U, ψ) is smoothly compatible with every (U_i, φ_i) , then $(U, \psi) \in \mathcal{A}$.

Remark 1.2.10 — Indeed, as the definition suggests, $\mathcal{A}$ is maximal if it is not contained in any larger smooth atlas.

Definition 1.2.11. A **smooth manifold** is a topological manifold with maximal smooth atlas.

Definition 1.2.12. A **smooth structure** on a topological manifold X is a maximal smooth atlas.

Remark 1.2.13 — We may just say that a smooth manifold is a topological manifold with a smooth structure. Further, all atlases will be smooth.

Remark 1.2.14 — We note that a chart of a smooth manifold is actually a diffeomorphism.

Proof. Let X be a smooth manifold, and (U, φ) a chart. Then, we have that $\varphi \circ \varphi^{-1} = \text{id}$. Therefore, $\text{id} \circ \varphi \circ \varphi^{-1} = \text{id}$, which is smooth. It follows, by definition of smoothness, that φ is smooth. Now, to check for φ^{-1} . For this, just take an arbitrary chart (V, ψ) . Then, $\psi \circ \varphi^{-1}$ is smooth, so this tells us that φ^{-1} is smooth. $\square$

Example 1.2.15

If X is covered by a single chart (U, φ) , then $\mathcal{A} = \{(U, \varphi)\}$ is a smooth atlas. In other words, if X is already homeomorphic to some open region in $\mathbb{R}^n$.

Example 1.2.16

Consider $\mathbb{R}$. We claim that $\mathcal{A} = \{(U_i = (a_i, b_i), \text{id})\}$ is a smooth atlas. We check this. We do this, by checking for each pair of charts $U_i = (a_i, b_i)$ and $U_j = (a_j, b_j)$ are smoothly compatible; indeed, this is true since the chart functions are identity, so the transition map is just the identity. We may also consider $\mathcal{B} = \{(\mathbb{R}, \text{id})\}$. This also forms a smooth atlas.

Now, we ask is $\mathcal{A}$ maximal? That is, do there exist charts that are not in $\mathcal{A}$ but are compatible with all charts in $\mathcal{A}$? No. We may consider disjoint unions of open intervals, such as $((a, b) \cup (c, d), \text{id})$; these are compatible with any (U_i, id) .

Remark 1.2.17 — In practice, it is very difficult to find the maximal atlas and so we may consider simpler atlases. We get around this issue via Proposition 1.2.22.

Example 1.2.18

Consider the set $\mathbb{S}^1$ defined by $\{(x_1, x_2) \in \mathbb{R}^2 : x_1^2 + x_2^2 = 1\}$ with the subspace topology. We define four charts on $\mathbb{S}^1$:

$$\begin{aligned} U_1^+ &:= \{(x_1, x_2) \in \mathbb{S}^1 : x_1 > 0\}, & \varphi_1^+(x_1, x_2) &:= x_2 \\ U_1^- &:= \{(x_1, x_2) \in \mathbb{S}^1 : x_1 < 0\}, & \varphi_1^-(x_1, x_2) &:= x_2 \\ U_2^+ &:= \{(x_1, x_2) \in \mathbb{S}^1 : x_2 > 0\}, & \varphi_2^+(x_1, x_2) &:= x_1 \\ U_2^- &:= \{(x_1, x_2) \in \mathbb{S}^1 : x_2 < 0\}, & \varphi_2^-(x_1, x_2) &:= x_1 \end{aligned}$$

We can directly check that the transition maps are diffeomorphisms. For instance, consider $\varphi_2^+ \circ (\varphi_1^+)^{-1}$, which is a map from $\varphi_1^+(U_1^+ \cap U_2^+) = x_2$ to $\varphi_2^+(U_1^+ \cap U_2^+) = x_1$. Direct computation shows that $\varphi_2^+ \circ (\varphi_1^+)^{-1}(x_2) = \sqrt{1 - (x_2)^2}$ which is a diffeomorphism from x_2 to x_1 . Moreover, these four subsets cover $\mathbb{S}^1$, so they form a smooth atlas on $\mathbb{S}^1$.

Example 1.2.19 (n -Spheres $\mathbb{S}^n$)

For each integer $n \geq 0$, the unit n -sphere $\mathbb{S}^n$ is Hausdorff and second-countable, for it is a topological subspace of $\mathbb{R}^{n+1}$. as usual, what's left is locally Euclidean-ness. For each index $i = 1, \dots, n+1$, we shall let U_i^+ and U_i^- to be subsets of $\mathbb{R}^{n+1}$ be the subset where the i -th coordinate is positive and negative, respectively:

$$U_i^+ := \{(x_1, \dots, x_{n+1}) \in \mathbb{R}^{n+1} : x_i > 0\}$$

$$U_i^- := \{(x_1, \dots, x_{n+1}) \in \mathbb{R}^{n+1} : x_i < 0\}$$

Now let $f: \mathbb{B}^n \rightarrow \mathbb{R}$ be the continuous function $f(u) = \sqrt{1 - |u|^2}$. Then, for each $i = 1, \dots, n+1$, we can check that $U_i^+ \cap \mathbb{S}^n$ is the graph of the function

$$x_i = f(x_1, \dots, \hat{x}_i, \dots, x_{n+1}),$$

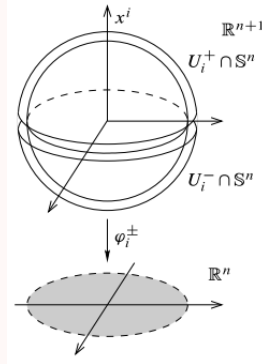
where the hat indicates that the x_i is omitted. Similarly, $U_i^- \cap \mathbb{S}^n$ is the graph of

$$x_i = -f(x_1, \dots, \hat{x}_i, \dots, x_{n+1}).$$

Then, each $U_i^\pm \cap \mathbb{S}^n$ is locally Euclidean of dimension n , and the maps $\varphi_i^\pm: U_i^\pm \cap \mathbb{S}^n \rightarrow \mathbb{B}^n$ given by

$$\varphi_i^\pm(x_1, \dots, x_{n+1}) = (x_1, \dots, \hat{x}_i, \dots, x_{n+1})$$

are graph coordinates for $\mathbb{S}^n$. Since each point of $\mathbb{S}^n$ is in the domain of at least one of these $2n+2$ charts, $\mathbb{S}^n$ is a topological n -manifold. We can further check that the transition maps are diffeomorphisms, so these charts form a smooth atlas on $\mathbb{S}^n$. the induced smooth structure is called the standard smooth structure on $\mathbb{S}^n$.

**Example 1.2.20** (Non-Example)

Let $X = \mathbb{R}$, with two charts $(U, \varphi) = (\mathbb{R}, \text{id})$ and $(V, \psi) = (\mathbb{R}, x^3)$. We first check that these are charts. Indeed, the maps id and x^3 are continuous with continuous inverse, so these are homeomorphisms (to $\mathbb{R}$). We claim that these two charts are not smoothly compatible. For this, we must check $\varphi \circ \psi^{-1}(x) = x^{\frac{1}{3}}$. However, this is not smooth; particularly, it is not smooth at 0. This tells us that (V, ψ) is not contained in the maximal atlas containing (U, φ) .

Remark 1.2.21 — This example motivates the question of uniqueness of atlases—naturally, about existence as well. Explicitly, we note that there exist some topological

manifold X which do not admit a smooth structure, i.e., a maximal atlas. Note that this is a nontrivial fact.

Proposition 1.2.22

Let X be a topological manifold.

- (i) Every smooth atlas $\mathcal{A}$ for X is contained in a unique maximal smooth atlas, called the **smooth structure determined by $\mathcal{A}$** . We may denote this as $\overline{\mathcal{A}}$.
- (ii) Two smooth atlases for X determine the same smooth structure if and only if their union is a smooth atlas.

Proof. (i): Let $\mathcal{A}$ be a smooth atlas for M ; and let $\overline{\mathcal{A}}$ denote the set of all charts that are smoothly compatible with every chart in $\mathcal{A}$.

To show that $\overline{\mathcal{A}}$ is a smooth atlas, we need to show that any two charts of $\overline{\mathcal{A}}$ are smoothly compatible with each other, which is to say that for any $(U, \varphi), (V, \psi) \in \overline{\mathcal{A}}$, the map $\psi \circ \varphi^{-1}: \varphi(U \cap V) \rightarrow \psi(U \cap V)$ is a diffeomorphism. Let $x = \varphi(p) \in \varphi(U \cap V)$ be arbitrary. Because the domains of the charts in $\mathcal{A}$ cover M ; there is some chart (W, θ) such that $p \in W$. Since every chart in $\overline{\mathcal{A}}$ is smoothly compatible with (W, θ) , both of the maps $\theta \circ \varphi^{-1}$ and $\psi \circ \theta^{-1}$ are diffeomorphisms where they are defined. Since $p \in U \cap V \cap W$, it follows that $\psi \circ \varphi^{-1} = (\psi \circ \theta^{-1}) \circ (\theta \circ \varphi^{-1})$ is a diffeomorphism on a neighborhood of x . Thus, $\psi \circ \varphi^{-1}$ is a diffeomorphism in a neighborhood of each point in $\varphi(U \cap V)$. Therefore, $\overline{\mathcal{A}}$ is a smooth atlas.

To check that it is maximal, just note that any chart that is smoothly compatible with every chart in $\overline{\mathcal{A}}$ must in particular be smoothly compatible with every chart in $\mathcal{A}$, so it is already in $\overline{\mathcal{A}}$. This proves the existence of a maximal smooth atlas containing $\mathcal{A}$.

Finally, we prove uniqueness. If $\mathcal{B}$ is any other maximal smooth atlas containing $\mathcal{A}$, each of its charts is smoothly compatible with each chart in $\mathcal{A}$, so $\mathcal{B} \subseteq \overline{\mathcal{A}}$. By maximality of $\mathcal{B}$, $\mathcal{B} = \overline{\mathcal{A}}$.

(ii): Suppose we have two smooth atlases $\mathcal{A}, \mathcal{B}$ on X . First, if $\mathcal{A} \cup \mathcal{B}$ is a smooth atlas, then any element in $\mathcal{A}$ is smoothly compatible with every element in $\mathcal{B}$, so $\mathcal{A} \subseteq \overline{\mathcal{B}}$. On the other hand, $\mathcal{A} \subseteq \overline{\mathcal{A}}$. By uniqueness $\overline{\mathcal{A}} = \overline{\mathcal{B}}$. On the other hand, if $\overline{\mathcal{A}} = \overline{\mathcal{B}}$, then we need to show that $\mathcal{A} \cup \mathcal{B}$ is a smooth atlas. Note that the assumption tells us that $\mathcal{B} \subseteq \overline{\mathcal{A}}$. Thus, any element in $\mathcal{B}$ is smoothly compatible with any element in $\mathcal{A}$. This implies that $\mathcal{A} \cup \mathcal{B}$ is a smooth atlas. $\square$

Remark 1.2.23 — Suppose we have two smooth atlases $\mathcal{A}$ and $\mathcal{B}$. The second statement is saying that $\overline{\mathcal{A}} = \overline{\mathcal{B}}$ if and only if $\mathcal{A} \cup \mathcal{B}$ is a smooth atlas.

Example 1.2.24

$(\mathbb{R}, \text{id} = x)$ and $(\mathbb{R}, 2x)$ determine the same atlas. Moreover, $(\mathbb{R}, \text{id})$ is determined by the same atlas, as long as our chart is some linear (affine) transformation.

§1.2.i More Examples

Example 1.2.25

Go back to the previous example on $\mathbb{R}$ with the same atlases. We have

$$\mathcal{A} = \{(a_i, b_i) : \text{id}\}, \quad \mathcal{B} = \{(\mathbb{R}, \text{id})\}.$$

By the previous proposition, there is a unique maximal smooth atlas for each: $\overline{\mathcal{A}}$ and $\overline{\mathcal{B}}$. To see whether these two maximal smooth atlases are equal, we need to see that $\mathcal{A} \cup \mathcal{B}$ is a smooth atlas. We note that this is indeed true, so we have that $\overline{\mathcal{A}} = \overline{\mathcal{B}}$.

Example 1.2.26

Take $\mathbb{R}$ again with the smooth atlases

$$\mathcal{B} = \{(\mathbb{R}, \text{id})\}, \quad \mathcal{C} = \{(\mathbb{R}, \varphi(x) = x^3)\}.$$

By the proposition, we check the union to see if they determine the same smooth structure or not. $\mathcal{B} \cup \mathcal{C}$, however, is not a smooth atlas because the transition map is not smoothly compatible in one direction.

Exercise 1.2.27. Construct infinitely many distinct smooth structures on $\mathbb{R}$. A more challenging one is to construct uncountably many distinct structures on $\mathbb{R}$.

Example 1.2.28

$\mathbb{R}^n$ with $\mathcal{A} = \{(\mathbb{R}^n, \text{id})\}$ is a smooth manifold. This is called the **standard smooth structure**.

Example 1.2.29 (Finite Dimensional Vector Spaces)

An immediate corollary of the example above is that any n -dimensional vector space E over $\mathbb{R}$ can be equipped with a smooth structure by identifying E with $\mathbb{R}^n$.

Example 1.2.30 (Open Submanifolds)

Suppose $(M, \mathcal{A})$ be a smooth manifold ($\mathcal{A}$ is the smooth atlas). Let U be an open subset of M . Then, U admits a smooth structure induced from M . Consider $\mathcal{B} = \{(V, \varphi) : (V, \varphi) \in \mathcal{A}, V \subseteq U\}$. We claim that $\mathcal{B}$ is a smooth atlas for U . First, we must show that the charts in $\mathcal{B}$ covers U . Take $p \in U$. Since $\mathcal{A}$ is an atlas, there exists $(\tilde{V}, \tilde{\varphi}) \in \mathcal{A}$ such that $p \in \tilde{V}$. Note that U is open, so we can take the intersection $(V := U \cap \tilde{V}, \varphi = \tilde{\varphi}|_V)$. This is still a chart, and $(V, \varphi) \in \mathcal{B}$. We can do this for any $p \in U$, so it is true that elements in $\mathcal{B}$ cover U . The transition maps are diffeomorphic because we're only taking intersections and restrictions as well, so $\mathcal{B}$ determines a smooth structure; this is called the **open submanifold** smooth structure. U is called an **open submanifold** of M .

Remark 1.2.31 — If U were not open, then we don't expect the above example to hold.

Example 1.2.32 (Non-Example)

Recall the manifold $M = \mathbb{R}^2$, and subspace $X = \{(x, y) \in \mathbb{R}^2 : xy = 0\}$. This is not an open submanifold. Everywhere besides the origin is 1-dimensional and we can find some homeomorphism to some open interval. However, around the origin, we cannot find an open interval such that it's homeomorphic to some open U containing the origin because if we remove the origin, then there are four connected components, while if we do the same for an interval, there should only be two.

Example 1.2.33 ($M_{n \times m}(\mathbb{R})$)

Consider the space of real valued matrices: $M_{n \times m}(\mathbb{R})$. This is a finite dimensional vector space over $\mathbb{R}$ (of dimension nm), so it follows that it is a smooth manifold.

Example 1.2.34 ($GL_n(\mathbb{R})$)

Consider now the space of invertible matrices $GL_n(\mathbb{R})$. We have that $GL_1(\mathbb{R}) \cong \mathbb{R}^\times$, which is an open subset. We can use the determinant map $\det: M_n(\mathbb{R}) \rightarrow \mathbb{R}$ to show that $GL_n(\mathbb{R})$ is open. $\det$ is a continuous map, and $\mathbb{R}^\times$ is an open set, so $GL_n(\mathbb{R})$ is an open subset of $M_n(\mathbb{R})$.

Example 1.2.35

The n -sphere $\mathbb{S}^n$ with the subspace topology from $\mathbb{R}^{n+1}$ is a smooth manifold.

Definition 1.2.36. For a positive integer n , the **n -torus** is the product space $\mathbb{T}^n = \mathbb{S}^1 \times \cdots \times \mathbb{S}^1$ (product taken n -times of course).

Example 1.2.37 (Tori)

The n -torus, by the previous example, is a topological n -manifold.

Example 1.2.38

Note that the disjoint union $\cdot \sqcup \cdot$ of two topological spaces has a natural topology. The disjoint union of two smooth manifolds with the same dimension is also a smooth manifold with the same dimension.

Example 1.2.39

In particular, we can see that $GL_n(\mathbb{R})$ is a disjoint union of two components, since we can see that it is the union of the positive part and negative part of the preimage of the determinant map.

Example 1.2.40 (The Graph)

let $F: U \rightarrow \mathbb{R}^m$ be a smooth map, where $U \subseteq \mathbb{R}^n$ is open. Consider the graph of F , defined as the set

$$G_F := \{(x, F(x)) : x \in U\} \subset \mathbb{R}^n \times \mathbb{R}^m$$

with the subset topology. Then, the map from before, $\varphi: G_F \rightarrow U$ defined by $\varphi((x, F(x))) := x$ is a homeomorphism, so we get a global chart; that is, the smooth atlas $\{(G_F, \varphi)\}$ makes G_F a smooth manifold.

Remark 1.2.41 — Note that this is not an open submanifold of dimension $n + m$, since G_F is not open in $\mathbb{R}^{n+m}$

Remark 1.2.42 — Note that we have a global chart, so there is no compatibility condition (between charts) needed to check. It follows that F can just be continuous; that is, smoothness is unnecessary. Indeed, F being continuous already implies that φ is a homeomorphism.

Example 1.2.43

Consider the graphs $F: \mathbb{R} \rightarrow \mathbb{R}$ given by $F(x) = x^2$ and $G: \mathbb{R} \rightarrow \mathbb{R}$ given by $G(x) = |x|$. As topological spaces, these two are homeomorphic. We can identify these with $\mathbb{R}$.

Proposition 1.2.44 (Products Smooth Structure)

Let X, Y be two smooth manifolds. Then, $X \times Y$ admits a product smooth structure. Further, $\dim(X \times Y) = \dim(X) + \dim(Y)$

Proof. Suppose that X is equipped with an atlas $\mathcal{A} = \{(U_i, \varphi_i)\}$ and Y is equipped with an atlas $\mathcal{B} = \{(V_j, \psi_j)\}$. Then, we claim that the collection $\mathcal{C} = \{(U_i \times V_j, \varphi_i \times \psi_j)\}$ is an atlas on $X \times Y$. We first check that $\varphi_i \times \psi_j: U_i \times V_j \rightarrow \varphi_i(U_i) \times \psi_j(V_j)$ is a homeomorphism and that $\cup_{i,j} U_i \times V_j$ covers $X \times Y$. Indeed, the maps $\varphi_i \times \psi_j$ are homeomorphisms since each φ_i and ψ_j are homeomorphisms. Further, $U_i \times V_j$ cover $X \times Y$ since U_i covers X and V_j covers Y . This tells us that $X \times Y$ is a topological manifold. Now, we check for smooth compatibility. Since

$$(\varphi_i \times \psi_j) \circ (\varphi_{i'} \times \psi_{j'})^{-1} = (\varphi_i \circ \varphi_{i'}^{-1}) \times (\psi_j \circ \psi_{j'}^{-1}),$$

so we have a product of smooth maps, so this is smooth. The inverse is smooth as well, by an analogous calculation, so we're done. Further, this calculation tells us the dimension of $X \times Y$.

$$\dim(X \times Y) = \dim(X) + \dim(Y).$$

□

Exercise 1.2.45. Let X be a topological manifold of dimension $n \geq 1$. If X admits a smooth structure, then show it admits uncountably many distinct smooth structures.

Solution: Consider the map $f_s: \mathbb{R}^n \rightarrow \mathbb{R}^n$ defined by

$$f_s(x) = |x|^{s-1}x,$$

where $s \in \mathbb{R}_{>0}$. Note that f_s is continuous and has inverse given by $f_{\frac{1}{s}}(x)$ as can be verified. This also is bijective, and so $f_s(x)$ is a homeomorphism. However, $f_s(x)$ is not a diffeomorphism unless $s = 1$ and $x \neq 0$.

Now, let $\mathcal{A}$ be a smooth atlas of X . After possible refining this atlas, suppose that $\mathcal{A}$ comprises of charts which (cover X) and each of which is homeomorphic to an open ball. Then, again after possible refining this atlas once more, we can find a chart $(U, \varphi) \in \mathcal{A}$ such that its center p is not contained in any other chart. Note that this refinement process does not change the smooth structure, as we are only restricting our chart maps or shrinking our open sets. Then, define $\varphi_s := f_s \circ \varphi$. Then, consider the smooth structure determined by the atlas $\mathcal{B} := \mathcal{A} \setminus \{(U, \varphi)\} \cup (U, \varphi_s)$. Note that this is indeed a smooth atlas because all the charts in $\mathcal{A} \setminus \{(U, \varphi)\}$ do not contain p , and so the transition maps are diffeomorphisms (as the intersection of open balls do not contain 0, so f_s is a diffeomorphism). The claim is then that $\mathcal{B} \cup \mathcal{A}$ doesn't determine a smooth structure. The reason for this is that if we consider the charts (U, φ) and (U, φ_s) , they of course have nonempty intersection, but their transition maps are not diffeomorphisms:

$$f_s = (f_s \circ \varphi) \circ \varphi^{-1}: \varphi^{-1}(U) \rightarrow (f_s \circ \varphi)(U)$$

Note that this is not a diffeomorphism because $p \in U$, so f_s is not diffeomorphic at $0 = \varphi(p)$. $\square$

2 Differentiation on Maps between Manifolds

Remark 2.0.1 — Our definition of Smooth Manifolds does not require an ambient space.

Recall that graph of a continuous function is a smooth manifold, and our example of $F(x) = x^2$ and $G(x) = |x|$. These were homeomorphic as topological manifolds, but we see that G admits a singularity at $x = 0$, while F admits no singularities. Is there a *difference* then between these two topological manifolds? In other words, how can we put one manifold into another? These are the questions that will motivate the study in this chapter.

§2.1 Smooth Maps

Definition 2.1.1. Let X, Y be smooth manifolds. A map $F: X \rightarrow Y$ is called **smooth** if for any $p \in X$, there exist a chart (U, φ) of X and a chart (V, ψ) of Y such that:

1. $p \in U$ and $F(U) \subset V$
2. $\psi \circ F \circ \varphi^{-1}$ is a smooth map from $\varphi(U)$ to $\psi(V)$

The map $\psi \circ F \circ \varphi^{-1}$ is called a **coordinate representative**.

Notation 2.1.2. The set of smooth maps from X to Y is denoted as $C^\infty(X, Y)$.

Remark 2.1.3 — In particular, the second condition is saying that we require the map $\psi \circ F \circ \varphi^{-1}$ between open sets of Euclidean spaces is smooth. It is a reoccurring theme that we use the chart maps to restrict to Euclidean spaces.

Notation 2.1.4. Particularly, when $Y = \mathbb{R}$, we write $C^\infty(X) := C^\infty(X, \mathbb{R})$.

Example 2.1.5

Let X, Y be smooth manifolds. Then, the two projection maps

$$\pi_X: X \times Y \rightarrow X, \quad \pi_Y: X \times Y \rightarrow Y$$

are smooth.

Solution: Let $(x, y) \in X \times Y$ and $U \times V$ be a chart for $X \times Y$ containing (x, y) , with chart maps $\varphi \times \psi$. Let $x' = \varphi(x), y' = \psi(y)$. Let θ be a chart containing x in X . Then,

$$\theta \circ \pi_X \circ (\varphi \times \psi)^{-1}(x', y') = \theta \circ \varphi^{-1}(x'),$$

which is then smooth. □

Exercise 2.1.6. Let X be a smooth manifold. Show that pointwise multiplications turns $C^\infty(X)$ into a commutative ring and a commutative and associative algebra over $\mathbb{R}$.

Solution: Clearly, the usual addition of functions forms an abelian group. Further, multiplication is associative and unit is just the constant 1 function. Distributivity is also clear. $\square$

Remark 2.1.7 — The definition of smoothness doesn't depend on the choices of charts, since if we chose different charts, the transition maps are smooth by definition.

Proposition 2.1.8

Every smooth map between manifolds is continuous.

Proof. Let M and N be smooth manifolds with or without boundary, and $F: M \rightarrow N$ is smooth. Given $p \in M$, the smoothness of F means there are smooth charts (U, φ) containing p and (V, ψ) containing $F(p)$, such that $F(U) \subseteq V$ and $\psi \circ F \circ \varphi^{-1}: \varphi(U) \rightarrow \psi(V)$ is smooth; thus, continuous. Since $\varphi: U \rightarrow \varphi(U)$ and $\psi: V \rightarrow \psi(V)$ are homeomorphisms, these imply that

$$F|_U = \psi^{-1} \circ (\psi \circ F \circ \varphi^{-1}) \circ \varphi: U \rightarrow V,$$

is a composition of continuous maps. Since F is continuous in a neighborhood of each point, it is continuous on M . $\square$

Exercise 2.1.9. Let $F: X \rightarrow Y$ be a smooth map. Then the restriction $F|_U: U \rightarrow Y$ is smooth for any open subset U , equipped with the open submanifold smooth structure.

Solution: Let $A \subset X$ and $B \subset Y$ be open charts (with chart maps φ and ψ , respectively) so that $\psi \circ F \circ \varphi^{-1}$ is smooth. By the open submanifold structure, $U \cap A$ is a chart for U , so $\psi \circ F \circ (\varphi|_{U \cap A})^{-1}$ is smooth, since it is only the restriction of the original map. $\square$

Example 2.1.10

Let X, Y be two smooth manifolds. Consider the two projection maps $\pi_X: X \times Y \rightarrow X$ and $\pi_Y: X \times Y \rightarrow Y$. Then, both π_X and π_Y are smooth.

Proof. Let X and Y be equipped with the atlas $\mathcal{A} = \{(U_i, \varphi_i)\}$ and $\mathcal{B} = \{(V_j, \psi_j)\}$, respectively. Then, we can define an atlas on $X \times Y$ by $\mathcal{C} = \{(U_i \times V_j, \varphi_i \times \psi_j)\}$. Consider the following:

$$\begin{array}{ccc} (a, b) \in X \times Y & \xrightarrow{\varphi_i \times \psi_j} & (\varphi_i(a), \psi_j(b)) \in \mathbb{R}^n \times \mathbb{R}^m \\ \downarrow \pi_X & & \downarrow \\ a \in X & \xrightarrow{\varphi_i} & \varphi_i(a) \in \mathbb{R}^n \end{array}$$

We ask if the dashed arrow is smooth, but this is just the projection map in Euclidean space, which we know is smooth. The same proof holds for π_Y . $\square$

Remark 2.1.11 — Let $F: X \rightarrow Y$ be a map. The smoothness of F does not depend on the choice of charts. That is, suppose we have (U, φ) a chart for a given point $p \in X$ and (V, ψ) a chart for Y and we know $\psi \circ F \circ \varphi^{-1}$ is smooth. Then, suppose we have another chart (U', φ') neighboring the same $p \in X$; then, $\psi \circ F \circ \varphi'^{-1}$ is smooth, because

$$(\psi \circ F \circ \varphi^{-1}) \circ (\varphi \circ \varphi'^{-1})$$

is smooth; indeed, $U \cap U' \neq \emptyset$ and the composition $\varphi \circ \varphi'^{-1}$ is a diffeomorphism.

Example 2.1.12

Let $X = (\mathbb{R}, \mathcal{A} = (\mathbb{R}, \varphi = \text{id}))$ and $Y = (\mathbb{R}, \mathcal{B} = (\mathbb{R}, \psi(x) = x^3))$ be two smooth manifolds. Construct a smooth map $F: X \rightarrow Y$ and $G: Y \rightarrow X$.

Solution: Indeed, we've seen that these do not determine the same smooth structure. We first see if $F(x) = x$ works. Then, the coordinate representative $\psi \circ F \circ \varphi^{-1}(x) = x^3$ is smooth! Let us try $G(x) = x$ for the backwards direction too. Then, we can see that $\varphi \circ G \circ \psi^{-1}(x) = \sqrt[3]{x}$, so this is not smooth. We can try $G(x) = x^3$ now. This works, as $\varphi \circ G \circ \psi^{-1}(x) = x$, and this is smooth. $\square$

Diffeomorphisms

Definition 2.1.13. Let $U \subseteq \mathbb{R}^n$ and $V \subseteq \mathbb{R}^m$ be open. Recall that a map $f: U \rightarrow V$ is a **diffeomorphism** if and only if f is bijective, smooth, and has a smooth inverse.

Remark 2.1.14 — Then, the Jacobian tells us that $n = m$ in this case.

Definition 2.1.15. Let X and Y be smooth manifolds. A smooth map $F: X \rightarrow Y$ is a **diffeomorphism** if it's bijective with a smooth inverse. We say that two manifolds are **diffeomorphic** if there exists a diffeomorphism between them.

Exercise 2.1.16. Check that being diffeomorphic is an equivalence relation between smooth manifolds.

Example 2.1.17

Let us construct a diffeomorphism between $(\mathbb{R}, \mathcal{A})$ and $(\mathbb{R}, \mathcal{B})$, where

$$\mathcal{A} = \{(U = \mathbb{R}, \varphi(x) = x)\}, \quad \mathcal{B} = \{(V = \mathbb{R}, \psi(x) = x^3)\}.$$

Previously, we showed that they give different smooth structure on $\mathbb{R}$. However, viewed as two different smooth manifolds, they are diffeomorphic. Consider the map

$$F: (\mathbb{R}, \mathcal{A}) \rightarrow (\mathbb{R}, \mathcal{B}), \quad x \mapsto x^{\frac{1}{3}}.$$

We have that $\psi \circ F \circ \varphi^{-1}(x) = x$ is smooth and bijective. On the other hand, $\varphi \circ F^{-1} \circ \psi^{-1}(x) = x$ is also smooth. Thus, F is a bijective smooth function with a smooth inverse.

Given this notion of diffeomorphism, it is natural to ask that, given a topological manifold, can one classify the smooth structures on it up to diffeomorphism? This question motivates many advances in modern research and is still open in general. Given the Euclidean space $\mathbb{R}^n$ with the standard topology:

- If $n \neq 4$, then there is a unique smooth structure on it, up to diffeomorphism.
- If $n = 4$, there exist uncountably many smooth structures, no two of which are diffeomorphic. This was proven by Simon Donaldson and Michael Freedman in 1984.

§2.1.i Gluing Smooth Maps

Definition 2.1.18. Let X be a smooth manifold. For a subset $K \subseteq X$, a function $F: K \rightarrow \mathbb{R}$ is called **smooth** if for any $x \in K$, there exists an open subset U_x of X containing x and a smooth function $\tilde{F}_x: U_x \rightarrow \mathbb{R}$ such that $\tilde{F}_x|_{U_x \cap K} = F$.

Remark 2.1.19 — When can generalize the above definition to $\mathbb{R}^n$.

Recall that if we have a smooth map $F: X \rightarrow Y$, then $F|_U: U \rightarrow Y$ is smooth, where U is open in X .

Lemma 2.1.20

Let X, Y be smooth manifolds. Let U, V be two open subsets of X whose union is X . Suppose we have two smooth maps

$$F_U: U \rightarrow Y, \quad F_V: V \rightarrow Y$$

such that $F_U = F_V$ on $U \cap V$. Then, there is a unique smooth map $F: X \rightarrow Y$ such that $F|_U = F_U$, $F|_V = F_V$.

Proof. The definition of F is clear since $F_U = F_V$ on $U \cap V$:

$$F(x) = \begin{cases} F_U(x), & x \in U \\ F_V(x) & x \in V \end{cases}$$

We first check that it is continuous. Choose an open subset B of Y . Then, we have

$$F^{-1}(B) = (F^{-1}(B) \cap U) \cup (F^{-1}(B) \cap V) = F_U^{-1}(B) \cup F_V^{-1}(B),$$

which is a union of two open subsets. Next, we show the smoothness of F . Pick $x \in X$, so that it is in U or V . Say $x \in U$. Since F_U is smooth, there exist charts (W, φ) of X and (W', φ') on Y such that $W \subseteq U$, $F_U(W) \subseteq W'$ such that $\varphi' \circ F_U \circ \varphi^{-1}$ is smooth. However, $\varphi' \circ F_U \circ \varphi^{-1} = \varphi' \circ F \circ \varphi^{-1}$, since $W \subseteq U$, so this is true. $\square$

Remark 2.1.21 — U, V being open in the proposition above is crucial, because otherwise, we can't choose an open subset B of Y to work with, or a chart W of U to work with.

Remark 2.1.22 — We note that the gluing of two smooth functions on closed subsets may not be smooth, as can be seen in the following example.

Example 2.1.23

Let $X = \mathbb{R}$, $U = (-\infty, 0]$, $V = [0, \infty)$, and $F_U(x) = -x$, $F_V(x) = x$. Then, $F(x) = |x|$ is continuous but not smooth.

Example 2.1.24

Let $X = \mathbb{R}$, and $U = (0, \infty)$, and $(-\infty, 0)$, then $F_U = 1$ and $F_V = 0$ has no way to extend it.

Lemma 2.1.25 (Cutoff Functions.)

Given any real number $r_1 < r_2$, there exists a smooth function $f: \mathbb{R} \rightarrow \mathbb{R}$ such that

- (i) $0 \leq f \leq 1$.
- (ii) f is non-increasing.
- (iii) $f(x) = 0$ when $x \geq r_2$.
- (iv) $f(x) = 1$ when $x \leq r_1$.
- (v) $0 < f(x) < 1$ when $r_1 < x < r_2$.

We call such a function a **smooth cutoff function**.

Proof. We can consider the function

$$g(x) = \begin{cases} e^{-\frac{1}{x}}, & x > 0 \\ 0, & x \leq 0 \end{cases}$$

and set

$$f(x) := \frac{g(r_2 - x)}{g(r_2 - x) + g(x - r_1)}.$$

By direct computation, we can check that this f is a function we want. $\square$

Remark 2.1.26 — The cutoff function is not unique, and f as above is an explicit example. Note that the product $f \cdot F$ of f and any smooth function $F: \mathbb{R} \rightarrow \mathbb{R}$ is a smooth function which equals F when $x \leq r_1$ and equals zero when $x \geq r_2$. Hence the name ‘cutoff’.

Notation 2.1.27. Let $r: \mathbb{R}^n \rightarrow \mathbb{R}$ denote the **radius function**:

$$r(x) := (x_1^2 + \cdots + x_n^2)^{\frac{1}{2}}.$$

Notation 2.1.28. A **closed ball** of radius $c > 0$ centered at the origin is

$$\overline{B}_c(0) = \{x \in \mathbb{R}^n : r(x) \leq c\}.$$

Notation 2.1.29. An **open ball** of radius $c > 0$ centered at the origin is

$$B_c(0) = \{x \in \mathbb{R}^n : r(x) < c\}.$$

Lemma 2.1.30

Given any positive real numbers $r_1 < r_2$, there exists a smooth function $h: \mathbb{R}^n \rightarrow \mathbb{R}$ such that

- (i) $h(x) = 1$ when $x \in \overline{B}_{r_1}(0)$
- (ii) $h(x) = 0$ when $x \notin B_{r_2}(0)$
- (iii) $0 < h(x) < 1$ when $r_1 < r(x) < r_2$

We call such a function a **smooth bump function**.

Proof. Set $h(x) = f(r(x))$, where f is the function we defined above. □

Remark 2.1.31 — Possibly, it may be that $h(x)$ has a bit of increasing, but what will really be useful is the assumption that $0 < h(x) < 1$ when $r_1 < r(x) < r_2$.

Remark 2.1.32 — The product of $h(x)$ in the lemma above with any smooth function is supported in $B_{r_2}(0)$.

§2.1.ii Partition of Unity

Definition 2.1.33. For any smooth function $f: X \rightarrow \mathbb{R}$ on a manifold, the **support** is defined as

$$\text{supp}(f) := \overline{\{x \in X : f(x) \neq 0\}}.$$

That is, the closure of the set of its nonzero points.

Definition 2.1.34. If $\text{supp } f$ is contained in some set $U \subset M$, we say that f is **supported** in U . If $\text{supp } f$ is also compact, then we say that f is **compactly supported**.

Remark 2.1.35 — It is clear that every function on a compact space is compactly supported.

Definition 2.1.36. Let $\mathcal{U} = \{U_\alpha\}_{\alpha \in A}$ be an open cover of a smooth manifold X . A **partition of unity subordinate to $\mathcal{U}$** is a collection of smooth functions $\{f_\alpha: X \rightarrow \mathbb{R}\}_{\alpha \in A}$ such that:

PU1 $0 \leq f_\alpha \leq 1$ for any $\alpha \in A$

PU2 $\text{supp}(f_\alpha) \subseteq U_\alpha$ for any $\alpha \in A$

PU3 Any point $x \in X$ admits a neighborhood which intersects $\text{supp}(f_\alpha)$ for only finitely many values of α

PU4 $\sum_{\alpha \in A} f_\alpha(x) = 1$ for any $x \in X$

Remark 2.1.37 — (PU3) is saying that there is a neighborhood for which there are finitely many functions in the collection which are nonzero. Therefore, the sum in (PU4) is actually a finite sum.

Example 2.1.38

A partition of unity for $\mathbb{S}^1$ is given by the functions $\{f_1 = \sin^2\left(\frac{\theta}{2}\right), f_2 = \cos^2\left(\frac{\theta}{2}\right)\}$ on $\{U_1 = (0, 2\pi), U_2 = (-\pi, \pi)\}$.

Theorem 2.1.39 (Existence of partition of unity)

For any open cover $\mathcal{U} = \{U_\alpha\}_{\alpha \in A}$ of a smooth manifold X , there exists a partition of unity subordinate to $\mathcal{U}$.

Proof. We will only prove the theorem when X is compact. The general case follows from similar constructions but with more involved topological modifications. For the general case, refer to Theorem 2.23 [1].

For any $x \in X$, it admits a coordinate chart (V_x, φ_x) . By shrinking it, we assume $V_x \subset U_\alpha$ for some α . Then we get another open cover $\mathcal{V} = \{V_x\}_{x \in X}$ of X . The compactness of X implies we have a finite subcover $\{V_{x_1}, \dots, V_{x_N}\}$. The image of any φ_{x_i} is an open ball $B_r(0)$ in $\mathbb{R}^n$. Now pick the function h in Lemma 2.1.30 which is positive on $B_r(0) = \varphi_{x_i}(V_{x_i})$ and zero outside. The new function $g_i := h \circ \varphi_{x_i}$ gives a smooth function on X which is positive on V_{x_i} and zero outside. Next we define $g := \sum_{i=1}^N g_i$ which is a smooth function on X . This is positive on X since any point x is contained in some V_{x_i} ; this enables us to define $f_i := \frac{g_i}{g}$. This then gives $\sum f_i = \sum \frac{g_i}{g} = 1$.

We now need to relabel so that the f_i are indexed by A . Note that for any i , the set V_{x_i} is contained in some U_α . We pick one of such U_α and write $F(i) = \alpha$. This choice is not unique but gives us a function $F : \{1, \dots, N\} \rightarrow A$. When we relabel, it may be that two f_i, f_j are indexed using the same α . In this case, we write $f_\alpha = f_i + f_j$; in other words, we make f_α the sum of such functions. To recheck the conditions using the relabeling, we get that for any $\alpha \in A$, define $f_\alpha := \sum_{F(i)=\alpha} f_i$. If α is not in the image of F , we set $f_\alpha = 0$. This collection $\{f_\alpha\}_{\alpha \in A}$ gives a partition of unity we are looking for. $\square$

Applications

Here is a basic application of the Existence of partition of unity.

Lemma 2.1.40 (Extension of Smooth Functions)

Let X be a smooth manifold and K be a closed subset. For any smooth function $F : K \rightarrow \mathbb{R}$, there exists a smooth function $G : X \rightarrow \mathbb{R}$ such that $G|_K = F$.

Proof. By the definition of F , any point $x \in K$, pick an open subset U_x containing x and a smooth function $G_x : U_x \rightarrow \mathbb{R}$ such that $G_x|_{U_x \cap K} = F$. Hence the collection

$\{U_x: x \in K\} \cup \{X \setminus K\}$ is an open cover of X . Let $\{f_x: x \in K\} \cup \{f_{X \setminus K}\}$ be a partition of unity subordinate to this open cover, which exists by the previous theorem. Next set

$$G(p) := \sum_{x \in K} f_x(p) \cdot G_x(p), \quad \forall p \in X.$$

By (PU3), the above sum is a finite sum at any x . Hence G is a well-defined smooth function. When $p \in K$, $\text{supp}(f_{X \setminus K}) \subset X \setminus K$, we have that $G_x(p) = F(p)$. Then, we can write the following:

$$\begin{aligned} G(p) &= \sum_{x \in K} f_x(p) G_x(p) \\ &= \underbrace{f_{X \setminus K}(p)}_{=0} F(p) + \sum_{x \in K} f_x(p) F(p) \\ &= F(p) \left(f_{X \setminus K}(p) + \sum_{x \in K} f_x(p) \right) \\ &= F(p), \end{aligned}$$

for all $p \in K$. Note that we can add in $f_{X \setminus K} G_x(p)$, since $f_{X \setminus K}(p) = 0$, as $\text{supp}(f_{X \setminus K}) \subset X \setminus K$. This implies $G|_K = F$, by (PU4). $\square$

Remark 2.1.41 — If f is smooth and supported in U and $g: U \rightarrow \mathbb{R}$ is smooth, then $f \cdot g$ is a globally defined smooth function on X .

Remark 2.1.42 — The K being closed is really crucial; in fact, it is easy to construct smooth functions which do not extend, such as the following.

Example 2.1.43

Consider $U = \mathbb{R} \setminus \{0\}$, which is an open subset of $\mathbb{R}$. Define

$$F(x) := \begin{cases} 1, & x > 0 \\ -1, & x < 0 \end{cases}.$$

This is a smooth function on U . However, it doesn't even admit a continuous extension.

Exercise 2.1.44. Show the following strong version. Let X be a smooth manifold, with $K \subseteq X$ be closed, and $F: K \rightarrow \mathbb{R}^n$ be a smooth function. Then, for any open U containing K , there exists a smooth function $G: X \rightarrow \mathbb{R}^n$ such that $G|_K = F$ and $\text{supp}(G) \subseteq U$.

Solution: Let $x \in K$, so that U_x is an open neighborhood of x . Replacing U_x with $U_x \cap U$ if necessary, we may assume that $U_x \subset U$. Then, there is some smooth function $g_x: U_x \rightarrow \mathbb{R}^n$ which agrees with F on $U_x \cap K$: $g_x|_{U_x \cap K} = F_x$. Then, $\{U_x: x \in K\} \cup \{X \setminus K\}$ is an open cover of X . Then, let $\{\rho_x: x \in K\} \cup \{\rho_{X \setminus K}\}$ be a partition of unity subordinate to this open cover. We shall note then that $\text{supp}(\rho_x) \subseteq U_x \subseteq U$. $\rho_x g_x$ is then smooth on U_x , and it has a smooth extension to all of X if we view it to be zero on $X \setminus \text{supp} \rho_x$. We define

$$G(p) = \sum_{x \in K} (\rho_x \cdot g_x)(p)$$

This is well defined, since the collection $\text{supp } \rho_x$ is locally finite. Then, note that $\text{supp } (\rho_{X \setminus K}) \subseteq X \setminus K$ and if $p \in K$, then $g_x(p) = F(p)$. It follows that then $G(p) = F(p)$, for all $p \in K$. To check the condition on the support, we have that

$$\text{supp } (G) = \overline{\bigcup_{x \in K} \text{supp } (\rho_x)} = \bigcup_{x \in K} \text{supp } (\rho_x) \subseteq U.$$

□

Given the extension above, we ask the following. Given manifolds X and Y , a closed subset $K \subseteq X$, and a smooth $F: K \rightarrow Y$, can we extend F to a smooth function globally on X ? The answer is in general no.

Example 2.1.45

Let $X = \mathbb{R}$, and $Y = \mathbb{R} \sqcup \mathbb{R} = \{(x, y): \mathbb{R}^2: y = \pm 1\}$. Consider $K = \{-1, 1\} \subseteq X$, and consider $F: K \rightarrow Y$ such that $F(-1) = (0, -1)$ and $F(1) = (0, 1)$. F is smooth, since we can take an open neighborhood around 1, say $(1 - \epsilon, 1 + \epsilon)$ and map it to a neighborhood of $(0, 1)$, say $U = ((-\delta, \delta), 1)$ (abuse of notation). Can we find a $G: \mathbb{R} \rightarrow \mathbb{R} \sqcup \mathbb{R}$ such that $G|_K = F$?

The answer is no. Smooth maps are continuous. $\mathbb{R}$ is connected, and -1 and 1 are mapped to distinct components in $\mathbb{R}^2$.

Example 2.1.46

Let $X = \mathbb{R}^2$ and $\mathbb{S}^1 = K \subseteq X$, and $Y = \mathbb{S}^1$. Let $F = \text{id}: K \subseteq Y$. This is a smooth map; however, it doesn't even admit a continuous extension to $\mathbb{R}^2$. This is due to the fact that the fundamental group of $\mathbb{R}^2$ is zero while the fundamental group of $\mathbb{S}^1$ is $\mathbb{Z}$ (non-examinable content).

Proposition 2.1.47 (Theorem 2.29 in [1])

Let K be a closed subset of $\mathbb{R}^n$. There exists a smooth nonnegative function $f: \mathbb{R}^n \rightarrow \mathbb{R}$ such that $f^{-1}(0) = K$.

Proof. For any $p \in \mathbb{R}^n - K$, there is an open ball $B_r(p) \subset \mathbb{R}^n - K$ for some $0 < r < 1$. All such open balls form a cover of $\mathbb{R}^n \setminus K$. Since $\mathbb{R}^n$ is second countable, it admits a countable subcover $\{B_{r_i}(x_i)\}$.

Fix a bump function $h: \mathbb{R}^n \rightarrow \mathbb{R}$ which is supported in $B_1(0)$ and is one on $B_{\frac{1}{2}}(0)$. For any $i \geq 1$ pick a constant $C_i > 1$ which is larger than all the partial derivatives of h up to order i . Next define

$$f: \mathbb{R}^n \rightarrow \mathbb{R}, \quad f(x) = \sum_{i=1}^{\infty} \frac{(r_i)^i}{2^i \cdot C_i} h\left(\frac{x - x_i}{r_i}\right).$$

The terms of the series are bounded in absolute value by those of the convergent series $\sum_i \frac{1}{2^i}$, so the entire series converges uniformly to a continuous function by the Weierstrass M -test. Because the i th term is positive exactly when $x \in B_{r_i}(x_i)$, it follows that f is zero on K and positive elsewhere.

It remains only to show that f is smooth. We have already shown that it is continuous, so suppose $k \geq 1$ and assume by induction that all partial derivatives of f of order less than k exist and are continuous. By the chain rule and induction, every k th partial derivative of the i th term in the series can be written in the form

$$\frac{(r_i)^{i-k}}{2^i C_i} D_k h\left(\frac{x - x_i}{r_i}\right),$$

where $D_k h$ is some k th partial derivative of h . By our choices of r_i and C_i , as soon as $i \geq k$, each of these terms is bounded in absolute value by $1/2^i$, so the differentiated series also converges uniformly to a continuous function. It then follows from Theorem C.31 that the k th partial derivatives of f exist and are continuous. This completes the induction, and shows that f is smooth. $\square$

Another application of partition of unity is the following.

Corollary 2.1.48

Let K be a closed subset of a smooth manifold M . There exists a nonnegative smooth function $f: M \rightarrow \mathbb{R}$ such that $f^{-1}(0) = K$.

Proof. Let $\{(U_\alpha, \varphi_\alpha)\}$ be an open cover of M such that $\varphi_\alpha(U_\alpha)$ is an open ball in $\mathbb{R}^n$. Since each open ball is diffeomorphic to $\mathbb{R}^n$, we can apply the previous result to say that there is a smooth function $f_\alpha: \varphi_\alpha(U_\alpha) \rightarrow \mathbb{R}^n$ with $f_\alpha^{-1}(0) = \varphi_\alpha(K \cap U_\alpha)$. This is because $K \cap U_\alpha$ is closed as a subset of U_α (with the subspace topology if you like), and φ_α is a homeomorphism, so the resulting set $\varphi_\alpha(K \cap U_\alpha)$ is closed as a subspace of $\varphi_\alpha(U_\alpha)$. Finally, pick a partition of unity $\{g_\alpha\}$ subordinate to $\{(U_\alpha, \varphi_\alpha)\}$. The function

$$f := \sum_{\alpha} g_{\alpha} \cdot (f_{\alpha} \circ \varphi_{\alpha})$$

is the function we want. If $x \in K$, then $f_{\alpha}(\varphi_{\alpha}(x)) = 0$, and if $x \notin K$, then $f_{\alpha}(\varphi_{\alpha}(x)) \neq 0$, and g_{α} are positive, so $f \neq 0$. Thus, we are done. $\square$

Corollary 2.1.49 (Problem 2.14 in [1])

Let K_0, K_1 be two disjoint closed subsets of X . Then, there exists f smooth, $f^{-1}(0) = K_0$, $f^{-1}(1) = K_1$.

Proof. Let f_0, f_1 be nonnegative smooth functions such that $f_0^{-1}(0) = K_0$, $f_1^{-1}(1) = K_1$. In other words, f_i is 0 on K_i , and positive elsewhere. Then,

$$f(x) = \frac{f_0(x)}{f_0(x) + f_1(x)}$$

clearly works. If $x \in K_0$, then $f(x) = 0$. If $x \in K_1$, then $f(x) = 1$. $\square$

§2.2 Tangent Space

A fundamental concept in calculus is *linearization*. That is, using linear objects to approximate nonlinear objects. One basic example is using the tangent line of a function to approximate a general function, at least locally. We generalize this idea to smooth manifolds.

§2.2.i Tangent vectors

Recall that in order to differentiate a function $F: \mathbb{R}^n \rightarrow \mathbb{R}$, one needs to first choose a direction. We first define a notion of direction on general smooth manifolds.

Definition 2.2.1. Let X be a smooth manifold and $p \in X$ be a point. A **tangent vector** at p is a map $v: C^\infty(X) \rightarrow \mathbb{R}$ which satisfies the following:

1. **Linearity:** $v(af + bg) = av(f) + bv(g)$ for all $a, b \in \mathbb{R}$ and $f, g \in C^\infty(X)$.
2. **Leibniz rule:** $v(fg) = f(p)v(g) + v(f)g(p)$ for all $f, g \in C^\infty(X)$.

Remark 2.2.2 — This was a formal definition, but there is a more geometric definition that we could've provided by considering the velocity along a curve. Given a manifold X , consider a smooth curve $\gamma: \mathbb{R} \rightarrow X$ such that $\gamma(0) = p$. Then, we can define a velocity vector at p by *pushing forward* at p .

Example 2.2.3 (Directional Derivatives)

Let $X = \mathbb{R}^2$. Directional derivatives are tangent vectors. We write

$$v = a \frac{\partial}{\partial x} + b \frac{\partial}{\partial y}.$$

For any $p \in \mathbb{R}^2$ set

$$v(f) := a \frac{\partial f}{\partial x}(p) + b \frac{\partial f}{\partial y}(p),$$

for all $f \in C^\infty(\mathbb{R}^2)$. We verify linearity:

$$v(cf + dg) = cv(f) + dv(g).$$

Similarly, the Leibniz rule follows from the product rule. Similarly, the directional derivative of any smooth function on $\mathbb{R}^n$ is a tangent vector.

Remark 2.2.4 — This example indicates that the notion of a *tangent vector* is designed as an abstract analogue of ‘direction’. Once it is applied to a smooth function, we get a ‘directional derivative’.

Exercise 2.2.5. For any tangent vector v at p , the following hold:

1. If f is a constant function, then $v(f) = 0$.
2. If $f(p) = g(p) = 0$, then $v(fg) = 0$.

Solution: 1. Let $f \equiv c$, for some $c \in \mathbb{R}$. First, note that $v(1) = 1 \cdot v(1) + v(1) \cdot 1$, so $v(1) = 0$. Therefore, $v(c) = c \cdot v(1) = 0$.

2. This is clear via Leibniz. □

Definition 2.2.6. Let X be a smooth manifold and $p \in X$ be a point. The **tangent space** of X at p is the set of all tangent vectors at p , written as $T_p X$.

Remark 2.2.7 — It is clear that the space of all tangent vectors at p form a vector space over $\mathbb{R}$. We define

$$(av + bu)(f) := av(f) + bu(f).$$

We also define the zero vector as $0(f) = 0$, for all f .

Remark 2.2.8 — We note that $C^\infty(\mathbb{R})$ is not finite dimensional over $\mathbb{R}$. In other words, for any positive integer N , we can find $f_1, \dots, f_{N+1}$ which are in $C^\infty(\mathbb{R})$ and are linearly independent. For instance, we can take the f_i to be bump functions with disjoint support. Thus, we see that $C^\infty(X)$ is definitely not finitely dimensional. However, we will see that the tangent space is finite dimensional.

Let us start with the case that $X = \mathbb{R}^n$.

Lemma 2.2.9

Let $X = \mathbb{R}^n$ and $p \in \mathbb{R}^n$ be a point. The map

$$F: T_p X \rightarrow \mathbb{R}^n, \quad F(v) := (v(x_1), \dots, v(x_n))$$

is a linear isomorphism (between finite dimensional vector spaces). Here $x_1, \dots, x_n$ are the coordinate functions on $\mathbb{R}^n$.

Proof. We start with linearity. Let $a, b \in \mathbb{R}$. Then

$$\begin{aligned} F(av_1 + bv_2) &= ((av_1 + bv_2)(x_1), \dots, (av_1 + bv_2)(x_n)) \\ &= (av_1(x_1) + bv_2(x_1), \dots, av_1(x_n) + bv_2(x_n)) = aF(v_1) + bF(v_2). \end{aligned}$$

Since we've verified that F is a linear map, it suffices to check bijectivity.

We check surjectivity first. Let $(a_1, \dots, a_n)$ be any element in $\mathbb{R}^n$. Set

$$v := a_1 \frac{\partial}{\partial x_1} \Big|_p + \dots + a_n \frac{\partial}{\partial x_n} \Big|_p.$$

Then v is a tangent vector and $F(v) = (v(x_1), \dots, v(x_n)) = (a_1, \dots, a_n)$.

What remains is injectivity. Let $v \in T_p X$ with $F(v) = 0$. We will show v is the zero tangent vector. That is, $v(f) = 0$ for any $f \in C^\infty(X)$. Consider the Taylor expansion of f at p :

$$f(x) = f(p) + \sum_{i=1}^n \frac{\partial f}{\partial x_i}(p) (x_i - p_i) + \sum_{i,j=1}^n (x_i - p_i)(x_j - p_j) \int_0^1 (1-t) \frac{\partial^2 f}{\partial x_i \partial x_j}(p + t(x-p)) dt.$$

The first term is a constant function, so its derivative under v is zero. The second term also becomes zero since $v(p_i) = 0$ and since we assume $v(x_i) = 0$ (from $F(v) = 0$). The third term is a sum of products of $(x_i - p_i)$ and hence vanishes under v by the exercise above. Therefore, $v(f) = 0$. Hence v is the zero tangent vector. This shows F is injective. $\square$

Remark 2.2.10 — We note that the tangent space at any point in $\mathbb{R}^n$ is a real n -dimensional vector space.

Corollary 2.2.11

The set

$$\left\{ \frac{\partial}{\partial x_1} \Big|_p, \dots, \frac{\partial}{\partial x_n} \Big|_p \right\}$$

form a basis for $T_p\mathbb{R}^n$.

Remark 2.2.12 — While the definition of a tangent vector uses functions which are globally on the manifold, the tangent vector is usually a local notion.

Lemma 2.2.13

Let $f, g \in C^\infty(X)$ which agree on a neighborhood U of $p \in X$, i.e. $f|_U = g|_U$. Then $v(f) = v(g)$ for any $v \in T_pX$.

Proof. Set $h := f - g$. There exists an open set U containing p such that $h|_U = 0$. Pick another open set V such that $p \in V \subset \bar{V} \subset U$, and a smooth bump function φ that is zero on V and is one on $X \setminus U$. Note that we can find $\bar{V}$ by the locally-Euclidean property of X . The function φh is globally smooth and equals h everywhere on X ; if $x \in U$, then $\varphi h(x) = 0 = h(x)$. If $x \notin U$, then $\varphi(x) = 1$, so $(\varphi h)(x) = h(x)$. Hence

$$v(f) - v(g) = v(h) = v(\varphi h) = 0.$$

The last equality holds because both φ and h vanish at p . □

§2.2.ii The Differential of a Smooth Map

Recall that for a smooth map between Euclidean spaces, the Jacobian matrix is a linear map that represents the ‘best linear approximation’ of this map. We now generalize this to smooth maps between smooth manifolds.

Definition 2.2.14. Let $F: X \rightarrow Y$ be a smooth map. For any point $p \in X$, we define a map

$$dF_p: T_pX \rightarrow T_{F(p)}Y,$$

called the **differential** of F at p as follows. For any $v \in T_pX$ and any $f \in C^\infty(Y)$, we define

$$dF_p(v)(f) := v(f \circ F).$$

Remark 2.2.15 — We may also call this the **tangent map** of F at p .

Exercise 2.2.16. Suppose M and N are smooth manifolds with or without boundary, and $F: M \rightarrow N$ is a smooth map. Show that

$$dF_p: T_pM \rightarrow T_{F(p)}N$$

is the zero map for each $p \in M$ if and only if F is constant on each component of M .

Solution: Let F be constant on each component of M . Let U_p be the component containing p , so that for $v \in T_p M$ and $f \in C^\infty(N)$, $f \circ F = f \circ F|_{U_p}$ on U_p . Then, $v(f \circ F) = v(f \circ F|_{U_p})$ for all $v \in T_p M$. Therefore,

$$dF_p(v)(f) = v(f \circ F) = v(f \circ F|_{U_p}) = 0.$$

Since this p was arbitrary and U_p was arbitrary, it follows that dF_p is the zero map.

For the converse, let $S := \{q \in M : F(p) = F(q)\}$. We show that this is a component. First, S is clearly closed because $S = F^{-1}(\{F(p)\})$. Let q be in S . We now show that this is open. Let U be a chart of q with coordinate functions $x_1, \dots, x_n$, and V be a chart of $F(q)$ with coordinates functions $y_1, \dots, y_m$. Then for any point $r \in U$, we get that $dF_r = 0$; in coordinates, if we write $F = (F_1(x_1, \dots, x_n), \dots, F_m(x_1, \dots, x_n))$, this conditions translates to

$$\frac{\partial F_i}{\partial x_j} = 0,$$

for all $i = 1, \dots, m$ and $j = 1, \dots, n$. This means that F is constant on U , so $U \subset S$; this implies that S is open. Because S is closed and open, it's a union of connected components, so is a component. $\square$

The following properties of the differential are direct consequences of the definition.

Proposition 2.2.17 (Properties of the Differential)

Let $F: X \rightarrow Y$, $G: Y \rightarrow Z$ be smooth maps and let $p \in X$ be a point.

1. $dF_p: T_p X \rightarrow T_{F(p)} Y$ is linear.
2. $d(G \circ F)_p = dG_{F(p)} \circ dF_p$.
3. $d(\text{id})_p = \text{id}: T_p X \rightarrow T_p X$.
4. If F is a diffeomorphism then dF_p is an isomorphism and $(dF_p)^{-1} = (dF^{-1})_{F(p)}$.

Proof. 1. $dF_p(av + bw)(f) = (av + bw)(f \circ F) = av(f \circ F) + bw(f \circ F) = a \cdot dF_p(v)(f) + b \cdot dF_p(w)(f)$.

$$2. d(G \circ F)_p(v)(f) = v(f \circ G \circ F) = dF_p(v)(f \circ G) = dG_{F(p)} \circ dF_p(v)(f).$$

$$3. d(\text{id})_p(v)(f) = v(f \circ \text{id}) = v(f).$$

4. If F is a diffeomorphism, then F^{-1} is smooth and exists. Thus, by 2 and 3,

$$\text{id} = d(F^{-1} \circ F)_p = (dF^{-1})_{F(p)} \circ dF_p,$$

$$\text{so } (dF_p)^{-1} = (dF^{-1})_{F(p)}.$$

$\square$

Note that we have shown that the tangent space at any point in $\mathbb{R}^n$ is an n -dimensional vector space.

Example 2.2.18

Let $F: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ map $(x, y) \mapsto (u, v)$. Compute dF_p . Then,

$$\begin{aligned} T_p \mathbb{R}^2 &= \left\{ \frac{\partial}{\partial x}, \frac{\partial}{\partial y} \right\} \\ T_p \mathbb{R}^2 &= \left\{ \frac{\partial}{\partial u}, \frac{\partial}{\partial v} \right\}. \end{aligned}$$

First, we have the following:

$$dF_p \left(\frac{\partial}{\partial x} \right) = a \frac{\partial}{\partial u} + b \frac{\partial}{\partial v} \in T_{F(p)} \mathbb{R}^2.$$

We have that $dF_p \left(\frac{\partial}{\partial x} \right) (f) = \frac{\partial}{\partial x} (f \circ F)$, and $F(x, y) = (u(x, y), v(x, y))$ and $(f \circ F)(x, y) = f((u(x, y), v(x, y)))$. Then,

$$a \frac{\partial}{\partial u} + b \frac{\partial}{\partial v} = \frac{\partial}{\partial x} (f \circ F) = \frac{\partial f}{\partial u} \frac{\partial u}{\partial x} + \frac{\partial f}{\partial v} \frac{\partial v}{\partial x}.$$

Therefore, $a = \frac{\partial u}{\partial x}$ and $b = \frac{\partial v}{\partial x}$. The claim is then that $dF_p(v)$ is the Jacobian matrix:

$$dF_p(v) = \begin{pmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{pmatrix}$$

Next we show it is also true for smooth manifolds.

Proposition 2.2.19

Let U be an open subset of a smooth manifold X and let $i: U \rightarrow X$ be the inclusion map. The differential $di_p: T_p U \rightarrow T_p X$ is an isomorphism for any $p \in U$.

Proof. First is injectivity. Suppose $di_p(v) = 0$. We need to show $v = 0 \in T_p U$, so suppose not. There exists $f \in C^\infty(U)$ with $v(f) \neq 0$. Pick a nonempty open subset V such that $p \in V \subset \bar{V} \subset U$. Then f is also smooth on $\bar{V}$. By the smooth extension lemma of smooth functions on closed sets, there is a smooth extension $\tilde{f} \in C^\infty(X)$ of $f|_{\bar{V}}$. We get two smooth functions $f, \tilde{f}$ on U , which agree on V . Hence we apply Lemma 2.2.13 to get

$$0 \neq v(f) = v(\tilde{f}|_U) = v(\tilde{f} \circ i) = di_p(v)(\tilde{f}) = 0,$$

a contradiction.

Next is surjectivity. For any $f \in C^\infty(U)$, we keep the above notation that $\tilde{f}$ is one smooth extension of $f|_{\bar{V}}$ for some $\bar{V}$. Next, for any $w \in T_p X$, we define a map

$$w': C^\infty(U) \rightarrow \mathbb{R}, \quad w'(f) := w(\tilde{f}).$$

Note that this is well defined even if we take two different extensions coming from different open subsets V and V' of U , then this well-definedness comes from taking a smaller open subset (contained by V and V'). Again, by Lemma 2.2.13 we have w' is well-defined. More precisely, different choices of extensions give the same $w'(f)$. Moreover, one can directly check that w' is linear and satisfies the Leibniz rule. Hence $w' \in T_p U$. Now for any $g \in C^\infty(X)$, we have

$$di_p(w')(g) = w'(g \circ i) = w'(g|_U) = w(g).$$

The last equality holds because g is a smooth extension of $g|_{\overline{V}}$. Hence $di_p(w') = w$. $\square$

Corollary 2.2.20

Let $p \in X$ be a point in an n -dimensional smooth manifold X . Then T_pX is an n -dimensional $\mathbb{R}$ -vector space.

Proof. Let (U, φ) be a smooth chart containing p . Then φ is a diffeomorphism and its differential is an isomorphism. Hence we get three isomorphisms $di_p: T_pU \rightarrow T_pX$, $d\varphi_p: T_pU \rightarrow T_{\varphi(p)}\varphi(U)$, and $di_{\varphi(p)}: T_{\varphi(p)}\varphi(U) \rightarrow T_{\varphi(p)}\mathbb{R}^n$. We already know that $T_{\varphi(p)}\varphi(U)$ is an n -dimensional vector space.

$$\begin{array}{ccc} T_pU & \xrightarrow[\sim]{d\varphi_p} & T_{\varphi(p)}\varphi(U) \\ \wr \downarrow di_p & & \wr \downarrow di_{\varphi(p)} \\ T_pX & \dashrightarrow & T_{\varphi(p)}\mathbb{R}^n \end{array}$$

$\square$

A Simple Description of Tangent Vectors

Now we arrive at a simpler description of tangent vectors, by using coordinate charts. Let X be a smooth manifold and let (U, φ) be a smooth chart containing $p \in X$. Since $\varphi(U)$ is an open subset of $\mathbb{R}^n$, the coordinate functions $x_1, \dots, x_n$ are smooth functions on U , and the directional derivative operators

$$\left. \frac{\partial}{\partial x_1} \right|_{\varphi(p)}, \dots, \left. \frac{\partial}{\partial x_n} \right|_{\varphi(p)}$$

form a basis of $T_{\varphi(p)}\mathbb{R}^n$. The above corollary showed that $d\varphi_p: T_pU \rightarrow T_{\varphi(p)}\varphi(U)$ is an isomorphism between vector spaces. In particular, we have

$$(d\varphi_p)^{-1} \left. \frac{\partial}{\partial x_1} \right|_{\varphi(p)}, \dots, (d\varphi_p)^{-1} \left. \frac{\partial}{\partial x_n} \right|_{\varphi(p)}$$

form a basis of T_pX . Note that we already identified T_pX with T_pU . By abusing the notation a bit, we write

$$\left. \frac{\partial}{\partial x_i} \right|_p := (d\varphi_p)^{-1} \left. \frac{\partial}{\partial x_i} \right|_{\varphi(p)}, \quad \forall 1 \leq i \leq n.$$

This is an explicit basis of T_pX by using the coordinate chart (U, φ) . In other words, any tangent vector $v \in T_pX$ can be written as a linear combination of them.

Exercise 2.2.21. Let (U, φ) and (V, ψ) be two coordinate charts containing $p \in X$. Then, we get another basis of T_pX . Work out the change-of-basis matrix from φ to ψ .

Solution: On one hand, we have $\left. \frac{\partial}{\partial x_1} \right|_p, \dots, \left. \frac{\partial}{\partial x_n} \right|_p$, and on the other, we have $\left. \frac{\partial}{\partial y_1} \right|_p, \dots, \left. \frac{\partial}{\partial y_n} \right|_p$. Pick $v \in T_pX$. Using φ , we have

$$v = \sum_{i=1}^n v_i \left. \frac{\partial}{\partial x_i} \right|_p$$

and using ψ , we get

$$v = \sum_{i=1}^n \tilde{v}_i \frac{\partial}{\partial y_i}.$$

Then, check

$$\frac{\partial}{\partial x_1} \mapsto a_1 \frac{\partial}{\partial y_1} + \cdots + a_n \frac{\partial}{\partial y_n}.$$

$\frac{\partial}{\partial x_1} \mapsto d(\psi \circ \varphi^{-1})\left(\frac{\partial}{\partial x}\right)$, so

$$\begin{aligned} d(\psi \circ \varphi^{-1})\left(\frac{\partial}{\partial x_1}\right)(f) &= \frac{\partial}{\partial x_1} f(y_1(x_1, \dots, x_n), \dots, y_n(x_1, \dots, x_n)) \\ &= \sum_{i=1}^n \frac{\partial f}{\partial y_i} \frac{\partial y_i}{\partial x_1}, \end{aligned}$$

Therefore, we have that

$$d(\psi \circ \varphi^{-1}): \frac{\partial}{\partial x_i} \mapsto \sum_{j=1}^n \frac{\partial y_j}{\partial x_i} \frac{\partial}{\partial y_j}.$$

In other words, the change of basis matrix is just the Jacobian of the transition map $\psi \circ \varphi^{-1}$. Therefore, we have $d(\psi \circ \varphi^{-1})$ maps $v_1 \frac{\partial}{\partial x_1} + \cdots + v_n \frac{\partial}{\partial x_n}$ to

$$\left(v_1 \frac{\partial y_1}{\partial x_1} + v_2 \frac{\partial y_1}{\partial x_2} + \cdots + v_n \frac{\partial y_1}{\partial x_n}\right) \frac{\partial}{\partial y_1} + \cdots + \left(v_1 \frac{\partial y_n}{\partial x_1} + v_2 \frac{\partial y_n}{\partial x_2} + \cdots + v_n \frac{\partial y_n}{\partial x_n}\right) \frac{\partial}{\partial y_n}.$$

□

§2.2.iii The Tangent Bundle

Definition 2.2.22. Given a smooth manifold X , we define the **tangent bundle** of X , denoted by TX , to be the disjoint union of the tangent spaces at all points of X :

$$TX := \bigsqcup_{p \in X} T_p X.$$

This is the first time we meet with the notion of a *vector bundle*. As the name suggests, it is a collection of vectors bundled together in a particular geometric way. We will present a more precise definition later. The tangent bundle is a collection of tangent vectors, and it carries rich geometric information of the underlying manifold. First, we show it is also a smooth manifold.

Theorem 2.2.23 (Topology on TX)

For any n -dimensional smooth manifold X , its tangent bundle TX has a natural topology and a smooth structure which make it into a $2n$ -dimensional smooth manifold.

Proof. Any element in TX can be written as a pair (p, v) where p is a point in X and $v \in T_p X$. Hence there is a natural projection $\pi: TX \rightarrow X$ which sends (p, v) to p . We will use this map π to pull back charts on X .

Let (U, φ) be a chart on X , note that $\pi^{-1}(U) \subset TX$ is the set of all tangent vectors to X at all points of U . Let $(x_1, \dots, x_n)$ denote the coordinate functions of φ , and define

$$\tilde{\varphi}: \pi^{-1}(U) \rightarrow \mathbb{R}^{2n}, \quad \tilde{\varphi} \left(p, \sum_{i=1}^n v_i \frac{\partial}{\partial x_i} \Big|_p \right) := (x_1(p), \dots, x_n(p), v_1, \dots, v_n).$$

The image of $\tilde{\varphi}$ is $\varphi(U) \times \mathbb{R}^n$, which is an open subset of $\mathbb{R}^{2n}$. Moreover, it is bijective. We define a subset of TX to be open if and only if it is generated by sets of the form $\tilde{\varphi}^{-1}(U')$ for some open $U' \subset \mathbb{R}^{2n}$. This gives a topology on TX , which is verified at the end.

Now let W be an open subset of TX . Then, note that for any chart (U, φ) , the image of $\tilde{\varphi}(\pi^{-1}(W)) = \varphi(U) \times \mathbb{R}^n$. By our definition of open subsets of TX , we have

$$\tilde{\varphi}: \pi^{-1}(U) \rightarrow \varphi(U) \times \mathbb{R}^n$$

is a homeomorphism, giving a topological atlas on TX . We now just check the charts are smooth. Let $(U, \varphi), (V, \psi)$ be two smooth charts on X and let $(\pi^{-1}(U), \tilde{\varphi}), (\pi^{-1}(V), \tilde{\psi})$ be the corresponding charts on TX . The transition map $\tilde{\psi} \circ \tilde{\varphi}^{-1}$ is

$$(x_1, \dots, x_n, v_1, \dots, v_n) \mapsto (y_1, \dots, y_n, \sum_{j=1}^n \frac{\partial y_1}{\partial x_j} v_j, \dots, \sum_{j=1}^n \frac{\partial y_n}{\partial x_j} v_j).$$

Here, $x_1, \dots, x_n$ are coordinates of φ and $y_1, \dots, y_n$ are coordinates of ψ . The first n coordinates is smooth, since $\psi \circ \varphi^{-1}$ is smooth. As for the last n coordinates, this is just the change of basis formula. Therefore, this map is smooth.

We verify that this gives a topology on TX by showing that these sets $\tilde{\varphi}^{-1}(U')$ form a basis for a topology. For this, we show the covering property and the intersection property. For any $v \in T_p X$, choose a chart (U, φ) containing p , so that $v \in \pi^{-1}(U)$. $\tilde{\varphi}$ is bijective onto $\varphi(U) \times \mathbb{R}^n$, so take a small open neighborhood U' around $\tilde{\varphi}(v)$ so that $v \in \tilde{\varphi}^{-1}(U')$. We now show that if $v \in \tilde{\varphi}_1^{-1}(U'_1) \cap \tilde{\varphi}_2^{-1}(U'_2)$, then there is some basis element which contains v . Consider the transition map $\tilde{\varphi}_2 \circ \tilde{\varphi}_1^{-1}: \varphi_1(U_1 \cap U_2) \times \mathbb{R}^n \rightarrow \varphi_2(U_1 \cap U_2) \times \mathbb{R}^n$

We verify that this gives a topology on TX by showing that these sets $\tilde{\varphi}^{-1}(U')$ form a basis for a topology. For this, we show the covering property and the intersection property. For any $v \in T_p X$, choose a chart (U, φ) containing p , so that $v \in \pi^{-1}(U)$. $\tilde{\varphi}$ is bijective onto $\varphi(U) \times \mathbb{R}^n$, so take a small open neighborhood U' around $\tilde{\varphi}(v)$ so that $v \in \tilde{\varphi}^{-1}(U')$. We now show that if $v \in \tilde{\varphi}_1^{-1}(U'_1) \cap \tilde{\varphi}_2^{-1}(U'_2)$, then there is some basis element which contains v . Consider the transition map $\tilde{\varphi}_2 \circ \tilde{\varphi}_1^{-1}: \varphi_1(U_1 \cap U_2) \times \mathbb{R}^n \rightarrow \varphi_2(U_1 \cap U_2) \times \mathbb{R}^n$. Because $v = \tilde{\varphi}_1(x)$, it follows that the intersection property $v \in \tilde{\varphi}_1^{-1}(U'_1) \cap \tilde{\varphi}_2^{-1}(U'_2)$ corresponds to

$$x \in U'_1 \cap (\varphi_1 \circ \varphi_2^{-1}(U'_2)),$$

which is open in $\mathbb{R}^{2n}$. Therefore, we may find some subset $V \subseteq U'_1 \cap (\varphi_1 \circ \varphi_2^{-1}(U'_2))$ containing x . Then, $v \in \varphi_1^{-1}(V) \subseteq \tilde{\varphi}_1^{-1}(U'_1) \cap \tilde{\varphi}_2^{-1}(U'_2)$, so we are done. $\square$

Corollary 2.2.24

Let X be an n -dimensional smooth manifold. If X can be covered by a global smooth chart, then TX is diffeomorphic to $X \times \mathbb{R}^n$ with the product smooth structure.

The above corollary gives us a special class of manifolds whose tangent bundle is *trivial*.

Definition 2.2.25. The tangent bundle TX of X is called **trivial** if TX is diffeomorphic to $X \times \mathbb{R}^n$.

Example 2.2.26

Let U be an open subset of $\mathbb{R}^n$. Then, TU is diffeomorphic to $U \times \mathbb{R}^n$. This diffeomorphism maps (p, v) to $(p, v_1, \dots, v_n)$, where $v_1, \dots, v_n$ are the coefficients of $v = \sum_{i=1}^n v_i \frac{\partial}{\partial x_i}$.

Definition 2.2.27. Let $F: X \rightarrow Y$ be a smooth map. The **total differential** $dF: TX \rightarrow TY$ is defined as the map whose restriction to any $p \in X$ is dF_p . The total differential is also called the **tangent map** of F .

Corollary 2.2.28

If we denote TX, TY as two smooth manifolds with the natural smooth structures, then dF is a smooth map.

Corollary 2.2.29

Let X, Y be two smooth manifolds. If $F: X \rightarrow Y$ is a diffeomorphism, then $dF: TX \rightarrow TY$ is a diffeomorphism.

Proof. Let $F: X \rightarrow Y$ be a diffeomorphism between X, Y . Then, $dF: TX \rightarrow TY$ is a smooth bijective map with a smooth inverse $d(F^{-1})$. $\square$

Remark 2.2.30 — The converse of the above corollary is not true. There exist manifolds X, Y which are not diffeomorphic but have diffeomorphic tangent bundles TX, TY . For instance, Lens spaces are such examples.

§2.2.iv Vector Fields

Definition 2.2.31. Let X be a smooth manifold and let TX be its tangent bundle. A **smooth vector field** on X is a smooth map $\sigma: X \rightarrow TX$ such that $\pi \circ \sigma$ is the identity map. Here, $\pi: TX \rightarrow X$ is the projection map. Note that we may also call σ a section of TX .

$$\begin{array}{ccc} X & \xrightarrow{\sigma} & TX \\ & \searrow \text{id} & \downarrow \pi \\ & & X \end{array}$$

In other words, $\sigma(p) \in T_p X$.

Remark 2.2.32 — In (local) coordinates, a vector field always looks like

$$v(p) = a_1(p) \frac{\partial}{\partial x_1} + \cdots + a_n(p) \frac{\partial}{\partial x_n}.$$

Example 2.2.33

If $X = \mathbb{R}^n$, then $TX \simeq X \times \mathbb{R}^n$, and the vector field is σ such that

$$\begin{array}{ccc} X & \xrightarrow{\sigma} & X \times \mathbb{R}^n \\ & \searrow \text{id} & \downarrow \pi \\ & & X \end{array}$$

commutes. Pick n smooth functions $f_1, \dots, f_n$ on $\mathbb{R}^n$. We then get a smooth vector field

$$\sigma(x) = f_1(x) \frac{\partial}{\partial x_1} + \cdots + f_n(x) \frac{\partial}{\partial x_n}$$

on $\mathbb{R}^n$. We reiterate that locally, a vector field will always look like this.

Exercise 2.2.34. Let X be a smooth manifold. Let $\mathfrak{X}(X)$ be the set of vector fields on X .

1. Show that $\mathfrak{X}(X)$ is a real vector space, see [1], Page 177.
2. For any smooth function f on X and $\sigma \in \mathfrak{X}(X)$, show that $(f \cdot \sigma)_p := f(p) \cdot \sigma(p)$ gives an element $f\sigma \in \mathfrak{X}(X)$.

Solution: This must be verified point-wise. Let σ, ν be two vector fields on X , and $a, b \in \mathbb{R}$. Then, we define

$$(a\sigma + b\nu)_p := a\sigma_p + b\nu_p.$$

Then, $a\sigma_p + b\nu_p$ is clearly, a section to $\pi: TX \rightarrow X$ again. The zero element is the zero vector field, which sends every $p \in X$ to the 0 vector in T_pX .

2. Again, $(f\sigma)_p$ is clearly a section to π because $f(p) \cdot \sigma(p)$ is contained in T_pX . We also have smoothness because f is smooth. $\square$

Remark 2.2.35 — We note that $\mathfrak{X}(\mathbb{R}^n) \simeq (C^\infty(X))^{\oplus n}$, and so $\mathfrak{X}(\mathbb{R}^n)$ is a free module over $C^\infty(X)$. The module action here is given in the second part in the exercise above.

Remark 2.2.36 — We have $\sigma(\gamma(t))(f) = \frac{d}{ds}(f \circ \gamma)|_{s=t}$, where γ is a curve.

One way we can construct vector fields is by considering curves. If γ passes through $p \in X$, we can then use this to determine the direction at the point (using the curve). This is now the goal.

Definition 2.2.37. Let X be a smooth manifold. A **(smooth) flow** on X is a smooth map $F: \mathbb{R} \times X \rightarrow X$ such that

1. $F(0, p) = p$ for any point $p \in X$.
2. $F(s + t, p) = F(s, F(t, p))$ for any $s, t \in \mathbb{R}$ and $p \in X$.

Remark 2.2.38 — This is the collection of curves going through p .

Proposition 2.2.39

Any flow on X determines a smooth vector field on X .

Proof. Let F be a flow on X . For any point $p \in X$ and any $f \in C^\infty(X)$, it makes sense to define

$$\sigma(p)(f) := \left. \frac{d}{ds}(f \circ F(s, p)) \right|_{s=0}.$$

It suffices then to verify that this map σ is a smooth vector field on X .

First, we show that $\sigma(p) \in T_p X$. In particular, we must show that it is linear and satisfies the Leibniz rule. First, linearity is clear. Let $a, b \in \mathbb{R}$ and $f, g \in C^\infty(X)$.

$$\begin{aligned} \sigma(p)(af + bg) &= \left. \frac{d}{ds}((af + bg) \circ F(s, p)) \right|_{s=0} \\ &= \left. \frac{d}{ds}(a \cdot f \circ F(s, p)) \right|_{s=0} + \left. \frac{d}{ds}(b \cdot g \circ F(s, p)) \right|_{s=0} = a\sigma(p)(f) + b\sigma(p)(g). \end{aligned}$$

Next, we show it satisfies the Leibniz rule. Let $f, g \in C^\infty(X)$. Then,

$$\begin{aligned} \sigma(p)(fg) &= \left. \frac{d}{ds}((fg) \circ F(s, p)) \right|_{s=0} \\ &= f'(F(0, p)) \cdot F'(0, p) \cdot g(F(0, p)) + f(F(0, p)) \cdot g'(F(0, p)) \cdot F'(0, p) \\ &= g(p) (f'(F(0, p)) \cdot F'(0, p)) + f(p) \cdot (g'(F(0, p)) \cdot F'(0, p)) \\ &= g(p) \left. \frac{d}{ds}(f \circ F(s, p)) \right|_{s=0} + f(p) \left. \frac{d}{ds}(g \circ F(s, p)) \right|_{s=0} \\ &= g(p)\sigma(p)(f) + f(p)\sigma(p)(g). \end{aligned}$$

Therefore, σ satisfies the Leibniz rule. Thus, $\sigma(p) \in T_p X$. Therefore, σ is a section to π .

Next, $f \circ F$ depends smoothly on (s, p) , so is the partial derivative with respect to s . Therefore, $\sigma(p)(f)$ depends smoothly on p , so σ is smooth. $\square$

Example 2.2.40

The map $F: \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$ defined by $F(s, x) := x + s$ is a flow on $\mathbb{R}$. What is the vector field determined by this flow? Take $f \in C^\infty(\mathbb{R})$.

$$\begin{aligned} \sigma(x)(f) &= \left. \frac{d}{ds}(f \circ (F(s, x))) \right|_{s=0} \\ &= \left. \frac{d}{ds}(f(s + x)) \right|_{s=0} \\ &= \frac{df}{dx}. \end{aligned}$$

Example 2.2.41

The map $F: \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$ defined by $F(s, x) := x + s^2$ is not a flow on $\mathbb{R}$.

Example 2.2.42

The map $F: \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$ defined by $F(s, x) := e^s x$ is a flow on $\mathbb{R}$. Let us now compute the vector field determined by this flow:

$$\sigma(p)(f) := \frac{d}{ds}(f \circ F(s, x)) \big|_{s=0}.$$

Therefore,

$$\sigma(p)(f) = \frac{d}{ds} f(e^s x) \big|_{s=0} = f'(e^s x) e^s x \big|_{s=0} = x \cdot \frac{df}{dx}.$$

Example 2.2.43

The map $F: \mathbb{R} \times \mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined by

$$F(s, x, y) := (e^s x, y + s)$$

is a flow on $\mathbb{R}^2$. The vector field determined by F is

$$\sigma(x, y) = \frac{\partial}{\partial x} f(e^s x, y + s) \cdot e^s x + \frac{\partial}{\partial y} f(e^s x, y + s) \cdot 1 \bigg|_{s=0} = x \frac{\partial f}{\partial x} + \frac{\partial f}{\partial y}.$$

Integral Curves

Definition 2.2.44. A **smooth curve** in a smooth manifold X is a smooth map $\gamma: (a, b) \rightarrow X$. Note that a, b can be $-\infty$ or ∞ , respectively.

Definition 2.2.45. The **velocity vector** of a curve $\gamma: (a, b) \rightarrow X$, $\gamma'(t_0)$, of γ at $t = t_0$ is the tangent vector

$$d\gamma_{t_0} \left(\frac{\partial}{\partial t} \right) \in T_{\gamma(t_0)} X.$$

Definition 2.2.46. Given a smooth vector field σ on X , an **integral curve** of σ is a curve γ satisfying the differential equation

$$\gamma'(t) = \sigma(\gamma(t)), \quad \forall t \in (a, b).$$

Example 2.2.47

Given a smooth curve $\gamma: (a, b) \rightarrow X$ with $\gamma(0) = p$, we can define a tangent vector $\gamma'(0) \in T_p X$ as follows:

$$\gamma'(0)(f) := \frac{d}{ds}(f \circ \gamma)'(s) \big|_{s=0} = (f \circ \gamma)'(0).$$

Example 2.2.48

Let (x, y) be standard coordinates in $\mathbb{R}^2$, and $\sigma(x, y)$ be the vector field $\frac{\partial}{\partial x}$. Let us find an integral curve $\gamma: (a, b) \rightarrow \mathbb{R}^2$ of σ . Write $\gamma(t) = (x(t), y(t))$. Then,

$$\gamma'(t) = x'(t) \frac{\partial}{\partial x} + y'(t) \frac{\partial}{\partial y}.$$

Therefore, we want $y'(t) = 0$ and $x'(t) = 1$. Therefore, our curve can be of the form

$$\gamma(t) = (t + a, b),$$

where a, b are just some constants.

Example 2.2.49

Let x be the coordinate of $\mathbb{R}$, and $\sigma(x) = x \frac{\partial}{\partial x}$. Let's find an integral curve. Writing $\gamma: (a, b) \rightarrow \mathbb{R}$ as $\gamma(t) = x(t)$, we get that $\gamma'(t) = x'(t) \frac{\partial}{\partial x}$. Therefore, we want

$$x'(t) = x(t).$$

Thus $x(t) = C \cdot e^t$ works, where C is some constant.

Example 2.2.50

Let x, y be coordinates of $\mathbb{R}^2$, $\sigma(x) = y \frac{\partial}{\partial x} + \frac{\partial}{\partial y}$. Let's find an integral curve which satisfies $\gamma(0) = (2, 1)$ written as $\gamma(t) = (x(t), y(t))$. We have $\gamma'(t) = x'(t) \frac{\partial}{\partial x} + y'(t) \frac{\partial}{\partial y}$, so we want

$$y = x'(t), \quad y'(t) = 1.$$

Then, $y(t) = t + a$, for some constant a . Then, $x'(t) = \frac{t^2}{2} + at + b$, where b is some constant. Thus, an integral curve for σ is

$$\gamma(t) = \left(\frac{t^2}{2} + at + b, t + a \right).$$

To satisfy the initial conditions, we set $a = 1, b = 2$.

Example 2.2.51

Consider the vector field on $\mathbb{R}^2$ defined by

$$\sigma(x, y) := -y \frac{\partial}{\partial x} + x \frac{\partial}{\partial y}.$$

We want to find an integral curve $\gamma: (a, b) \rightarrow \mathbb{R}^2$, and suppose we consider the integral curve with $\gamma(0) = (1, 0)$. If $\gamma(t) = (x(t), y(t))$, then

$$\gamma'(t_0) = d\gamma_{t_0} \left(\frac{\partial}{\partial t} \right) = x'(t_0) \frac{\partial}{\partial x} + y'(t_0) \frac{\partial}{\partial y}.$$

We want this to equal $-y \frac{\partial}{\partial x} + x \frac{\partial}{\partial y}$. Thus, we get

$$x'(t_0) = -y(t_0), \quad y'(t_0) = x(t_0),$$

for all $t_0 \in (a, b)$. Then, γ is given by

$$\gamma(t) := (c \cos t - d \sin t, c \sin t + d \cos t), \quad t \in \mathbb{R}.$$

Specifying to our specific initial condition, we must have $c = 1, d = 0$.

Example 2.2.52

A general smooth vector field on $\mathbb{R}^2$ is

$$\sigma(x, y) = f(x, y) \frac{\partial}{\partial x} + g(x, y) \frac{\partial}{\partial y}$$

for some smooth functions f, g . Suppose $\gamma(t) = (\gamma_1(t), \gamma_2(t))$ is an integral curve of σ , then we have

$$\gamma'_1(t) = f(\gamma_1(t), \gamma_2(t)), \quad \gamma'_2(t) = g(\gamma_1(t), \gamma_2(t)).$$

This is a system of ordinary differential equations. By the local existence theorem for ODEs, there always exists some solution $\gamma(t)$ for $t \in (-\epsilon, \epsilon)$ for some $\epsilon > 0$. By the uniqueness of solutions, two such integral curves $\gamma(t), \tilde{\gamma}(t)$ for $t \in (-\epsilon, \epsilon)$ are the same.

By using coordinate charts, the above computation generalizes to the following lemma:

Lemma 2.2.53

Let σ be a smooth vector field on X . For any $p \in X$, there exists $\epsilon > 0$ and an integral curve

$$\gamma: (-\epsilon, \epsilon) \rightarrow X, \quad \gamma(0) = p.$$

Two such integral curves are the same.

Proof. This follows by the local existence of solutions of ODEs, and uniqueness of solutions. $\square$

Example 2.2.54

Consider the vector field $\sigma(x) = x^2 \frac{\partial}{\partial x}$ on $\mathbb{R}$. Show that it does not have an integral curve defined for all time.

Remark 2.2.55 — With a further study of local existence, one can show that a smooth vector field on a compact manifold has integral curves which are defined globally (that is, on all of $\mathbb{R}$). Refer to Chapter 9 in [1] for a detailed discussions.

Definition 2.2.56. Let $F: X \rightarrow Y$ be a diffeomorphism and σ be a vector field on X . We define the **push forward** of σ to be $F_*(\sigma)(q) := dF_p(\sigma(p))$, where $q = F(p)$. This is a vector field on Y .

Notation 2.2.57. Sometimes, we may write $F_*\sigma = dF(\sigma)$.

Exercise 2.2.58. Do concrete calculations.

Remark 2.2.59 — Let X, Y be smooth manifolds. If $F: X \rightarrow Y$ is a diffeomorphism, then $dF: TX \rightarrow TY$ is also a diffeomorphism.

Exercise 2.2.60. Let M be the open submanifold of $\mathbb{R}^2$ where both x and y are positive, and let $F: M \rightarrow M$ be the map

$$F(x, y) = (xy, \frac{y}{x}).$$

Show that F is a diffeomorphism, and compute F_*X and F_*Y , where

$$X = x \frac{\partial}{\partial x} + y \frac{\partial}{\partial y}, \quad Y = y \frac{\partial}{\partial x}.$$

Solution: The easiest way to show F is a diffeomorphism is to compute the inverse. It is

$$F^{-1}(x, y) = (\sqrt{\frac{x}{y}}, \sqrt{xy}).$$

Thus, F is a bijection and F and F^{-1} are clearly smooth (because x, y are positive), so F is a diffeomorphism. Now, dF takes the matrix

$$dF = \begin{pmatrix} y & x \\ -\frac{y}{x^2} & \frac{1}{x} \end{pmatrix}.$$

Therefore, at $q = F(p)$, we have that

$$F_*(X)(q) = dF_p \cdot (x, y)^T = (2xy, 0) = 2xy \frac{\partial}{\partial u} = 2u \frac{\partial}{\partial u}$$

Further,

$$F_*(Y)(q) = dF_p \cdot (y, 0)^T = (y^2, -\frac{y^2}{x^2}) = y^2 \frac{\partial}{\partial u} - \frac{y^2}{x^2} \frac{\partial}{\partial v} = uv \frac{\partial}{\partial u} - v^2 \frac{\partial}{\partial v}.$$

□

Exercise 2.2.61. For each of the following vector fields on the plane, compute its coordinate representation in polar coordinates on the right half-plane $\{(x, y) : x > 0\}$.

1. $X = x \frac{\partial}{\partial x} + y \frac{\partial}{\partial y}$.
2. $Y = x \frac{\partial}{\partial y} - y \frac{\partial}{\partial x}$.
3. $Z = (x^2 + y^2) \frac{\partial}{\partial x}$.

Solution: Let H be the right half-plane in consideration. We first need to construct a diffeomorphism from (H, x, y) to (H, r, θ) . Let $F: H \rightarrow \mathbb{R} \times (-\frac{\pi}{2}, \frac{\pi}{2})$ be the usual map:

$$F: (x, y) \mapsto (\sqrt{x^2 + y^2}, \arctan(\frac{y}{x})).$$

The inverse of this is F^{-1} defined by

$$F^{-1}: (r, \theta) \mapsto (r \cdot \cos \theta, r \cdot \sin \theta).$$

These maps are inverses to each other, so is a bijection. Further, these are clearly smooth, since $x > 0$. Since we want to find the polar coordinate representation of the vector fields above, we calculate the Jacobian for the inverse map, since we want to find vector fields in polar coordinates that push forward to X, Y, Z .

$$dF^{-1} = \begin{pmatrix} \cos \theta & -r \sin \theta \\ \sin \theta & r \cos \theta \end{pmatrix}$$

Thus, $dF_*^{-1}(\frac{\partial}{\partial r}) = \cos \theta \frac{\partial}{\partial x} + \sin \theta \frac{\partial}{\partial y}$ and $dF_*^{-1}(\frac{\partial}{\partial \theta}) = -r \sin \theta \frac{\partial}{\partial x} + r \cos \theta \frac{\partial}{\partial y}$. Given this, we now write X, Y, Z using $\frac{\partial}{\partial r}$ and $\frac{\partial}{\partial \theta}$. First, we have that

$$X = r \frac{\partial}{\partial r}.$$

Next, we can find that

$$Y = \frac{\partial}{\partial \theta}.$$

Finally, we can see that

$$Z = r \cos \theta \frac{\partial}{\partial r} - r \sin \theta \frac{\partial}{\partial \theta}.$$

Of course, we could’ve pushed it along F_* , but I wanted to do it a different way. $\square$

§2.3 More on Smooth Maps

In calculus, the tangent line of a function provides a “linear approximation” of the function. Now we apply this philosophy to the study of smooth maps between manifolds. That is, the total differential of a smooth map provides us with much information about the map itself.

§2.3.i Immersions and Submersions

An important invariant of a linear map is its rank:

Definition 2.3.1. Let $F: X \rightarrow Y$ be a smooth map. We define the **rank** of F at $p \in X$ as the rank of the linear map

$$dF_p: T_p X \rightarrow T_{F(p)} Y.$$

A smooth map $F: X \rightarrow Y$ is said to have **constant rank** if it has the same rank at every point $p \in X$.

Example 2.3.2

Diffeomorphisms have constant rank, since they induce isomorphisms on the tangent spaces.

Definition 2.3.3. A smooth map $F: X \rightarrow Y$ is called a **smooth immersion** if dF_p is injective at any point $p \in X$.

Definition 2.3.4. A smooth map $F: X \rightarrow Y$ is called a **smooth submersion** if dF_p is surjective at any point $p \in X$.

Remark 2.3.5 — The map F itself need not be injective, surjective, or bijective for these conditions to hold; only the behavior of the derivative is important.

Example 2.3.6

Suppose m, n are two positive integers and $m > n$. The map

$$\mathbb{R}^m \rightarrow \mathbb{R}^n, \quad (x_1, \dots, x_n, x_{n+1}, \dots, x_m) \mapsto (x_1, \dots, x_n)$$

is a submersion, usually called a **projection**. The map

$$\mathbb{R}^n \rightarrow \mathbb{R}^m, \quad (x_1, \dots, x_n) \mapsto (x_1, \dots, x_n, 0, \dots, 0)$$

is an immersion, usually called an **embedding**.

Example 2.3.7

A diffeomorphism is both an immersion and a submersion.

Exercise 2.3.8. Consider the function

$$f: \mathbb{R} \rightarrow \mathbb{R}^2, \quad f(t) = (t^2, t^3 - t).$$

Show it is an immersion but not a submersion.

Solution: It is never a submersion due to dimension reasons. It is an immersion because the velocity vector, $f'(t) = (2t, 3t^2 - 1)$ is never zero. In other words, $\frac{\partial}{\partial t} \mapsto 2t \frac{\partial}{\partial x} + (3t^2 - 1) \frac{\partial}{\partial y}$, but not here that we must actually write $2t$ and $3t^2 - 1$ in terms of x and y . We've really abused notation here for illustrative purposes. $\square$

Exercise 2.3.9. Consider the function

$$f: \mathbb{R} \rightarrow \mathbb{R}^2, \quad f(t) = (t^3 - 4t, t^2 - 4).$$

Show it is an immersion but not injective.

Solution: We get that $f'(t) = (3t^2 - 4, 2t)$. Thus, the velocity vector is never zero and f is an immersion. If $t = \pm 2$, there is a self-intersection, so not injective. $\square$

Example 2.3.10

An example of a submersion which is not an immersion is

$$f: \mathbb{R}^2 \rightarrow \mathbb{R}, \quad (x, y) \mapsto x.$$

df can be calculated to be $(1, 0)$, so this is surjective onto $\mathbb{R}$, since $df_p(a, b) = a$. This is not injective, since $df_p(0, 1) = 0$.

Example 2.3.11

An example of a non-surjective submersion is

$$F: \mathbb{R}^2 \rightarrow \mathbb{R}, \quad (x, y) \mapsto e^x.$$

We can see that $dF_{(x,y)}(a, b) = e^x \cdot a$, so a submersion. It is clearly not surjective, since the image of f is the positive reals.

Example 2.3.12

In polar coordinates, the unit circle is $\mathbb{S}^1 = \{(r, \theta) \mid r = 1\} \subset \mathbb{R}^2$. We claim that the map $F: \mathbb{S}^1 \rightarrow \mathbb{S}^1$ defined by $F(r, \theta) = (r, 2\theta)$ is smooth and both an immersion and a submersion, but not a diffeomorphism.

Solution: The claim is that $dF_{(1,\theta)}(\frac{\partial}{\partial\theta}) = 2\frac{\partial}{\partial\theta}$. We note that F itself is not even injective, so not a diffeomorphism. $\square$

This example motivates the notion of local diffeomorphisms.

Definition 2.3.13. A smooth map $F: X \rightarrow Y$ is called a **local diffeomorphism** if for any $p \in X$ there is an open neighborhood U of p such that $F(U)$ is open in Y and $F|_U: U \rightarrow F(U)$ is a diffeomorphism.

Remark 2.3.14 — A local diffeomorphism is not necessarily a (global) diffeomorphism, as the above example.

We recall the following inverse function theorem in several variable calculus, which will give a key property of local diffeomorphisms.

Theorem 2.3.15

Suppose U, V are open subsets of $\mathbb{R}^n$, and $F: U \rightarrow V$ be a smooth function. If the Jacobian matrix of F is invertible at some $p \in U$, then there exists an open set $U_0 \subset U$ which contains p , an open set $V_0 \subset V$ which contains $F(p)$, and $F|_{U_0}: U_0 \rightarrow V_0$ is a diffeomorphism.

Corollary 2.3.16 (Inverse Function Theorem)

Suppose X, Y are smooth manifolds and $F: X \rightarrow Y$ be a smooth map. If dF_p is an invertible linear map at some point $p \in X$, then there exist open neighborhood U of p and open neighborhood V of $F(p)$ such that $F|_U: U \rightarrow V$ is a diffeomorphism.

Proof. Pick two smooth charts containing p and $F(p)$ respectively, and apply the inverse function theorem for functions. $\square$

Exercise 2.3.17. A bijective local diffeomorphism is a (global) diffeomorphism.

Solution: We really only need to show that F^{-1} is smooth. This follows from the fact that F is a local diffeomorphism and that smoothness is a local property. That is, F^{-1} is smooth at every point in Y , so this is true. $\square$

The following is a sufficient condition for a smooth map being a diffeomorphism.

Corollary 2.3.18

Let X, Y be two smooth manifolds with the same dimension. A bijective submersion $F: X \rightarrow Y$ is a (global) diffeomorphism.

Proof. We just need to check F^{-1} is smooth. Since F is bijective, for any $q \in Y$ there is $p \in X$ such that $F(p) = q$. Since X, Y have the same dimension and F is a submersion, we have dF_p invertible at p . By the inverse function theorem, $F|_U$ is a diffeomorphism on neighborhood U of p . In particular, F^{-1} is smooth on $F(U)$. Then, the bijectivity of F allows us to conclude that F^{-1} is smooth at every point, which provides a global diffeomorphism. $\square$

The Rank Theorem

An important corollary or consequence of the inverse function theorem is the rank theorem, which we have already seen in some basic examples. We first prove some basic versions.

Lemma 2.3.19

Let $U \subset \mathbb{R}^r$ be an open neighborhood of the origin. Let $f: U \rightarrow \mathbb{R}^n$ be a smooth map of constant rank r with $f(0) = 0$. Then there exists an open neighborhood $V \subset \mathbb{R}^n$ containing the origin and a smooth map $g: V \rightarrow \mathbb{R}^n$ which is a diffeomorphism from V to $g(V)$, such that

$$g \circ f(x_1, \dots, x_r) = (x_1, \dots, x_r, 0, \dots, 0).$$

Note that $r \leq n$ here.

Proof. The function $f(x) = (f_1(x), \dots, f_n(x))$ has constant rank r . Considering the Jacobian, the rank of this is r , so after rescaling the coordinates on $\mathbb{R}^n$, we assume that the matrix $\left(\frac{\partial f_i}{\partial x_j}\right)_{1 \leq i, j \leq r}$ is invertible. Consider the function $F: U \times \mathbb{R}^{n-r} \rightarrow \mathbb{R}^n$ defined by

$$\begin{aligned} F(x_1, \dots, x_r, x_{r+1}, \dots, x_n) &= f(x_1, \dots, x_r) + (0, \dots, 0, x_{r+1}, \dots, x_n) \\ &= (f_1, f_2, \dots, f_r, x_{r+1}, \dots, x_n) \end{aligned}$$

The Jacobian of F is then a $n \times n$, and is invertible, since the new ‘added on’ part has derivatives 1, giving a I_{n-r} matrix on the last $(n-r)$ coordinates. Then function F has

constant rank n , by our previous assumption. Applying the inverse function theorem to F , we attain is a local smooth inverse g of F . This means that

$$\begin{aligned} g \circ F(x_1, \dots, x_r, x_{r+1}, \dots, x_n) &= g(f(x_1, \dots, x_r) + (0, \dots, 0, x_{r+1}, \dots, x_n)) \\ &= (x_1, \dots, x_r, x_{r+1}, \dots, x_n) \end{aligned}$$

for any $(x_1, \dots, x_r, x_{r+1}, \dots, x_n)$. Particularly, setting $x_{r+1} = \dots = x_n = 0$ we get

$$g \circ f(x_1, \dots, x_r) = (x_1, \dots, x_r, 0, \dots, 0).$$

□

Example 2.3.20

Consider the map $f: \mathbb{R} \rightarrow \mathbb{R}^2$ given by $t \mapsto (\sin(t), \cos(t))$. This is an immersion, so the lemma above tells us that there is a diffeomorphism g , such that the second coordinate is 0. We can actually do this concretely by considering $g: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ where $(x, y) \mapsto (t, r - 1)$, where $r = \sqrt{x^2 + y^2}$ is the radius. The lemma tells us we can do this more generally.

Lemma 2.3.21

Let $U \subset \mathbb{R}^r$ be an open neighborhood of the origin. Let $f: U \rightarrow \mathbb{R}^n$ be a smooth map of constant rank n with $f(0) = 0$. Then there exists an open neighborhood $V \subset \mathbb{R}^r$ containing the origin and a smooth map $g: V \rightarrow \mathbb{R}^r$ which is a diffeomorphism from V to $g(V)$, such that

$$f \circ g(x_1, \dots, x_n, x_{n+1}, \dots, x_r) = (x_1, \dots, x_n).$$

Note that $r \geq n$ here.

Proof. The Jacobian matrix of $f = (f_1, \dots, f_n)$ has constant rank n . Analogous to the previous lemma, we may assume that $\left(\frac{\partial f_i}{\partial x_j}\right)_{1 \leq i, j \leq n}$ is invertible. Otherwise, just perform a change of coordinate on U . Consider the function $F: U \rightarrow \mathbb{R}^r$ defined by

$$F(x_1, \dots, x_r) = (f_1(x_1, \dots, x_r), \dots, f_n(x_1, \dots, x_r), x_{n+1}, \dots, x_r),$$

which has constant rank r , since f has constant rank n . By the inverse function theorem, there is a local inverse $g = (g_1, \dots, g_n)$ of F . This means that

$$\begin{aligned} F \circ g(x_1, \dots, x_r) &= (f_1(g_1, \dots, g_r), \dots, f_n(g_1, \dots, g_r), g_{n+1}, \dots, g_r) \\ &= (x_1, \dots, x_r). \end{aligned}$$

Therefore, focusing on the first n coordinates, we get $f \circ g(x_1, \dots, x_n) = (x_1, \dots, x_n)$. □

Example 2.3.22

Consider a map $f: \mathbb{R}^2 \rightarrow \mathbb{R}$ defined by $(x, y) \mapsto x^2 + y^2$. This is not a submersion, since $df_{(x,y)}(a, b) = 2xa + 2yb$, but at (x, y) , $df_{(0,0)} = 0$. Away from $(0, 0)$, $df_{(x,y)}$ is nonzero, so surjective.

Theorem 2.3.23 (Rank theorem)

Let M, N be smooth manifolds with dimensions m, n respectively. Let $F: M \rightarrow N$ be a smooth map with constant rank r . For any $p \in M$ there exist charts (U, φ) containing p and a smooth chart (V, ψ) containing $F(p)$ such that $F(U) \subset V$ and

$$\hat{F} = \psi \circ F \circ \varphi^{-1}: \mathbb{R}^m \rightarrow \mathbb{R}^n$$

has the form

$$\hat{F}(x_1, \dots, x_m) = (x_1, \dots, x_r, 0, \dots, 0).$$

Proof. Because the theorem is local, after choosing smooth coordinates we can replace M and N by open subsets $U \subseteq \mathbb{R}^m$ and $V \subseteq \mathbb{R}^n$. The fact that $DF(p)$ has rank r implies that its matrix has some $r \times r$ submatrix with nonzero determinant. By reordering the coordinates, we may assume that it is the upper left submatrix, $\left(\frac{\partial F^i}{\partial x_j}\right)$ for $i, j = 1, \dots, r$. Let us relabel the standard coordinates as

$$(x, y) = (x_1, \dots, x_r, y_1, \dots, y_{m-r}) \text{ in } \mathbb{R}^m$$

and

$$(v, w) = (v_1, \dots, v_r, w_1, \dots, w_{n-r}) \text{ in } \mathbb{R}^n.$$

By initial translations of the coordinates, we may assume without loss of generality that $p = (0, 0)$ and $F(p) = (0, 0)$. If we write

$$F(x, y) = (Q(x, y), R(x, y))$$

for some smooth maps $Q: U \rightarrow \mathbb{R}^r$ and $R: U \rightarrow \mathbb{R}^{n-r}$, then our hypothesis is that $\left(\frac{\partial Q^i}{\partial x_j}\right)$ is nonsingular at $(0, 0)$.

Define $\varphi: U \rightarrow \mathbb{R}^m$ by

$$\varphi(x, y) = (Q(x, y), y).$$

Its total derivative at $(0, 0)$ is

$$D\varphi(0, 0) = \begin{pmatrix} \frac{\partial Q^i}{\partial x_j}(0, 0) & \frac{\partial Q^i}{\partial y_j}(0, 0) \\ 0 & \delta_j^i \end{pmatrix},$$

where we have used the following standard notation: for positive integers i and j , the symbol δ_j^i , called the Kronecker delta, is defined by

$$\delta_j^i = \begin{cases} 1 & \text{if } i = j, \\ 0 & \text{if } i \neq j. \end{cases}$$

The matrix $D\varphi(0, 0)$ is nonsingular by virtue of the hypothesis. Therefore, by the inverse function theorem, there are neighborhoods U_0 of $(0, 0)$ and $\tilde{U}_0$ of $\varphi(0, 0) = (0, 0)$ such that $\varphi: U_0 \rightarrow \tilde{U}_0$ is a diffeomorphism. By shrinking U_0 and $\tilde{U}_0$ if necessary, we may assume that $\tilde{U}_0$ is an open cube.

Writing the inverse map as $\varphi^{-1}(x, y) = (A(x, y), B(x, y))$ for some smooth functions $A: \tilde{U}_0 \rightarrow \mathbb{R}^r$ and $B: \tilde{U}_0 \rightarrow \mathbb{R}^{m-r}$, we compute

$$(x, y) = \varphi(A(x, y), B(x, y)) = (Q(A(x, y), B(x, y)), B(x, y)).$$

Comparing components shows that $B(x, y) = y$, and therefore φ^{-1} has the form $\varphi^{-1}(x, y) = (A(x, y), y)$. On the other hand, $\varphi \circ \varphi^{-1} = \text{Id}$ implies $Q(A(x, y), y) = x$, and therefore $F \circ \varphi^{-1}$ has the form

$$F \circ \varphi^{-1}(x, y) = (x, \tilde{R}(x, y)),$$

where $\tilde{R}: \tilde{U}_0 \rightarrow \mathbb{R}^{n-r}$ is defined by $\tilde{R}(x, y) = R(A(x, y), y)$. The Jacobian matrix of this composite map at an arbitrary point $(x, y) \in \tilde{U}_0$ is

$$D(F \circ \varphi^{-1})(x, y) = \begin{pmatrix} \delta_j^i & 0 \\ \frac{\partial \tilde{R}^i}{\partial x_j}(x, y) & \frac{\partial \tilde{R}^i}{\partial y_j}(x, y) \end{pmatrix}.$$

Since composing with a diffeomorphism does not change the rank of a map, this matrix has rank r everywhere in $\tilde{U}_0$. The first r columns are obviously linearly independent, so the rank can be r only if the derivatives $\frac{\partial \tilde{R}^i}{\partial y_j}$ vanish identically on $\tilde{U}_0$, which implies that $\tilde{R}$ is actually independent of $(y_1, \dots, y_{m-r})$. Thus, if we let $S(x) = \tilde{R}(x, 0)$, then we have

$$F \circ \varphi^{-1}(x, y) = (x, S(x)). \quad (2.1)$$

To complete the proof, we need to define an appropriate smooth chart in some neighborhood of $(0, 0) \in V$. Let $V_0 \subseteq V$ be the open subset defined by

$$V_0 = \{(v, w) \in V : (0, 0) \in \tilde{U}_0\}.$$

Then V_0 is a neighborhood of $(0, 0)$. Because $\tilde{U}_0$ is a cube and $F \circ \varphi^{-1}$ has the form (2.1), it follows that $F \circ \varphi^{-1}(\tilde{U}_0) \subseteq V_0$, and therefore

$$F(U_0) \subseteq V_0.$$

Define $\psi: V_0 \rightarrow \mathbb{R}^n$ by

$$\psi(v, w) = (v, w - S(v)).$$

This is a diffeomorphism onto its image, because its inverse is given explicitly by

$$\psi^{-1}(s, t) = (s, t + S(s));$$

thus (v, w) is a smooth chart. It follows from (2.1) that

$$\psi \circ F \circ \varphi^{-1}(x, y) = \psi(x, S(x)) = (x, S(x) - S(x)) = (x, 0),$$

which was to be proved. □

Remark 2.3.24 — The rank theorem describes the local behavior of a smooth map with constant rank.

§2.3.ii Embeddings

Definition 2.3.25. Let X, Y be two topological spaces. A map $F: X \rightarrow Y$ is called a **topological embedding** if F is a homeomorphism from X to $F(X)$, where the latter set is equipped with the subspace topology from Y .

Example 2.3.26

Consider the set

$$N := \{0, 1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \dots\}$$

equipped with the discrete topology. The inclusion map $i: N \rightarrow \mathbb{R}$ is not a topological embedding. This is because in the with respect to the subspace topology, 0 is a limit point while in the discrete topology, it is not.

Definition 2.3.27. Let X, Y be two smooth manifolds. A map $F: X \rightarrow Y$ is called a **smooth embedding** if F is both a topological embedding and an immersion.

Remark 2.3.28 — An injective immersion between smooth manifolds need not be a topological embedding. An example of this would be $f: (-\pi, \pi) \rightarrow \mathbb{R}^2$ which maps $t \mapsto (\sin 2t, \sin t)$. This is clearly an immersion and is injective (when restricted to this domain). However, this is not a topological embedding because the image is compact, while the domain is not. To see this, we use sequential compactness. If we have a sequence a_n in the domain, since a_n is bounded, there is some convergent subsequence, say a_{n_i} which converges to some point a in $[-\pi, \pi]$. By continuity, $f(a_{n_i})$ converges to $f(a)$, which is equal to $f(a)$ is $a \in (-\pi, \pi)$ and $f(0) = (0, 0)$ otherwise. Therefore, every sequence in the image has a convergent subsequence, while this is not the case in the domain. Therefore, f cannot be a homeomorphism onto its image.

Example 2.3.29

Let U be an open subset of a smooth manifold X . It inherits a smooth manifold structure from X (the open subset manifold structure), which makes the inclusion map $i: U \rightarrow X$ a smooth embedding.

Example 2.3.30

Let X, Y be two smooth manifolds. For any $p \in Y$, the inclusion map

$$i_p: X \rightarrow X \times Y, \quad i_p(x) = (x, p)$$

is a smooth embedding. This is like a *fixed slice*.

Example 2.3.31

Another famous example is the so-called figure-eight curve. Consider the map $f: (-\pi, \pi) \rightarrow \mathbb{R}^2$ defined by $f(t) = (\sin 2t, \sin t)$. First, note that f is smooth since the components are smooth. Next, we claim that this is an immersion. We need to check the velocity vector is never zero. We get

$$f'(t) = (2 \cos 2t, \cos t),$$

which is never zero. Next, we claim that this is not a topological embedding. If take an open ball around $(0, 0)$ and consider its inverse, then it contains more than just the open interval around 0 (in $\mathbb{R}$). This leads to a contradiction, as if we use f^{-1} to map an open set around 0 to an open set around $(0, 0)$, there aren't enough components.

Example 2.3.32

The map

$$\mathbb{R} \rightarrow \mathbb{R}^2, \quad t \mapsto (t^2, t^3)$$

is a topological embedding and a smooth map. But it is not an immersion since the velocity vector $h'(t) = (2t, 3t^2)$ is zero at $t = 0$.

Exercise 2.3.33. Give an example of a smooth embedding that is neither an open map nor a closed map.

Solution: The map $f: (0, 1) \rightarrow \mathbb{R}^2$ defined by $t \mapsto (t, 0)$ is a smooth embedding which is neither an open map nor a closed map. $\square$

Theorem 2.3.34 (Local Embedding Theorem)

Suppose that X, Y are smooth manifolds and $F: X \rightarrow Y$ is a smooth map. Then F is a smooth immersion if and only if every point in X has a neighborhood $U \subset X$ such that $F|_U: U \rightarrow Y$ is a smooth embedding.

Proof. ($\Leftarrow$): If every point in X has a neighborhood $U \subset X$ such that $F|_U: U \rightarrow Y$ is a smooth embedding then the derivative of F is an injection at every point, hence F is an immersion.

($\Rightarrow$): Suppose F is an immersion, then dF has constant rank at every point $p \in X$. By the constant rank theorem, there exists a neighborhood U of p and two charts containing $p, F(p)$ respectively such that F is in the standard form:

$$\psi \circ F|_U \circ \varphi^{-1}(x_1, \dots, x_r) = (x_1, \dots, x_r, 0, \dots, 0)$$

The standard form is a smooth embedding. $\square$

§2.3.iii Submanifolds

Definition 2.3.35. Let X be a smooth manifold. An **embedded submanifold** of X is a subset S of X that is a manifold in the subspace topology, endowed with a smooth structure with respect to which the inclusion map is a smooth embedding.

Definition 2.3.36. If S is an embedded submanifold of X , the difference $\dim X - \dim S$ is called the **codimension** of S in X .

Example 2.3.37

The empty set is an embedded submanifold, by convention.

Remark 2.3.38 — One can relax the embedded condition to be immersed, to define immersed submanifolds. In this course we only focus on the embedded ones.

Proposition 2.3.39

Let X be a smooth manifold. The codimension-zero embedded submanifolds of X are exactly the open submanifolds of X , i.e. the open subsets of X with the open subset manifolds structure.

Proof. If U is an open subset of X , then it inherits a smooth manifold structure which makes the inclusion map a smooth embedding. If we have a codimension-zero embedded submanifold Y of X , then the inclusion map is smooth embedding. Particularly, the inclusion map is a local diffeomorphism by the inverse function theorem. This shows that Y is an open subset of X . $\square$

Exercise 2.3.40. Let X, Y be two smooth manifolds. Then for any $p \in X$, the slice $\{p\} \times Y$ is an embedded submanifold in $X \times Y$, which is diffeomorphic to Y .

Solution: First, we show that the map $\iota: \{p\} \times Y \rightarrow X \times Y$ defined by $(p, q) \mapsto (p, q)$ is a smooth embedding. This map is clearly continuous, with continuous inverse. This map is an injection, so is homeomorphic to its image. Further, since inclusion maps are immersions, this is a smooth immersion. Therefore, this map is a smooth embedding, allowing us to conclude that $\{p\} \times Y$ is an embedded submanifolds of $X \times Y$. The diffeomorphism is obvious: $(p, q) \mapsto q$, with inverse $q \mapsto (p, q)$. This is clearly bijective, and is smooth with smooth inverse. $\square$

Exercise 2.3.41. Let X, Y be two smooth manifolds and $F: X \rightarrow Y$ be a smooth map. The graph

$$\Gamma(F) := \{(p, F(p)) \mid p \in X\}$$

is an embedded submanifold of $X \times Y$. What's its dimension?

Solution: We show that the map $G: X \rightarrow X \times Y$ defined by $x \mapsto (x, F(x))$ is a smooth embedding. The image of G is $\Gamma(F)$. Furthermore, G is injective because if $G(x) = G(y)$, then $x = y$. It is clearly smooth as well (so continuous as well). Its inverse from the image is $\pi: \Gamma(F) \rightarrow X$ defined by $(x, F(x)) \mapsto x$, which is continuous. Thus, we have a homeomorphism. We finally check that dG_p is injective. Indeed, this is the case, because $dG_p: v \mapsto (v, dF_p(v))$, so if $dG_p(v) = dG_p(w)$, then $v = w$. It follows that G is a smooth embedding. Now, if $\tilde{G}: X \rightarrow \Gamma(F)$ is the same map, then we see that $G = \iota \circ \tilde{G}$. It follows then that ι is a smooth embedding as G is one as well, so $\Gamma(F)$ is an embedded submanifold of $X \times Y$. Note that $T_{x, F(x)} \cong T_x X \oplus T_{F(x)} Y$ (as real vector spaces). We shall also note that

$$\text{cod } \Gamma(F) = \dim(X \times Y) - \dim(\Gamma(F)) = (\dim X + \dim Y) - \dim X = \dim Y$$

□

Example 2.3.42

Let us illustrate the above exercise concretely. Let $\gamma: \mathbb{R} \rightarrow \mathbb{R}$ be a smooth function, and let $I: \mathbb{R} \rightarrow \mathbb{R}^2$ be defined by $I(t) = (t, \gamma(t))$. Then, we consider the tangent map: $dI_p \left(\frac{d}{dt} \right)$. To check that this is nonzero, we must check that the velocity vector $1 \frac{\partial}{\partial x} + \gamma'(p) \frac{\partial}{\partial y}$ is nonzero.

Definition 2.3.43. A continuous map $f: X \rightarrow Y$ is **proper** if the preimage of any compact subset is compact.

Remark 2.3.44 — The following result is useful in solving problems about proper maps.

Theorem 2.3.45 (Proper Continuous Maps Are Closed)

Suppose X is a topological space and Y is a locally compact Hausdorff space. Then every proper continuous map $F: X \rightarrow Y$ is closed.

Proof. Let $K \subseteq X$ be a closed subset. To show that $F(K)$ is closed in Y , we show that it contains all of its limit points. Let y be a limit point of $F(K)$, and let $\bar{U}$ be a precompact neighborhood of y . Then y is also a limit point of $F(K) \cap \bar{U}$. Because F is proper, $F^{-1}(\bar{U})$ is compact, which implies that $K \cap F^{-1}(\bar{U})$ is compact. Because F is continuous,

$$F(K \cap F^{-1}(\bar{U})) = F(K) \cap \bar{U}$$

is compact and therefore closed in Y . In particular,

$$y \in F(K) \cap \bar{U} \subseteq F(K),$$

so $F(K)$ is closed. □

Definition 2.3.46. A **properly embedded submanifold** is an embedded submanifold such that the inclusion map is a proper map.

Example 2.3.47

The subset $\{x > 0 \mid x \in \mathbb{R}\}$ of $\mathbb{R}$ is an embedded submanifold but not a properly embedded submanifold.

Solution: This subset is an open submanifold of codimension 0 so is an embedded submanifold of codimension 0. This is not proper because $[0, 1]$ is compact in $\mathbb{R}$, but its preimage is $(0, 1]$, which is not compact. □

As next theorem will show, embedded submanifolds are modeled locally on the standard embedding of $\mathbb{R}^k$ into $\mathbb{R}^n$:

$$(x_1, \dots, x_k) \mapsto (x_1, \dots, x_k, x_{k+1} = 0, \dots, x_n = 0).$$

The image of this embedding is called a **k -slice**.

Theorem 2.3.48 (Local Slice Criterion for Embedded Submanifolds; Theorem 5.8 in [1])

Let X be a smooth manifold of dimension n and S be an embedded submanifold of dimension k . Then for any $p \in S$ there exists a chart (U, φ) for X containing p such that $\varphi(U \cap S)$ is a k -slice. Conversely, if S is a subset of X and for any $p \in S$ there exists a chart (U, φ) for X containing p such that $\varphi(U \cap S)$ is a k -slice, then S , with respect to the subspace topology, admits a smooth structure which makes S an embedded submanifold of dimension k .

Proof. First suppose that $S \subseteq M$ is an embedded k -dimensional submanifold. Since the inclusion map $S \hookrightarrow M$ is an immersion, the rank theorem shows that for any $p \in S$ there are smooth charts (U, φ) for S (in its given smooth manifold structure) and (V, ψ) for M , both centered at p , in which the inclusion map $\iota_U: U \rightarrow V$ has the coordinate representation

$$(x_1, \dots, x_k) \mapsto (x_1, \dots, x_k, 0, \dots, 0).$$

Choose $\varepsilon > 0$ small enough that both U and V contain coordinate balls of radius ε centered at p , and denote these coordinate balls by $U_0 \subseteq U$ and $V_0 \subseteq V$. It follows that $U_0 = \iota(U_0)$ is exactly a single slice in V_0 . Because S has the subspace topology, the fact that U_0 is open in S means that there is an open subset $W \subseteq M$ such that $U_0 = W \cap S$. Setting $V_1 = V_0 \cap W$, we obtain a smooth chart $(V_1, \psi|_{V_1})$ for M containing p such that $V_1 \cap S = U_0$, which is a single slice of V_1 .

Conversely, suppose S satisfies the local k -slice condition. With the subspace topology, S is Hausdorff and second-countable, because both properties are inherited by subspaces. To see that S is locally Euclidean, we construct an atlas. The basic idea of the construction is that if $(x_1, \dots, x_n)$ are slice coordinates for S in M , we can use $(x_1, \dots, x_k)$ as local coordinates for S .

For this proof, let $\pi: \mathbb{R}^n \rightarrow \mathbb{R}^k$ denote the projection onto the first k coordinates. Let (U, φ) be any slice chart for S in M , and define

$$V = U \cap S, \quad \widehat{V} = \pi \circ \varphi(V), \quad \psi = \pi \circ \varphi|_V: V \rightarrow \widehat{V}.$$

By definition of slice charts, $\varphi(V)$ is the intersection of $\varphi(U)$ with a certain k -slice $A \subseteq \mathbb{R}^n$ defined by setting $x_{k+1} = c_{k+1}, \dots, x_n = c_n$, and therefore $\varphi(V)$ is open in A . Since $\pi|_A$ is a diffeomorphism from A to $\mathbb{R}^k$, it follows that $\widehat{V}$ is open in $\mathbb{R}^k$. Moreover, ψ is a homeomorphism because it has a continuous inverse given by $\varphi^{-1} \circ j|_{\widehat{V}}$, where $j: \mathbb{R}^k \rightarrow \mathbb{R}^n$ is the map

$$j(x_1, \dots, x_k) = (x_1, \dots, x_k, c_{k+1}, \dots, c_n).$$

Thus S is a topological k -manifold, and the inclusion map $\iota: S \hookrightarrow M$ is a topological embedding.

To put a smooth structure on S , we need to verify that the charts constructed above are smoothly compatible. Suppose (U, φ) and (U', φ') are two slice charts for S in M , and let (V, ψ) , (V', ψ') be the corresponding charts for S . The transition map in S is given by $\psi' \circ \psi^{-1} = \pi \circ \varphi' \circ \varphi^{-1} \circ j$, which is a composition of four smooth maps. Thus the atlas we have constructed is in fact a smooth atlas, and it defines a smooth structure on S . In terms of a slice chart (U, φ) for M and the corresponding chart (V, ψ) for S , the inclusion map $S \hookrightarrow M$ has a coordinate representation of the form

$$(x_1, \dots, x_k) \mapsto (x_1, \dots, x_k, c_{k+1}, \dots, c_n),$$

which is a smooth immersion. Since the inclusion is a smooth immersion and a topological embedding, S is an embedded submanifold. $\square$

Level Sets and Regular Values

A useful way to construct properly embedded submanifolds is to consider level sets.

Theorem 2.3.49 (Level Set Theorem)

Let X, Y be two smooth manifolds and $F: X \rightarrow Y$ be a smooth map. If F has constant rank for any $p \in X$, then for any $q \in Y$, the level set $F^{-1}(q) \subset X$ is a properly embedded submanifold of X .

Proof. By the constant rank theorem, for any $p \in F^{-1}(q)$ one can find local charts around p and q such that the level set $F^{-1}(q)$ is a slice with respect to these charts; that is, $\psi \circ F \circ \varphi^{-1}(x_1, \dots, x_r, 0, \dots, 0)$. Then, we just apply the slice theorem to deduce that it is an embedded submanifold.

Next we show properness. For all C compact in X , we get that $i^{-1}(C)$ is compact in S , where we define $S = F^{-1}(q)$ and i is the inclusion map from $S \hookrightarrow X$. Note that S is closed in X . $i^{-1}(C) = C \cap F^{-1}(q)$, and C is compact and $F^{-1}(q)$ is closed, so this is compact. The intersection of a compact set with a closed set is compact. $\square$

Remark 2.3.50 — We can compare this with the following: any closed subset is $f^{-1}(0)$ for some f smooth.

Definition 2.3.51. Let $F: X \rightarrow Y$ be a smooth map. A point $p \in X$ is called **regular** if dF_p is surjective. Otherwise it is called **critical**. A point $q \in Y$ is called a **regular value** of F if $F^{-1}(q)$ only contains regular points.

Remark 2.3.52 — Note that if $F^{-1}(q) = \emptyset$, then we also say that it is regular, by convention.

Exercise 2.3.53 (Proposition 4.1 in [1]). For a smooth map $F: X \rightarrow Y$, if dF_p is surjective at some $p \in X$ then there exists an open neighborhood U of p such that dF_q is surjective for any $q \in U$. In other words, the subset of regular points of F is open in X .

Solution: Suppose that dF_p is surjective at $p \in X$. This means that the Jacobian matrix of F in coordinates is of maximal rank. The set of $m \times n$ matrices of full rank is an open subset of $M_{m \times n}(\mathbb{R})$, so by continuity, the Jacobian has full rank of some neighborhood of p . $\square$

Corollary 2.3.54 (Regular Level Set Theorem)

Let $F: X \rightarrow Y$ be a smooth map. If q is a regular value of F then the level set $F^{-1}(q)$ is a properly embedded submanifold of X . Moreover, we have $\dim(F^{-1}(q)) = \dim X - \dim Y$.

Proof. By the above exercise, there exists an open set U containing $F^{-1}(q)$ such that $F|_U$ is a submersion. Then we apply the level set theorem to $F|_U$. It gives us $F^{-1}(q)$ is a properly embedded submanifold of U . Since U is open in X , $F^{-1}(q)$ is also a properly embedded submanifold of X . To get the dimension, we note that $F|_U$ looks like a $\dim Y$ -slice. Since these charts map q to 0 in the Euclidean plane (or chart for Y), the $\dim Y$ -slice has $\dim X - \dim Y$ free coordinates. $\square$

Remark 2.3.55 — Note that this theorem does not say that if we have a critical value, then the preimage is not a properly embedded submanifold of X .

Example 2.3.56

Consider $F: \mathbb{R}^2 \rightarrow \mathbb{R}$ defined by $F(x, y) = xy$. Take some point $0 \neq q \in \mathbb{R}$. Then, the preimage is the graph

$$F^{-1}(q) = \Gamma_q = \{(x, y) \in \mathbb{R}^2 : xy = q\}.$$

Each of these are a properly embedded submanifold of X .

However, if we choose $q = 0$, then $xy = 0$, which is not a submanifold of $\mathbb{R}^2$, as discussed previously. Therefore, 0 is a critical value.

Exercise 2.3.57. Any submersion $F: X \rightarrow Y$ is an open map.

Solution: Let $p \in X$, and take a chart (U, φ) around p , and let $q = F(p)$. We want to show that $F(U)$ is open. The local structure theorem of a submersion tells us that we can choose a chart (V, ψ) of q such that

$$\psi \circ F \circ \varphi^{-1}: (x_1, \dots, x_n, \dots, x_m) \mapsto (x_1, \dots, x_n, 0, \dots, 0)$$

This tells us that F is an open map, since projections are open. $\square$

Exercise 2.3.58. Show that there is no submersion from $\mathbb{S}^2 \rightarrow \mathbb{R}$.

Solution: Suppose we have a submersion $F: \mathbb{S}^2 \rightarrow \mathbb{R}$ is a submersion. Then, it must be an open map. We note that $\mathbb{S}^2$ is compact. Then, $F(\mathbb{S}^2)$ is compact in $\mathbb{R}$ and $F(\mathbb{S}^2)$ is open in $\mathbb{R}$, which is a contradiction. $\square$

Remark 2.3.59 — We can generalize this argument. Then, there is no submersion from a compact manifolds M to $\mathbb{R}^n$.

Exercise 2.3.60. Let M be a nonempty smooth compact manifold. Show that there is no smooth submersion $F: M \rightarrow \mathbb{R}^n$ for any $n > 0$.

Solution: If we have a submersion, then it is an open map. Therefore, $F(M)$ is compact, so closed, and open. Therefore, $F(M)$ must be a component of $\mathbb{R}^n$, but $\mathbb{R}^n$ is connected, so it must be the entirety. However, $\mathbb{R}^n$ is not compact. $\square$

Exercise 2.3.61. What if we replace $\mathbb{R}$ with $\mathbb{S}^2$?

Solution: We have the identity map from $\mathbb{S}^2$ to $\mathbb{S}^2$, which is a diffeomorphism, which is a submersion. This doesn't contradict the previous exercise because the target $\mathbb{S}^2$ can have both open and compact sets. $\square$

Exercise 2.3.62. There is no submersion $\mathbb{S}^n$ to itself that is not surjective.

Solution: Let F be our map. Since F is smooth, so continuous, and $\mathbb{S}^n$ is compact, $F(\mathbb{S}^n)$ is also compact, so closed. However, $F(\mathbb{S}^n)$ is also open because F is an open map. $\mathbb{S}^n$ is connected, so it follows that $F(\mathbb{S}^n) = \mathbb{S}^n$, so surjective. $\square$

Exercise 2.3.63. Consider the map $\Phi: \mathbb{R}^4 \rightarrow \mathbb{R}^2$ defined by

$$\Phi(x, y, s, t) = (x^2 + y, x^2 + y^2 + s^2 + t^2 + y).$$

Show that $(0, 1)$ is a regular value of Φ , and that the level set $\Phi^{-1}(0, 1)$ is diffeomorphic to $\mathbb{S}^2$.

Solution: We calculate $d\Phi$ first:

$$d\Phi = \begin{pmatrix} 2x & 1 & 0 & 0 \\ 2x & 2y + 1 & 2s & 2t \end{pmatrix}.$$

We show that $(0, 1)$ is a regular value. Note that we must have $x^2 + y = 0$ and $x^2 + y^2 + s^2 + t^2 + y = 1$. This implies that $x^4 + s^2 + t^2 = 1$. Then, $d\Phi$ is surjective at all preimages of $(0, 1)$ because the rank of the matrix for $d\Phi$ is 2. We now wish to show that Φ^{-1} is diffeomorphic to $\mathbb{S}^2$. First, note that $\Phi^{-1}(0, 1)$ is diffeomorphic to the set $S := \{(x, s, t) \in \mathbb{R}^3 : x^4 + s^2 + t^2 = 1\}$. This is via the mapping $(x, y, s, t) \mapsto (x, s, t)$ with inverse $(x, y, z) \mapsto (x, -x^2, s, t)$. We now show that S and $\mathbb{S}^2$ are diffeomorphic. The map we use is

$$(x, s, t) \mapsto \frac{(x, s, t)}{\sqrt{x^2 + s^2 + t^2}},$$

and this is a diffeomorphism (check required). $\square$

Example 2.3.64

Consider the curve $\beta: (-\pi, \pi) \rightarrow \mathbb{R}^2$ defined by

$$\beta(t) = (\sin 2t, \sin t).$$

Its image is a set that looks like a figure-eight in the plane (sometimes called a *lemniscate*). (It is the locus of points (x, y) where $x^2 = 4y^2(1 - y^2)$.) β is an injective smooth immersion because $\beta'(t)$ never vanishes; but it is not a topological embedding, because its image is compact in the subspace topology, while its domain is not.

However, this is not an embedded submanifold of $\mathbb{R}^2$. This is because around $(0, 0)$, there are four connected components after removing the origin, while everywhere else, has two after such a procedure. Thus, not a manifold.

§2.3.iv The Tangent Space to a Submanifold

If S is a smooth submanifold of $\mathbb{R}^n$, we intuitively think of the tangent space $T_p S$ at a point of S as a subspace of the tangent space $T_p \mathbb{R}^n$. Similarly, the tangent space to a smooth submanifold of an abstract smooth manifold can be viewed as a subspace of the tangent space to the ambient manifold, once we make appropriate identifications.

Let M be a smooth manifold and let S be an embedded submanifold. This gives a smooth inclusion map $i : S \rightarrow M$. In particular, the tangent map of $di_p : T_p S \rightarrow T_p X$ is injective, i.e. a smooth immersion. Then, we identify $T_p S$ with its image under di_p , $di_p(T_p S) \subseteq T_p X$. Particularly, it works in the following way

$$di_p(v)(f) := v(f \circ i) = v(f|_S).$$

From now on we identify $T_p S$ with its image under this map, thereby thinking $T_p S$ as a linear subspace of $T_p M$. Now we prove several properties of this identification.

Lemma 2.3.65

Let f be a smooth function on X which is constant on S . Then $v(f) = 0$ for any $v \in \text{Im}(di_p) \subseteq T_p X$.

Proof. We have that

$$0 = v(f|_S) = v(f \circ i) = di_p(v)(f).$$

and that tangent vector vanishes on constant functions. □

The above lemma admits a suitable converse, see Proposition 5.37 in [1].

Proposition 2.3.66

Let $F : M \rightarrow N$ be a smooth map and $q \in N$ is a regular value of F . Let S be the submanifold given by $F^{-1}(q)$. For any $p \in S$, we have

$$T_p S = \ker(dF_p).$$

Proof. Pick a chart (U, ϕ) of M containing p and a chart (V, ψ) containing $F(p) = q$, such that $\psi \circ F \circ \phi^{-1}$ is of the standard form as in the constant rank theorem:

$$\psi \circ F \circ \phi^{-1} : (x_1, \dots, x_n, x_{n+1}, \dots, x_m) \rightarrow (x_1, \dots, x_n), \quad \phi(p) = 0, \quad \psi(q) = 0.$$

Then the coordinate components $x_1, \dots, x_n$ of $\psi \circ F \circ \phi^{-1}$ are constant functions along $\phi(S \cap U) = \{x_1 = \dots = x_n = 0\} \cap \phi(U)$. Hence $T_p S$ is generated by $\left\{ \frac{\partial}{\partial x_{n+1}}, \dots, \frac{\partial}{\partial x_m} \right\}$, which are all in $\ker(dF_p)$; thus, we have an inclusion $T_p S \subseteq \ker(dF_p)$. On the other hand, we know dF_p is surjective since q is a regular value, and we can calculate that the dimension of $T_p S$ and $\ker(dF_p)$ are the same. This implies that $T_p S = \ker(dF_p)$. □

We work out one example here.

Example 2.3.67

Let $S = \{x^2 + y^2 = 1\} \subset \mathbb{R}^2$ be the unit sphere. It is the level set $F^{-1}(1)$ where $F: \mathbb{R}^2 \rightarrow \mathbb{R}, F(x, y) = x^2 + y^2$. Then at any point $p = (x_0, y_0) \in S$, we have

$$T_p S = \ker(dF_p) = \left\{ a \cdot \left(-y_0 \frac{\partial}{\partial x} + x_0 \frac{\partial}{\partial y} \right) : a \in \mathbb{R} \right\} \subset T_p \mathbb{R}^2.$$

More generally, we can take the n -sphere and Consider $F: \mathbb{R}^{n+1} \rightarrow \mathbb{R}$ defined by $F(x_1, \dots, x_n) = x_1^2 + \dots + x_n^2$. Then, 1 would be a regular value.

We can say more; in particular, about TS . We see that there is a nowhere vanishing vector field $-y \frac{\partial}{\partial x} + x \frac{\partial}{\partial y}$ on S , which acts is just a nowhere vanishing section to the projection. First, note that $dF_{(x,y)}(-y, x) = 0$, so this is clearly in the tangent space of any point. Therefore, TS is a trivial vector bundle, so $TS \cong S \times \mathbb{R}^1$.

Exercise 2.3.68. More generally, for a smooth embedding $F: S \rightarrow M$, show that the differential $dF: TS \rightarrow TM$ is a smooth embedding.

Solution: First, recall that the differential dF is defined as $dF(x, v) = (F(x), dF_x(v))$. This is clearly smooth at x to begin with. We now must compute $d(dF)_{(x,v)}$ to show that this is an immersion. Now we use local coordinates. In coordinates, let $F(x) = (F_1(x), \dots, F_n(x))$. Then, we have that

$$dF(x, v) = \left(F_1(x), \dots, F_n(x), \sum_j \frac{\partial F_1}{\partial x_j} v_j, \dots, \sum_j \frac{\partial F_n}{\partial x_j} v_j \right).$$

Then, $d(dF)_{(x,v)}$ is just the Jacobian of dF evaluated at (x, v) , applied to a tangent vector $(\xi, \eta) \in \mathbb{R}^m \times \mathbb{R}^m$. The first component is just the usual Jacobian of F evaluated at x , applied to ξ : $dF_x(\xi)$. The second component is slightly more tricky. Because there are mixed terms, we must use the product rule. At the i -th position, we have the following:

$$d \left(\sum_j \frac{\partial F_i}{\partial x_j} v_j \right) \cdot (\xi, \eta) = \sum_{j,k} \frac{\partial^2 F_i}{\partial x_j \partial x_k} \xi_k v_j + \sum_j \frac{\partial F_i}{\partial x_j} \eta_j$$

Now suppose that $d(dF)_{(x,v)}(\xi, \eta) = 0$. This implies that $dF_x(\xi) = 0$. However, dF_x is injective because F is a smooth embedding, so $\xi = 0$. This implies that the part for second partials vanish, so we are left with $dF_x(\eta) = 0$. By the same reason, $\eta = 0$ then.

Next, we show that dF is a topological embedding. We show that dF is injective. If $dF(x, v) = dF(y, w)$, then since F is bijective, $F(x) = F(y)$ implies $x = y$. Next, $dF_x = dF_y$ is injective, so $dF_x(v) = dF_y(w)$ implies $v = w$, so dF is injective. Therefore, dF is bijective onto its image. The inverse of dF (from its image is) $(y, w) \mapsto (F^{-1}(y), dF_x^{-1}(w))$. F^{-1} exists and is continuous, since F is homeomorphic to its image. Further, dF_x is injective, so dF_x^{-1} exists and is also continuous, since it is simply the Jacobian (after passing to coordinates). Therefore, dF is a topological embedding.

We are done! □

Example 2.3.69

Let $S = \{x^2 + 2y^2 = 1\} \subseteq \mathbb{R}^2$.

1. Describe $T_p S$ in coordinates.
2. Find a vector field V on S such that $V(f) = g$, where $f(x, y) = 2x^2$ and $g(x, y) = 8xy$.

Solution: We note that $S = F^{-1}(1)$, where $F(x, y) = x^2 + 2y^2$. Then, $dF = 2xdx + 4ydy$. Suppose $v \in T_p S = \ker(dF_p)$. Then, we can write

$$v = a \frac{\partial}{\partial x} + b \frac{\partial}{\partial y}.$$

Then, $x_0 a + b 2y_0 = 0$, and

$$T_p S = \left\{ \lambda \left(y_0 \frac{\partial}{\partial x} - \frac{1}{2} x_0 \frac{\partial}{\partial y} \right) : \lambda \in \mathbb{R} \right\}.$$

Now, we can let

$$V = \left(\lambda y_0 \frac{\partial}{\partial x} - \frac{1}{2} \lambda x_0 \frac{\partial}{\partial y} \right) (2x_0^2) = \lambda y_0 4x_0.$$

Thus, $\lambda = 2$, since we want this to equal $g(x, y) = 8xy$. $\square$

Example 2.3.70

Let $F: M \rightarrow N$ be smooth.

1. If F is injective with constant rank, then show that F is an immersion.
2. Given one example where q is not a regular value, but $F^{-1}(q)$ is an embedded submanifold.

Solution: 1. Constant rank tells us that if we have charts (U, φ) for M and (V, ψ) for N , then

$$\psi^{-1} \circ F \circ \varphi(x_1, \dots, x_r, x_{r+1}, \dots, x_m) = (x_1, \dots, x_r, 0, \dots, 0).$$

Now, if $r < m$, then F essentially forgets about whatever $x_{r+1}, \dots, x_m$ are, so contradicts injectivity. Thus, $x = m$, where $m = \dim M$. This implies that dF_p has rank m . By rank nullity, $\ker dF_p = 0$, so is an immersion.

2. Consider $F: \mathbb{R} \rightarrow \mathbb{R}$, with $F(x) = x^3$. Then, $q = 0$ is not a regular value, but $F^{-1}(0)$ is an embedded submanifold because it's a 0-dimensional manifold $\square$

Exercise 2.3.71. Show that the set $S = \{[x, y, z] \in \mathbb{RP}^2 : 2x^2 + y^2 + z^2 = 0\}$ is an embedded submanifold.

Solution: Let $f: \mathbb{RP}^2 \rightarrow \mathbb{R}$ be the map $[x : y : z] \mapsto 2x^2 + y^2 + z^2$. Firstly, this map is clearly smooth. To be precise, we should check that $f \circ \varphi^{-1}$ is smooth, for all charts φ of $\mathbb{RP}^2$, but this will only make x, y , or z be 1, so it's trivial. Then, we claim that 0 is a regular value. We have that

$$df_{[x:y:z]} = \begin{pmatrix} 4x & 2y & 2z \end{pmatrix},$$

so $df_{[x:y:z]}$ is surjective if and only if x, y, z are not all zero. As a priori, this is not possible since $[x : y : z] \in \mathbb{RP}^2$. Thus, $S = f^{-1}(0)$ is a regular level set, so an embedded submanifold of $\mathbb{RP}^2$. Furthermore, the dimension of S is $\dim \mathbb{RP}^2 - \dim \mathbb{R} = 2 - 1 = 1$. $\square$

Exercise 2.3.72. Let $F: M \rightarrow X$. $G: N \rightarrow X$ be smooth maps, and submersions. Show that

$$\{(m, n) \in M \times N: F(m) = G(n)\}$$

is a submanifold of $M \times N$. *Hint: Use local k -slice lemma.*

Solution: We can rewrite this set as

$$\{(m, n) \in M \times N: H(m, n) = 0\},$$

where $H(m, n) = F(m) - G(n)$. Note that H is clearly smooth. Then, this space is just $H^{-1}(0)$, and we claim that 0 is a regular value of H . For $(m, n) \in M \times N$, we have that

$$dH_{(m,n)} = \begin{pmatrix} dF_m & 0 \\ 0 & -dG_n \end{pmatrix}.$$

Since dF_m and dG_n are of full rank, it follows that the above matrix is also of full rank, so surjective. Therefore, 0 is indeed a regular value, and so by the regular level set theorem, this space is a submanifold of $M \times N$. $\square$

§2.4 Exercises from [1]

Exercise 2.4.1. Prove that $T\mathbb{S}^1$ is diffeomorphic to $\mathbb{S}^1 \times \mathbb{R}^1$.

Solution: Note that $\mathbb{S}^1 = \{x^2 + y^2 - 1 = 0\}$. Let $F = x^2 + y^2 - 1$. Let $p = (p_1, p_2) \in \mathbb{S}^1$. Thus, if $v \in T_p\mathbb{S}^1$, then $v(F) = 0$. Via the canonical inclusion $\iota: \mathbb{S}^1 \hookrightarrow \mathbb{R}^2$, we attain a smooth embedding $d\iota_p: T_p\mathbb{S}^1 \hookrightarrow T_p\mathbb{R}^2$. As this is injective and continuous, it is clearly a topological embedding. Now, we want to show that ι_p is injective, for all p . Suppose that $d\iota_p(v) = 0$; this means $d\iota_p(v)(f) = 0$, for all $f \in C^\infty(\mathbb{R}^2)$. Therefore, $v(f|_{\mathbb{S}^1}) = 0$. However, every smooth $f \in C^\infty(\mathbb{S}^1)$ is the restriction of some smooth function in $C^\infty(\mathbb{R}^2)$, so in fact $v(f) = 0$ for all $f \in C^\infty(\mathbb{S}^1)$. Thus, $v = 0$.

We showed this because then $T_p\mathbb{S}^1 \subseteq T_p\mathbb{R}^2$, so that we can use the standard basis $\frac{\partial}{\partial x}, \frac{\partial}{\partial y}$ on $T_p\mathbb{S}^1$. Let $v \in T_p\mathbb{S}^1$. Then, on $\mathbb{S}^1$, F is constant, so $v(F) = 0$.

$$0 = v(F) = v_1 \frac{\partial F}{\partial x} \Big|_p + v_2 \frac{\partial F}{\partial y} \Big|_p = v_1 \cdot 2p_1 + v_2 \cdot 2p_2.$$

Thus,

$$T_p\mathbb{S}^1 = \{(v_1, v_2) \in T_p\mathbb{R}^2: v_1 \cdot p_1 + v_2 \cdot p_2 = 0\} = \{t(p_2, -p_1) \in \mathbb{R}^2: t \in \mathbb{R}\}.$$

We must now construct the diffeomorphism. We consider the map $\Phi: \mathbb{S}^1 \times \mathbb{R}^1$ which maps

$$(p, t) \mapsto (p, t(p_2, -p_1)), \quad p = (p_1, p_2).$$

This is also well defined by the construction above. It is also clearly smooth. By construction, it is also bijective. The inverse is the mapping

$$(p, (v_1, v_2)) \mapsto (p, v_1 p_2 - v_2 p_1).$$

This indeed gives t because we know that $(v_1, v_2) = t(p_2, -p_1)$, so we just take the dot product with $(p_2, -p_1)$. This map is also clearly smooth. Therefore, we have constructed a valid diffeomorphism. $\square$

Exercise 2.4.2 (Theorem A.38 in [1]). Suppose X and Y are topological spaces, and $F: X \rightarrow Y$ is a continuous map that is either open or closed.

1. If F is surjective, then it is a quotient map.
2. If F is injective, then it is a topological embedding.
3. If F is bijective, then it is a homeomorphism.

Solution: Suppose first that F is surjective. If it is open, it certainly takes saturated open subsets to open subsets. Similarly, if it is closed, it takes saturated closed subsets to closed subsets. Thus it is a quotient map (by Exercise A.29 in [1]).

Now suppose F is open and injective. Then $F: X \rightarrow F(X)$ is bijective, so $F^{-1}: F(X) \rightarrow X$ exists by elementary set-theoretic considerations. If $U \subseteq X$ is open, then

$$(F^{-1})^{-1}(U) = F(U)$$

is open in Y by hypothesis, and therefore is also open in $F(X)$ by definition of the subspace topology on $F(X)$. This proves that F^{-1} is continuous, so that F is a homeomorphism onto its image. If F is closed, the same argument goes through with ‘open’ replaced by ‘closed’. This proves part 2, and part 3 is just the special case of 2 in which $F(X) = Y$. $\square$

Exercise 2.4.3 (Proposition 4.22 in [1]). Suppose M and N are smooth manifolds with or without boundary, and $F: M \rightarrow N$ is an injective smooth immersion. If any of the following holds, then F is a smooth embedding.

1. F is an open or closed map.
2. F is a proper map.
3. M is compact.

Solution: If F is open or closed, then it is a topological embedding by Exercise 2.4.2, so it is a smooth embedding. Either 2. or 3. implies that F is closed: if F is proper, then it is closed by Theorem 2.3.45, and if M is compact, then F is closed by the closed map lemma. $\square$

3 Vector Bundles

§3.1 Vector Bundles

We have seen that the tangent bundle TX of a smooth manifold X has a natural smooth manifold structure. The coordinates we constructed on TX make it look, locally, like the product of an open subset of X with $\mathbb{R}^n$. This kind of structure arises quite frequently: a collection of vector spaces, one for each point in X , glued together in a way that looks locally like a product, but globally may be twisted. Such structures are called vector bundles, and are the main subject of this section.

Definition 3.1.1. Let X be a smooth manifold. A smooth **vector bundle** of rank k over X is a smooth manifold E with a surjective smooth map $\pi : E \rightarrow X$ satisfying the following conditions

1. For each $p \in X$, the **fiber** $E_p := \pi^{-1}(p)$ over p is a k -dimensional real vector space.
2. For each $p \in X$, there exists a neighborhood U of p in X and a diffeomorphism $\Phi : \pi^{-1}(U) \rightarrow U \times \mathbb{R}^k$, called the **local trivialization** of E over U , such that
 - $\pi_U \circ \Phi = \pi$, where $\pi_U : U \times \mathbb{R}^k \rightarrow U$ is the projection.
 - for each $q \in U$, the restriction of Φ to E_q is a vector space isomorphism from E_q to $\{q\} \times \mathbb{R}^k$.

Exercise 3.1.2. Suppose E is a smooth vector bundle over M . Show that the projection map $\pi : E \rightarrow M$ is a surjective smooth submersion.

Solution: It is already known to be a surjection and smooth. Thus, we show it's a submersion. For this, we must show that $d\pi_p$ is surjective. Let $p \in X$. Then, we know there is a neighborhood for p , say U in X and a diffeomorphism $\Phi : \pi^{-1}(U) \rightarrow U \times \mathbb{R}^k$ such that $\pi_U \circ \Phi = \pi$. Then, to show that $d\pi_p$ is surjective, we just want to show that $d(\pi_U \circ \Phi)_p$ is surjective. Note that this is equal to $d(\pi_U)_{\Phi(p)} \circ d\Phi_p$. Since Φ is a diffeomorphism, $d\Phi_p$ is an isomorphism, so surjective. As for $d\pi_U$, this is imply projection onto the first factor, which we know is a submersion. As a composition of two surjections, $d\pi_p$ is surjective. Our p was arbitrary and the existence of the open neighborhood U is ensured at any p , so we are done. $\square$

Example 3.1.3

For any smooth manifold X , consider the product $X \times \mathbb{R}^k$ together with $\pi : X \times \mathbb{R}^k \rightarrow X$, $(p, v) \mapsto p$, is a rank k vector bundle over X , for any k , called the **product bundle**. Note that we can take U to be the whole of X , hence the product bundle admits a global trivialization. A vector bundle that admits a global trivialization is called a **trivial vector bundle**.

Proposition 3.1.4 (The Tangent Bundle is a Vector Bundle)

Let X be a smooth manifold of dimension n and let TX be its tangent bundle. With its standard smooth structure, vector space structure on each fiber, and the projection map, TX is a smooth vector bundle of rank n over X .

Proof. Given any smooth chart (U, ϕ) for X with coordinate functions $x_1, \dots, x_n$, define a map $\Phi: \pi^{-1}(U) \rightarrow U \times \mathbb{R}^n$ by

$$\Phi \left(\sum_{i=1}^n v^i \frac{\partial}{\partial x_i} \Big|_p \right) := (p, (v_1, \dots, v_n)).$$

Then one can check Φ satisfies the conditions in the definition. $\square$

In the above proposition, if we take another smooth chart (V, ψ) of X , then we can consider the overlap $U \cap V$. Let $\Psi: \pi^{-1}(V) \rightarrow V \times \mathbb{R}^k$.

$$\begin{array}{ccc} & \pi^{-1}(U \cap V) & \\ \swarrow \Phi & & \searrow \Psi \\ U \cap V \times \mathbb{R}^k & \xleftarrow{\Phi \circ \Psi^{-1}} & U \cap V \times \mathbb{R}^k \end{array}$$

Note that $\Phi \circ \Psi^{-1}$ is smooth. Then, fixing p , we get that

$$\begin{aligned} \Phi \circ \Psi^{-1}: U \cap V \times \mathbb{R}^k &\rightarrow U \cap V \times \mathbb{R}^k \\ (p, a_1, \dots, a_k) &\mapsto (p, b_1, \dots, b_k). \end{aligned}$$

Moreover, the coordinate change of the fiber direction is given by the Jacobian of the coordinate change from ϕ to ψ . In other words, it is a smooth map from $U \cap V$ to the space of invertible $k \times k$ real matrices, which now we write as $\text{GL}(k, \mathbb{R})$. This fact also works for general vector bundles.

Lemma 3.1.5

Let $\pi: E \rightarrow X$ be a vector bundle of rank k . Let $\Phi: \pi^{-1}(U) \rightarrow U \times \mathbb{R}^k$ and $\Psi: \pi^{-1}(V) \rightarrow V \times \mathbb{R}^k$ be two local trivializations of E . If $U \cap V$ is not empty, then there exists a smooth map $\tau: U \cap V \rightarrow \text{GL}(k, \mathbb{R})$ such that the composition $\Phi \circ \Psi^{-1}: (U \cap V) \times \mathbb{R}^k \rightarrow (U \cap V) \times \mathbb{R}^k$ has the form

$$\Phi \circ \Psi^{-1}(p, v) = (p, \tau(p) \cdot v).$$

This map τ is called the local **transition map** or **transition matrix**.

Proof. By the definition of vector bundles, the map $\Phi \circ \Psi^{-1}(p, v) = (p, \sigma(p, v))$ for some smooth map $\sigma: U \cap V \times \mathbb{R}^k \rightarrow \mathbb{R}^k$. Note that for any fixed p , the map $\sigma(p, \cdot): \mathbb{R}^k \rightarrow \mathbb{R}^k$ is a linear vector space isomorphism. Hence $\sigma(p, v) = \tau(p) \cdot v$ for some $\tau(p) \in \text{GL}(k, \mathbb{R})$. Then we need to verify that τ depends on p smoothly. This follows from the smoothness of σ . If we let $e_1, \dots, e_k$ denote the standard basis vectors for $\mathbb{R}^k$, then $\sigma(p, e_j) = \tau(p) \cdot e_j$, so the j -th column of $\tau(p)$ is exactly $\sigma(p, e_j)$. As for the column, writing $\sigma(p, e_j) = (\sigma^1(p, e_j), \dots, \sigma^k(p, e_j))$, we get that $\tau_j^i(p) = \sigma^i(p, e_j)$. Now, σ is smooth as a function of (p, v) , so this coordinate representation of τ tells us that every coordinate $\sigma^i(p, e_j)$ is smooth in p . Thus, every entry $\tau_j^i(p)$ is smooth. $\square$

Exercise 3.1.6. Continue from Example 2.3.67. Consider the map $j: TS \times \mathbb{R}$ defined by $(p, v) \mapsto (p, a(v))$. Show it is a global trivialization hence S has a trivial tangent bundle.

Like the tangent bundle, smooth vector bundles are often easier to be described by giving a collection of vector spaces. The next lemma provides a sufficient condition to construct a vector bundle structure from smooth transition functions.

Lemma 3.1.7 (Vector Bundle Chart Lemma)

Let X be a smooth manifold. Suppose that for any $p \in X$ it is given a real vector space E_p of fixed dimension k . Set $E := \bigsqcup_{p \in X} E_p$. Let $\pi: E \rightarrow X$ be the map which sends each element of E_p to p . Suppose we furthermore have the following data:

- an open cover $\{U_\alpha\}_{\alpha \in A}$ of X
- for each $\alpha \in A$, a bijective map $\Phi_\alpha: \pi^{-1}(U_\alpha) \rightarrow U_\alpha \times \mathbb{R}^k$ whose restriction to each E_p is a vector space isomorphism from E_p to $\{p\} \times \mathbb{R}^k \cong \mathbb{R}^k$.
- for each $\alpha, \beta \in A$ with $U_\alpha \cap U_\beta \neq \emptyset$, a smooth map $\tau_{\alpha\beta}: U_\alpha \cap U_\beta \rightarrow GL_k(\mathbb{R})$ such that $\Phi_\alpha \circ \Phi_\beta^{-1}$ has the form

$$\Phi_\alpha \circ \Phi_\beta^{-1}(p, v) = (p, \tau_{\alpha\beta}(p) \cdot v).$$

Then E has a unique topology and smooth structure making it into a smooth manifold and a smooth rank- k vector bundle over X , with π as projection and $\{(U_\alpha, \Phi_\alpha)\}$ as smooth local trivializations.

Proof. For each point $p \in M$, choose some U_α containing p ; choose a smooth chart (V_p, φ_p) for M such that $p \in V_p \subseteq U_\alpha$; and let $\widehat{V}_p = \varphi_p(V_p) \subseteq \mathbb{R}^n$, where n is the dimension of M . Define a map $\tilde{\varphi}_p: \pi^{-1}(V_p) \rightarrow \widehat{V}_p \times \mathbb{R}^k$ by $\tilde{\varphi}_p = (\varphi_p \times \text{Id}_{\mathbb{R}^k}) \circ \Phi_\alpha$:

$$\pi^{-1}(V_p) \xrightarrow{\tilde{\varphi}_p} V_p \times \mathbb{R}^k \xrightarrow{\varphi_p \times \text{Id}_{\mathbb{R}^k}} \widehat{V}_p \times \mathbb{R}^k.$$

We will show that the collection of all such charts $\{(\pi^{-1}(V_p), \tilde{\varphi}_p): p \in M\}$ satisfies the conditions of the smooth manifold chart lemma (Lemma 1.1.32), and therefore gives E the structure of a smooth manifold.

As a composition of bijective maps, $\tilde{\varphi}_p$ is bijective onto an open subset of either $\mathbb{R}^n \times \mathbb{R}^k = \mathbb{R}^{n+k}$. For any $p, q \in M$, it is easy to check that

$$\tilde{\varphi}_p(\pi^{-1}(V_p) \cap \pi^{-1}(V_q)) = \varphi_p(V_p \cap V_q) \times \mathbb{R}^k,$$

which is open because φ_p is a homeomorphism onto an open subset of $\mathbb{R}^n$. Wherever two such charts overlap, we have

$$\tilde{\varphi}_p \circ \tilde{\varphi}_q^{-1} = (\varphi_p \times \text{Id}_{\mathbb{R}^k}) \circ \Phi_\alpha \circ \Phi_\beta^{-1} \circ (\varphi_q \times \text{Id}_{\mathbb{R}^k})^{-1}.$$

Since $\varphi_p \times \text{Id}_{\mathbb{R}^k}$, $\varphi_q \times \text{Id}_{\mathbb{R}^k}$, and $\Phi_\alpha \circ \Phi_\beta^{-1}$ are diffeomorphisms, the composition is a diffeomorphism. Thus, conditions (i) to (iii) of the smooth manifold chart lemma are satisfied. Because the open cover $\{V_p: p \in M\}$ has a countable subcover, (iv) is satisfied as well.

To check the Hausdorff condition (v), just note that any two points in the same space E_p lie in one of the charts we have constructed; while if $\xi \in E_p$ and $\eta \in E_q$ with $p \neq q$, we can choose V_p and V_q to be disjoint neighborhoods of p and q , so that the sets $\pi^{-1}(V_p)$ and $\pi^{-1}(V_q)$ are disjoint coordinate neighborhoods containing ξ and η , respectively. Thus we have given E the structure of a smooth manifold.

With respect to this structure, each of the maps Φ_α is a diffeomorphism, because in terms of the coordinate charts $(\pi^{-1}(V_p), \tilde{\varphi}_p)$ for E and $(V_p \times \mathbb{R}^k, \varphi_p \times \text{Id}_{\mathbb{R}^k})$ for $V_p \times \mathbb{R}^k$, the coordinate representation of Φ_α is the identity map. The coordinate representation of π , with respect to the same chart for E and the chart (V_p, φ_p) for M , is $\pi(x, v) = x$, so π is smooth as well. Because each Φ_α maps E_p to $\{p\} \times \mathbb{R}^k$, it is immediate that $\pi \circ \Phi_\alpha = \pi$, and Φ_α is linear on fibers by hypothesis. Thus, Φ_α satisfies all the conditions for a smooth local trivialization.

The fact that this is the unique such smooth structure follows from the requirement that the maps Φ_α be diffeomorphisms onto their images: any smooth structure satisfying the same conditions must include all of the charts we constructed, so it is equal to this one. $\square$

Remark 3.1.8 — This lemma is saying that if we have a set of matrices, then we can make a vector bundle. It is like a converse to the statement above the lemma.

Exercise 3.1.9. Let $\pi: E \rightarrow M$ be a smooth vector bundle of rank k over a smooth manifold M with or without boundary. Suppose that $\{U_\alpha\}_{\alpha \in A}$ is an open cover of M , and for each $\alpha \in A$ we are given a smooth local trivialization $\Phi_\alpha: \pi^{-1}(U_\alpha) \rightarrow U_\alpha \times \mathbb{R}^k$ of E . For each $\alpha, \beta \in A$ such that $U_\alpha \cap U_\beta \neq \emptyset$, let $\tau_{\alpha\beta}: U_\alpha \cap U_\beta \rightarrow \text{GL}(k, \mathbb{R})$ be the transition function defined as in the vector bundle chart lemma. Show that the following identity is satisfied for all $\alpha, \beta, \gamma \in A$:

$$\tau_{\alpha\beta}(p)\tau_{\beta\gamma}(p) = \tau_{\alpha\gamma}(p), \quad p \in U_\alpha \cap U_\beta \cap U_\gamma.$$

Solution: This follows by the fact that

$$\Phi_\alpha \circ \Phi_\gamma(p, v) = \Phi_\alpha \circ \Phi_\beta^{-1} \circ \Phi_\beta \circ \Phi_\gamma^{-1}(p, v).$$

Fiber-wise, the LHS has $\tau_{\alpha\gamma}(p)$ while the RHS has $\tau_{\alpha\beta}(p)\tau_{\beta\gamma}(p)$. Then, since the local trivializations are necessarily bijective, it follows that these two are equal. $\square$

Exercise 3.1.10 (Vector Bundle Construction Theorem). Let M be a smooth manifold with or without boundary, and let $\{U_\alpha\}_{\alpha \in A}$ be an open cover of M . Suppose for each $\alpha, \beta \in A$ we are given a smooth map

$$\tau_{\alpha\beta}: U_\alpha \cap U_\beta \rightarrow \text{GL}(k, \mathbb{R})$$

such that

$$\tau_{\alpha\beta}(p)\tau_{\beta\gamma}(p) = \tau_{\alpha\gamma}(p)$$

for all $\alpha, \beta, \gamma \in A$ and all $p \in U_\alpha \cap U_\beta \cap U_\gamma$. Then there is a smooth rank- k vector bundle $\pi: E \rightarrow M$ with smooth local trivializations

$$\Phi_\alpha: \pi^{-1}(U_\alpha) \rightarrow U_\alpha \times \mathbb{R}^k$$

whose transition functions are the given maps $\tau_{\alpha\beta}$. *Hint.* Define an appropriate equivalence relation on $\bigsqcup_{\alpha \in A} (U_\alpha \times \mathbb{R}^k)$, and use the vector bundle chart lemma.

Solution: As usual, we want to apply the vector bundle chart lemma, so we have to come up with good fibers and local trivializations. First, we follow the hint. Define $E := \bigsqcup_{\alpha \in A} (U_\alpha \times \mathbb{R}^k) / \sim$, where $(\alpha, p, v) \sim (\beta, q, w)$ if and only if $p = q$ (so $U_\alpha \cap U_\beta \neq \emptyset$) and $v = \tau_{\alpha\beta}(p)w$. This is indeed an equivalence relation, as it is reflexive, symmetric, and transitive. For symmetry, we note that $\tau_{\alpha\beta}(p)^{-1} = \tau_{\beta\alpha}(p)$. We are also equipped with a cover, so this assumption is satisfied. Let $\pi: E \rightarrow M$ be the natural projection. Then, we define $E_p := \pi^{-1}(p)$. Note that E_p is a collection of equivalent classes, and this forms a vector space by defining the following:

$$\begin{aligned} [\alpha, p, v] + [\beta, q, w] &:= [\alpha, p, v + \tau_{\alpha\beta}(p)w] \\ \lambda[\alpha, p, v] &:= [\alpha, p, \lambda v]. \end{aligned}$$

Now, we define the local trivializations $\Phi_\alpha: \pi^{-1}(U_\alpha) \rightarrow U_\alpha \times \mathbb{R}^k$ via

$$[\alpha, p, v] \mapsto (p, v).$$

This is clearly bijective. Restricting to E_p , this is clearly a vector space isomorphism since E_p and $\{p\} \times \mathbb{R}^k$ are both k -dimensional vector spaces and Φ_α is bijective. Now, let α, β be two indices which satisfy $U_\alpha \cap U_\beta \neq \emptyset$. We calculate the transition map $\Phi_\alpha \circ \Phi_\beta^{-1}$:

$$\Phi_\alpha \circ \Phi_\beta^{-1}(p, w) = \Phi_\alpha([\beta, p, w]) = \Phi_\alpha([\alpha, p, \tau_{\alpha\beta}(p)w]) = (p, \tau_{\alpha\beta}(p)w).$$

We know that these are smooth by assumption, so applying the vector bundle chart lemma, we are done. In particular, the transition maps are given by the $\tau_{\alpha\beta}$. $\square$

Example 3.1.11

We first let $E = \mathbb{R}^2 / \sim$ be the Möbius band, where $(x, y) \sim (x', y') \iff (x', y') = (x + n, (-1)^n y)$. I believe in the lectures, we took $n = 1$, but here I take general fixed n . Then, we define the Möbius bundle using the map π appearing in the following commutative diagram:

$$\begin{array}{ccc} \mathbb{R}^2 & \xrightarrow{q} & E \\ \downarrow \pi_1 & & \downarrow \\ \mathbb{R} & \xrightarrow{\epsilon} & \mathbb{S}^1 \end{array}$$

Then, π is defined on E by mapping on equivalence class $[t, x]$ to $e^{2\pi i t}$. Here, I've adapted the notion that $[\cdot, \cdot]$ represents an equivalence class in E .

Then, we define our fibers to be $E_p := \pi^{-1}(p)$, where $p \in \mathbb{S}^1$, and E is then the coproduct of E_p , indexed by p . Let U be an arc in $\mathbb{S}^1$, and define $\theta: U \rightarrow \mathbb{R}$ to be the map which takes the angle of $z \in U$. This is locally bijective. We now let U_1, U_2 be an open cover of $\mathbb{S}^1$. Using this, we define our local trivialization $\Phi_i: \pi^{-1}(U_i) \rightarrow U_i \times \mathbb{R}$ to be defined as $[\theta(z), x] \mapsto (z = e^{2\pi i \theta(z)}, x)$. We note that Φ_i are bijective, since θ is bijective on U_i . Now, the transition map is calculated as follows:

$$\Phi_2 \circ \Phi_1^{-1}(z, x) = \Phi_2([\theta_1(z), x]) = \Phi_2([\theta_2(z), (-1)^n x]) = (z, (-1)^n x)$$

Note that our transition map is in $\text{GL}_1(\mathbb{R})$ and depends smoothly on z . The assumptions of the vector bundle chart lemma are now fully satisfied, so this defines a vector bundle on $\mathbb{S}^1$.

Let n be odd now. To show this is nontrivial, we show there is no nowhere vanishing section $\sigma: \mathbb{S}^1 \rightarrow E$. If some σ did exist, then $\sigma(e^{2\pi i t}) = [t, f(t)]$, where f is some smooth function on $\mathbb{R}$ such that $f(t) \neq 0$, for all t . n is odd, so $[t + n, f(t + 1)] = [t, (-1) \cdot f(t)]$, so we must have that $f(t + 1) = -f(t)$. This contradicts the intermediate value theorem.

If n is even, then the transition maps are the identity, which glue together to provide a global chart. We've seen such vector bundles are trivial, so the trivial/non-trivial property of this vector bundle depends on the parity of n .

Definition 3.1.12. A **smooth section** of a vector bundle is a smooth map

$$\sigma: X \rightarrow E, \quad \pi \circ \sigma = \text{Id}.$$

For any subset $U \subseteq X$, one can also define a **local section** as a smooth map

$$\sigma: U \rightarrow \pi^{-1}(U), \quad \pi \circ \sigma = \text{Id}.$$

Example 3.1.13

A section of TX is a vector field.

Example 3.1.14

For any smooth vector bundle $\pi: E \rightarrow X$, the **zero section** is defined by $\sigma(p) = 0 \in E_p \forall p \in X$.

Example 3.1.15

If σ is a section of the Möbius bundle, then $\exists p \in X = \mathbb{S}^1$ such that $\sigma(p) = 0$. We prove this below.

Assume that σ has no zeros. Pick $p \in I_+$ and $q \in I_-$. Then, $\sigma|_U(p) = \sigma_U(p)$, since the identification on I_+ is the identity. On the other hand, we have $\sigma|_U(q) = -\sigma|_V(q)$. However, this contradicts the intermediate value theorem.

Given the claim, we can see that the Möbius bundle is not the trivial bundle, since the trivial bundle $\mathbb{S}^1 \times \mathbb{R}$ has nonzero sections.

Remark 3.1.16 — We can go an extra step and say that whether the bundle is trivial or not depends on the parity of the number of ‘flips’.

§3.1.i Comparing Vector Bundles

Definition 3.1.17. Given two vector bundles E and E' (with maps π and π' respectively), a smooth map $F: E \rightarrow E'$ is called a **bundle homomorphism** over M if $\pi' \circ F = \pi$ and $F|_{E_p}: E_p \rightarrow E'_p$ is linear for all $p \in M$.

$$\begin{array}{ccc} E & \xrightarrow{F} & E' \\ & \searrow \pi & \swarrow \pi' \\ & M & \end{array}$$

Moreover, if a bundle homomorphism is a diffeomorphism and $F^{-1}: E' \rightarrow E$ is also a bundle homomorphism, then we say that F is a **bundle isomorphism**, and E and E' are isomorphic.

Exercise 3.1.18. Show that a vector bundle is trivial if and only if it is isomorphic to the product bundle.

Solution: Suppose that $\pi: E \rightarrow X$ is trivial. Then, it admits a global trivialization $\Phi: \pi^{-1}(X) \rightarrow X \times \mathbb{R}^k$. In particular, Φ is a diffeomorphism. Then, Φ is a bundle homomorphism from E to $X \times \mathbb{R}^k$ because by definition, $\pi_X \circ \Phi = \pi$, where $\pi_X: X \times \mathbb{R}^k \rightarrow X$. Since the inverse, Φ^{-1} is also a bundle homomorphism, we have that E and $X \times \mathbb{R}^k$ are isomorphic bundles.

Conversely, if a vector bundle $\pi: E \rightarrow X$ is isomorphic to the product bundle, there exists a diffeomorphism $\Phi: E \rightarrow X \times \mathbb{R}^k$ such that $\pi = \pi_X \circ \Phi$. Then, Φ exists and is also a bundle homomorphism, so we get a global trivialization. $\square$

Definition 3.1.19. A C^∞ collection of smooth sections $\sigma_1, \dots, \sigma_k$ is called **linearly independent** at p if $\sigma_1(p), \dots, \sigma_k(p)$ are linearly independent vectors for any $p \in X$. For a rank- k smooth vector bundle, a **frame** is a collection of k linearly independent sections.

Proposition 3.1.20

A vector bundle $\pi: E \rightarrow M$ is trivial if and only if it admits a frame.

Proof. First suppose we have a trivial bundle. By the exercise above, we get that $E = \mathbb{R}^k \times M \xrightarrow{\pi} M$. A section is a right inverse of π , so we just need $\sigma_1, \dots, \sigma_k$ that are linearly independent. For this, pick a basis $\{e_1, \dots, e_k\}$ of $\mathbb{R}^k$ and define $\sigma_i(p) = (e_i, p)$, for all $p \in M$. Then, $\sigma_1, \dots, \sigma_k$ is a frame.

Now suppose that $\{\sigma_1, \dots, \sigma_k\}$ is a frame of E . Define a $\Psi: \mathbb{R}^k \times M \rightarrow E$ defined by

$$(a_1, \dots, a_k, p) \mapsto \left(\sum a_i \sigma_i(p), p \right).$$

This is bijective, so we prove smoothness. Pick a local trivialization $(U, \Phi: \pi^{-1}(U) \rightarrow \mathbb{R}^k \times U)$.

$$\begin{array}{ccccc} \mathbb{R}^k \times U & \xrightarrow{\Psi} & \pi^{-1}(U) & \xrightarrow{\Phi} & \mathbb{R}^k \times U \\ & & \downarrow \pi & & \\ & & M & & \end{array}$$

σ_i (dashed arrow from $\mathbb{R}^k \times U$ to M)

Thus, we can compose the two maps. For each σ_i , $\Phi \circ \sigma_i$ is a smooth map. Thus there exist smooth functions $\sigma_i^1, \dots, \sigma_i^k: U \rightarrow \mathbb{R}$ such that

$$\Phi(\sigma_i(p), p) = (\sigma_i^1(p), \dots, \sigma_i^k(p), p).$$

Therefore the composition $\Phi \circ \Psi$ has the form

$$(a_1, \dots, a_k, p) \mapsto \left(\sum_i a_i \sigma_i^1(p), \dots, \sum_i a_i \sigma_i^k(p), p \right).$$

This is smooth since σ_i^j are smooth. Hence $\Phi \circ \Psi$ is smooth. By bijectivity, by considering maps in the opposite direction we can show $(\Phi \circ \Psi)^{-1}$ is smooth. Therefore Ψ is a local diffeomorphism. $\square$

Example 3.1.21

If σ is a section of the Möbius bundle, then $\exists p \in X = \mathbb{S}^1$ such that $\sigma(p) = 0$, so the proposition tells us that this is nontrivial. For a trivial vector bundle $E = X \times \mathbb{R}^k$, let $e_1, \dots, e_k$ be a basis of $\mathbb{R}^k$. Then

$$e_i(p) = (p, (0, \dots, 1, \dots, 0)), \quad i = 1, \dots, k,$$

form a frame. The converse of this is also true.

Now what if there are two ‘twists’? On each of the local trivializations, these can be nonzero (constant), but they agree when identified together. This leads us to the claim that trivial bundle on $X = \mathbb{S}^1$ depends on the parity of these twists. One thing to note is that we can’t ‘untwist’ the ‘two-twist’ into the trivial bundle, despite these two being diffeomorphic. As abstract manifolds, these are diffeomorphic, but as embeddings in $\mathbb{R}^3$, these are different (non-isotopic)

Corollary 3.1.22

If a smooth manifold X of dimension k admits k smooth vector fields which are linearly independent at every point, then the tangent bundle of X is trivial.

Example 3.1.23

Consider the unit circle $S = \{x^2 + y^2 = 1\} \subset \mathbb{R}^2$ and a smooth map

$$F: \mathbb{R} \rightarrow S, \quad F(t) = (\cos(t), \sin(t)).$$

It is an immersion, hence the nowhere vanishing vector field $\left(\frac{\partial}{\partial t}\right)$ on $\mathbb{R}$ is pushed to a nowhere vanishing vector field on S . That is,

$$F_* \left(\frac{\partial}{\partial t} \right) = dF_t \left(\frac{\partial}{\partial t} \right) = (-\sin t, \cos t).$$

Thus, it is clear that $F_* \left(\frac{\partial}{\partial t} \right)$ is nowhere vanishing. Therefore S has a trivial tangent bundle.

There is a deep and extensive body of theory about vector bundles on manifolds, which we cannot even touch. We will mainly focus on the tangent bundles, and its dual, the cotangent bundle bundles.

§3.1.ii The Cotangent Bundle

Definition 3.1.24. Let V be a finite dimensional real vector space. We define a **covector** on V to be a real-valued linear map $\omega: V \rightarrow \mathbb{R}$. The space of all covectors on V is itself a real vector space, denoted by V^* and called the **dual space** of V .

Recall in linear algebra class, we have:

Proposition 3.1.25

Let V be a finite dimensional vector space. Given any basis $(e_1, \dots, e_n)$ for V , let $\epsilon_1, \dots, \epsilon_n \in V^*$ be the covectors defined by

$$\epsilon_i(e_j) = \delta_{ij}.$$

Then, the covectors $\epsilon_1, \dots, \epsilon_n$ form a basis for V^* , called the **dual basis** of $(e_1, \dots, e_n)$.

Definition 3.1.26. Let X be a smooth manifold. For any $p \in X$, define the **cotangent space** at p , denoted by T_p^*X , to be the dual space to T_pX . Elements of T_p^*X are called **tangent covectors** at p , or just **covectors** at p .

Definition 3.1.27. Let X be a smooth manifold, its **cotangent bundle** is defined as

$$T^*X := \bigsqcup_{p \in X} T_p^*X.$$

We note that the cotangent bundle is indeed a smooth manifold. Let (U, φ) be a smooth chart on X with coordinate functions $x_1, \dots, x_n$. Recall for each $p \in U$ we have a basis

$$\left. \frac{\partial}{\partial x_1} \right|_p, \dots, \left. \frac{\partial}{\partial x_n} \right|_p$$

for $T_p X$ by using the chart. Let $\lambda_1|_p, \dots, \lambda_n|_p$ be the dual basis. It follows then that any covector $\omega \in T_p^* X$ can be written as

$$\omega = \sum_i \omega_i \lambda_i|_p, \quad \omega_i = \omega \left(\frac{\partial}{\partial x_i} \Big|_p \right) \in \mathbb{R}.$$

Remark 3.1.28 — This tells us that the same chart (U, φ) provides coordinate expressions for covectors.

Suppose (V, ψ) is another chart containing p , with coordinate functions $\tilde{x}_1, \dots, \tilde{x}_n$. The coordinate vector transforms as follows

$$\frac{\partial}{\partial x_i} \Big|_p = \sum_j \frac{\partial \tilde{x}_j}{\partial x_i}(p) \frac{\partial}{\partial \tilde{x}_j} \Big|_p.$$

Here, we use p to denote either a point in X or its coordinate representation as appropriate. Writing ω in both coordinates as

$$\omega = \sum_i \omega_i \lambda_i|_p = \sum_j \tilde{\omega}_j \tilde{\lambda}_j|_p,$$

the components ω_i transforms as follows

$$\omega_i = \sum_j \frac{\partial \tilde{x}_j}{\partial x_i} \tilde{\omega}_j.$$

Now, we have that $\sum_j \frac{\partial \tilde{x}_j}{\partial x_i}$ is smooth. By using this coordinate change formula, we can verify that $T^* X$ is a smooth manifold.

Proposition 3.1.29

Let X be a smooth manifold with dimension n . Its cotangent bundle $T^* X$ admits a smooth manifold structure with dimension $2n$.

Proof. Let (U, ϕ) be a chart on X and let $\pi: T^* X \rightarrow X$ be the projection. The map

$$\tilde{\phi}: \pi^{-1}(U) \rightarrow \mathbb{R}^{2n}, \quad \tilde{\phi}(p, \omega) = (\phi(p), \omega_1, \dots, \omega_n)$$

is a homeomorphism. The transition maps between different charts are computed as above, which are smooth. $\square$

Proposition 3.1.30

The cotangent bundle $\pi: T^* X \rightarrow X$ is a vector bundle over X with rank equal to $\dim X$.

Proof. Similar to showing that the tangent bundle was a vector bundle, we must define the local trivializations. We note that a chart (U, φ) provides a coordinate basis for the covectors. Thus, given some $p \in M$ and chart (U, φ) , let $\Phi: \pi^{-1}(U) \rightarrow U \times \mathbb{R}^k$ be the map defined by

$$\left(p, \sum_{j=1}^n \omega_j \lambda_j \Big|_p \right) \mapsto (p, \omega_1, \dots, \omega_n).$$

Checking that this actually works is analogous to Proposition 3.1.4. $\square$

Exercise 3.1.31. Let $\pi: E \rightarrow X$ be a vector bundle. Show that:

1. $E^* := \coprod_{p \in M} E_p^* \xrightarrow{\pi} X$ is a vector bundle, where E_p^* is the dual space to E_p .
2. Compute the transition matrix of E^* in terms of those of E .

Solution: 1. We first need to understand what E_p^* looks locally. Particularly, I want to show that a chart (U, φ) can be used to provide coordinate representations for the dual space. Let $p \in X$. Then, E_p is a n -dimensional vector space, say spanned by $v_1, \dots, v_n$. We can find a dual basis to this, say $\lambda_1, \dots, \lambda_n$. Therefore, E_p^* is also a k -dimensional $\mathbb{R}$ -vector space. This tells us that any dual vector $\omega \in E_p^*$ can be written as

$$\omega = \sum_{i=1}^n \underbrace{\omega(v_i)}_{=: \omega_i} \lambda_i |_p,$$

We now need to define local trivializations $\Phi: \pi^{-1}(U) \rightarrow U \times \mathbb{R}^n$. This will be defined as

$$(q, \omega) \mapsto (q, \omega_1, \dots, \omega_n), \quad \forall q \in U.$$

This is a valid local trivialization. This is because we clearly have $\pi_U \circ \Phi = \pi$, where $\pi_U: U \times \mathbb{R}^n \rightarrow U$ is the projection. We also have that for each $q \in U$, Φ restricted to the fiber E_q is a vector space isomorphism from E_q to $\{q\} \times \mathbb{R}^n$, since E_q and E_q^* are isomorphic (because they are finite dimensional isomorphisms). As for the transition maps, they depend smoothly on p as it is some change of basis matrix. This is concretely computed below.

2. Let $(U, \varphi), (V, \psi)$ be two charts with nonempty intersection that induce two different bases on X , $(x_1, \dots, x_n)$ on U and $(y_1, \dots, y_n)$ on V . These also induce the dual bases $(\lambda_1, \dots, \lambda_n)$ and $(\omega_1, \dots, \omega_n)$; let these be the canonical ones. Further, these come with local trivializations $\Phi: \pi^{-1}(U) \rightarrow U \times \mathbb{R}^n$ and $\Psi: \pi^{-1}(V) \rightarrow V \times \mathbb{R}^n$. Let $p \in U \cap V$. Then, since E_p is an n -dimensional vector space with two different bases, one using x and one using y , we get that

$$x_i = \sum_{j=1}^n a_{i,j} y_j.$$

In matrix form, I write this as $x = Ay$, where x, y are $n \times 1$ matrices with x_i in each entry and A has entries $a_{i,j}$. Note that we can also write

$$\omega_i = \sum_{j=1}^n c_{i,j} \lambda_j.$$

If we apply this to x_k , we get that $\omega_i(x_k) = c_{i,k}$. Now, apply ω_i to y_j . This gives:

$$\delta_{ij} = \omega_i(y_j) = \omega_i \left(\sum_{\ell=1}^n (A^{-1})_{j\ell} x_\ell \right) = \sum_{\ell=1}^n (A^{-1})_{j\ell} \omega_i(x_\ell) = \sum_{\ell=1}^n (A^{-1})_{j\ell} c_{i,\ell}.$$

Solving this, we get that $c_{i,\ell} = (A^{-1})_{\ell,i}$. Thus,

$$\omega = \lambda(A^{-1})^T,$$

so the transition matrix is A^{-1T} , where A is the transition matrix for $\Phi \circ \Psi^{-1}$. It follows that $\tau(p)^* = (\tau(p)^{-1})^T$. $\square$

Exercise 3.1.32. Let E, F be two isomorphic vector bundles over a manifold M . Then, their dual bundles are also isomorphic.

Solution: Let $\Phi: E \rightarrow F$ be an isomorphism of vector bundles. Then, we show this induces a vector bundle isomorphism $\Phi^*: F^* \rightarrow E^*$. We define this fiber-wise in the following way:

$$\Phi_p^*: F_p^* \rightarrow E_p^*, \quad \Phi_p^*(\lambda)(v) = \lambda(\Phi_p(v)).$$

First of all, at each fiber, this is clearly a linear map. Further, this is an isomorphism because Φ_p is an isomorphism. Now we must check this and its inverse are smoothly dependent on p . If Φ takes the form $A(p)$, for some matrix A smoothly dependent on p , then the transition map/ matrix is given by $A(p)^T$ (if we act on λ from the right). This is still smoothly dependent on p . As for the inverse, it will be $(A(p)^T)^{-1}$, which is also smoothly dependent, so we are done. $\square$

Exercise 3.1.33. Show that T^*M is trivial if and only if TM is trivial.

Solution: We note that $M \times \mathbb{R}^k$ is isomorphic to its dual bundle (as above), which is just itself. Therefore, T^*M is trivial if and only if TM is trivial. $\square$

Definition 3.1.34. Similar to vector fields, we can define the notion of **covector field** is a section of T^*X :

$$\sigma: X \rightarrow T^*X, \quad \pi \circ \sigma = \text{Id}.$$

Remark 3.1.35 — In other words, a covector field is a smooth section of the cotangent bundle.

Definition 3.1.36. Given any smooth function $f: X \rightarrow \mathbb{R}$, it gives a covector field df by

$$(df)_p(v) := v(f) \mid_p, \quad \forall p \in X, v \in T_pX.$$

Let (U, ϕ) be a chart with coordinate functions $x_1, \dots, x_n$. Let $\lambda_1, \dots, \lambda_n$ be the dual basis to $\frac{\partial}{\partial x_1}, \dots, \frac{\partial}{\partial x_n}$. Then one can directly compute that

$$(df)_p \left(\frac{\partial}{\partial x_i} \right) = \frac{\partial f}{\partial x_i}(p),$$

so

$$df_p = \sum_i \frac{\partial f}{\partial x_i}(p) \lambda_i \mid_p.$$

In particular, we get $(dx_i)_p = \lambda_i \mid_p$ for any i . From now on, we will also use $dx_1 \mid_p, \dots, dx_n \mid_p$ as the dual basis to $\frac{\partial}{\partial x_1}, \dots, \frac{\partial}{\partial x_n}$, once a chart is chosen.

Remark 3.1.37 — Consider $f: X \rightarrow \mathbb{R}$ and $g: \mathbb{R} \rightarrow \mathbb{R}$. The push forward is defined as $df(v)(g) = v(g \circ f)$. Let $v = \sum a_i \frac{\partial}{\partial x_i}$. Then,

$$(df)(v) = \sum a_i \frac{\partial f}{\partial x_i},$$

in the notation just introduced. However, the push forward $(df)(v)(x)$ is equal to

$$\left(\sum a_i \frac{\partial}{\partial x_i} \right) (f) = \sum a_i \frac{\partial f}{\partial x_i},$$

so they agree: $(df)(v)(x) = (df)(v)$ (x here is the identity map on $\mathbb{R}$), where the left is the push forward and the right is the new definition just introduced. There is some coinciding of notation here but these mean different things.

Remark 3.1.38 — Note that to push forward a vector field, we really require the map to be a diffeomorphism. Indeed, injectivity is necessary for the map to be well defined, and we need the surjectivity since we want every point to be covered.

§3.1.iii Pullbacks of Covector Fields

Notation 3.1.39. Let $\mathfrak{X}^*(X)$ be the space of covector fields on X .

Similar to the space of vector fields on X , it is a vector space over real numbers. Moreover, for any smooth function f on X and $\sigma \in \mathfrak{X}^*(X)$, we get a new covector field $f \cdot \sigma \in \mathfrak{X}^*(X)$, by setting

$$(f \cdot \sigma)_p := f(p) \cdot \sigma(p).$$

Recall if we have a smooth map $F: X \rightarrow Y$ between two smooth manifolds then we can define a map $dF: TX \rightarrow TY$. When F is a diffeomorphism, we can push forward a vector field from X to Y . Now we define a dual notion.

Definition 3.1.40. Let $F: X \rightarrow Y$ be a smooth map. The **pullback** by F at $p \in X$, or the **cotangent map** of F is defined as

$$dF_p^*: T_{F(p)}^*Y \rightarrow T_p^*X, \quad dF_p^*(\omega)(v) := \omega(dF_p(v)).$$

(Not similar to vector fields, covector fields always pull back to covector fields; without assuming the map is a diffeomorphism.)

Proposition 3.1.41

Let $F: X \rightarrow Y$ be a smooth map. For any $\omega \in \mathfrak{X}^*(Y)$, define

$$(F^*\omega)_p := dF_p^*(\omega_{F(p)}).$$

Then $F^*\omega$ is a smooth covector field on X . That is, the pullback map $F^*\omega$ is always a covector field on X .

Proof. The proof follows from computation in coordinates, based on the lemma below. Let $p \in X$ and V be a smooth chart on Y containing $F(p)$. Let $(y_1, \dots, y_n)$ be the coordinate functions on V . Then we can write ω in coordinates as $\omega = \sum_j \omega_j dy_j$. Since ω is smooth, the functions ω_j 's are smooth. Apply the following lemma to the smooth map $F|_{F^{-1}(V)}: F^{-1}(V) \rightarrow V$, we get

$$F^*\omega = F^*\left(\sum_j \omega_j dy_j\right) = \sum_j (\omega_j \circ F) F^*(dy_j) = \sum_j (\omega_j \circ F) d(y_j \circ F).$$

Since ω_j, y_j, F are smooth, the pull back is smooth. □

Lemma 3.1.42 (Proposition 11.25 in [1])

Let $F: X \rightarrow Y$ be a smooth map.

- (i) For $\omega, \eta \in \mathfrak{X}^*(Y)$, we have $F^*(\omega + \eta) = F^*\omega + F^*\eta$.
- (ii) For $f \in C^\infty(Y)$ and $\omega \in \mathfrak{X}^*(Y)$, we have $F^*(f \cdot \omega) = (f \circ F) \cdot F^*\omega$.
- (iii) For $f \in C^\infty(Y)$, $F^*(df) = d(f \circ F)$.

Proof. 1. This follows by the linearity of the push forward map.

2. We have as follows:

$$\begin{aligned} F^*(f \cdot \omega)|_p(v) &= (f \cdot \omega)|_{F(p)}(dF_p(v)) \\ &= f(F(p)) \cdot \omega|_{F(p)}(dF_p(v)) \\ &= (f \circ F(p)) \cdot F^*\omega(v). \end{aligned}$$

3. We have as follows:

$$\begin{aligned} F^*(df)(v) &= dF_p^*(df_{F(p)})(v) \\ &= df_{F(p)}(dF_p(v)) \\ &= d(f \circ F)_p(v). \end{aligned}$$

□

§3.2 Differential Forms

This section is mostly about (multi-)linear algebras. Starting with the tangent bundle of a smooth manifold, we have performed the dual operation to get its cotangent bundle. Now we continue to perform algebraic operations on cotangent bundles to get a powerful invariant of a smooth manifold, the algebra of differential forms, which are, in some rough sense, sections of some vector bundles.

Let V be a finite dimensional real vector space and let V^* be its dual space. We summarize several results from multi-linear algebra. For background and detailed discussion we refer to chapter 12 in [1].

Let V be a real vector space and $\omega, \eta \in V^*$. Then, $\omega \otimes \eta$ is a bilinear map

$$\begin{aligned} \omega \otimes \eta: V \times V &\rightarrow \mathbb{R} \\ (v_1, v_2) &\mapsto \omega(v_1) \cdot \eta(v_2). \end{aligned}$$

We can make this *multi-linear* by tensoring more covectors, or maps in the dual space.

Notation 3.2.1. For any positive integer p , let $T^p V^*$ be the tensor product of p copies of V^* ; that is,

$$T^p V^* := \underbrace{V^* \otimes V^* \otimes \cdots \otimes V^*}_p.$$

When $p = 0$, we define $T^0 V^* := \mathbb{R}$. Each element of $T^p V^*$ is a multi-linear function on the product of p copies of V .

Definition 3.2.2. An element $\omega \in T^p V^*$ is called a **degree- p alternating element**, or a **p -form**, if

$$\omega(v_1, \dots, v_i, \dots, v_j, \dots, v_p) = -\omega(v_1, \dots, v_j, \dots, v_i, \dots, v_p).$$

That is, when we swap two entries we get a minus sign.

Exercise 3.2.3. Let V be a real vector space of dimension n . Then the set of degree- p alternating elements form a vector subspace of $T^p V^*$, denoted by $A^p V^*$. If $p > n$ or $p < 0$, then we define $A^p V^* := \emptyset$.

Solution: Let $\omega, \nu \in A^p V^*$. We check that $a \cdot \omega + b \cdot \nu \in A^p V^*$. It is clear then that we have

$$(a\omega + b\nu)(v_1, \dots, v_i, \dots, v_j, \dots, v_p) = -(a\omega + b\nu)(v_1, \dots, v_j, \dots, v_i, \dots, v_p).$$

Thus, it forms a vector space. □

Remark 3.2.4 — Recall that any permutation σ of the set $\{1, \dots, p\}$ has a decomposition into the product of transpositions. This implies the following sign formula for alternating elements:

$$\omega(v_1, \dots, v_p) = \text{sgn}(\sigma) \omega(v_{\sigma(1)}, \dots, v_{\sigma(p)}).$$

Here $\text{sgn}(\sigma)$ is the sign of the permutation σ .

Definition 3.2.5. The **wedge product** $\wedge: A^p V^* \times A^q V^* \rightarrow A^{p+q} V^*$, defined as

$$(\omega \wedge \eta)(v_1, \dots, v_{p+q}) := \sum_{\sigma} \text{sgn}(\sigma) \omega(v_{\sigma(1)}, \dots, v_{\sigma(p)}) \cdot \eta(v_{\sigma(p+1)}, \dots, v_{\sigma(p+q)}).$$

Here the sum is over all permutations σ of the set $\{1, \dots, p+q\}$ such that $\sigma(1) < \dots < \sigma(p)$ and $\sigma(p+1) < \dots < \sigma(p+q)$.

Remark 3.2.6 — An equivalent expression of the wedge product above is

$$\omega \wedge \eta(v_1, \dots, v_{p+q}) = \frac{1}{p!q!} \sum_{\sigma} \text{sgn}(\sigma) \omega(v_{\sigma(1)}, \dots, v_{\sigma(p)}) \cdot \eta(v_{\sigma(p+1)}, \dots, v_{\sigma(p+q)}),$$

where the sum is over all permutations σ of the set $\{1, \dots, p+q\}$. Compare this definition with Equation (14.3) in [1]. The $p!$ and $q!$ essentially takes care of the double counting.

Remark 3.2.7 — It is clear that the wedge product is bilinear.

Exercise 3.2.8. For any covectors $\omega^1, \dots, \omega^k$ and vectors $v_1, \dots, v_k$,

$$\omega^1 \wedge \dots \wedge \omega^k(v_1, \dots, v_n) = \det(\omega^j(v_i)).$$

Solution: First, note that covectors are just 1-forms. By the definition of the wedge product,

$$(\omega^1 \wedge \cdots \wedge \omega^k)(v_1, \dots, v_k) = \sum_{\sigma} \text{sgn}(\sigma) \omega^1(v_{\sigma(1)}) \cdots \omega^k(v_{\sigma(k)}).$$

On the other hand, recall that the determinant of the matrix $A = (a_{i,j})_{i,j}$ is given by the Leibniz formula:

$$\det(A) = \sum_{\sigma \in S_k} \text{sgn}(\sigma) a_{1,\sigma(1)} \cdots a_{n,\sigma(n)}.$$

Setting $a_{i,j} = \omega^j(v_i)$, the Leibniz formula tells us that

$$\det(\omega^j(v_i)) = \sum_{\sigma \in S_k} \text{sgn}(\sigma) \omega^1(v_{\sigma(1)}) \cdots \omega^k(v_{\sigma(k)}).$$

The two expressions are clearly equal, so we conclude that

$$\omega^1 \wedge \cdots \wedge \omega^k(v_1, \dots, v_k) = \det(\omega^j(v_i)).$$

□

Example 3.2.9

Let $\omega \in A^1V^*$ and $\eta \in A^2V^*$. Then we have

$$\begin{aligned} \omega \wedge \eta(v_1, v_2, v_3) &= \frac{1}{1!2!} (\omega(v_1)\eta(v_2, v_3) - \omega(v_1)\eta(v_3, v_2) - \omega(v_2)\eta(v_1, v_3) + \omega(v_2)\eta(v_3, v_1) \\ &\quad + \omega(v_3)\eta(v_1, v_2) - \omega(v_3)\eta(v_2, v_1)). \end{aligned}$$

It is equivalent to

$$\omega \wedge \eta(v_1, v_2, v_3) = \omega(v_1)\eta(v_2, v_3) - \omega(v_2)\eta(v_1, v_3) + \omega(v_3)\eta(v_1, v_2).$$

Remark 3.2.10 — Note that if $p = 1$ then $A^1V^* = V^*$. In other words, a one-form is just a covector. In higher dimensions we have an explicit basis for A^pV^* .

We next need to introduce some terminology to prove Proposition 3.2.14; this is not part of the course, so jump to Proposition 3.2.14 if you want to be speedy.

Let V be an n -dimensional vector space, and suppose $(\varepsilon^1, \dots, \varepsilon^n)$ is any basis for V^* . For each multi-index $I = (i_1, \dots, i_k)$ of length k such that $1 \leq i_1, \dots, i_k \leq n$, define a covariant k -tensor $\varepsilon^I = \varepsilon^{i_1 \cdots i_k}$ by

$$\varepsilon^I(v_1, \dots, v_k) = \det \begin{pmatrix} \varepsilon^{i_1}(v_1) & \cdots & \varepsilon^{i_1}(v_k) \\ \vdots & \ddots & \vdots \\ \varepsilon^{i_k}(v_1) & \cdots & \varepsilon^{i_k}(v_k) \end{pmatrix} = \det \begin{pmatrix} v_1^{i_1} & \cdots & v_k^{i_1} \\ \vdots & \ddots & \vdots \\ v_1^{i_k} & \cdots & v_k^{i_k} \end{pmatrix}. \quad (3.1)$$

Remark 3.2.11 — In other words, if v denotes the $n \times k$ matrix whose columns are the components of the vectors $v_1, \dots, v_k$ with respect to the basis (E_i) dual to (ε^i) , then $\varepsilon^I(v_1, \dots, v_k)$ is the determinant of the $k \times k$ submatrix consisting of rows $i_1, \dots, i_k$ of v .

Because the determinant changes sign whenever two columns are interchanged, it is clear that ε^I is an alternating k -tensor. We call ε^I an *elementary alternating tensor* or *elementary k -covector*. Now, to simplify computations with the elementary k -covectors, we introduce the following Kronecker-delta analogue for multi-indices. If I and J are multi-indices of length k , we define

$$\delta_J^I = \det \begin{pmatrix} \delta_{j_1}^{i_1} & \cdots & \delta_{j_k}^{i_1} \\ \vdots & \ddots & \vdots \\ \delta_{j_1}^{i_k} & \cdots & \delta_{j_k}^{i_k} \end{pmatrix}.$$

Exercise 3.2.12. Show that

$$\delta_J^I = \begin{cases} \text{sgn } \sigma & \text{if neither } I \text{ nor } J \text{ has a repeated index and } J = I_\sigma \text{ for some } \sigma \in S_k, \\ 0 & \text{if } I \text{ or } J \text{ has a repeated index or } J \text{ is not a permutation of } I. \end{cases}$$

Here, $I_\sigma = (i_{\sigma(1)}, \dots, i_{\sigma(k)})$.

Solution: First suppose that I or J has a repeated index. If $i_a = i_b$, then the rows a and b are equal, implying that the determinant is zero; same goes for J . Now suppose that there are no repeated indices in I and J . Suppose that J is not a permutation of I . Then, $\{i_1, \dots, i_k\} \neq \{j_1, \dots, j_k\}$ as sets. It follows then that there exists some row which is entirely zero. This is because there exists some i_ℓ which is not contained in $\{j_1, \dots, j_k\}$, so the Kronecker delta at that row is just full of 0s. Thus, the determinant is zero in this case. Now suppose that I and J are equal up to some permutation; that is, $J = I_\sigma$. Then, $j_b = i_{\sigma(b)}$, but then because

$$\delta_{j_b}^{i_a} = \delta_{i_{\sigma(b)}}^{i_a}$$

and all the i_a are distinct, this equals 1 if and only if $i_{\sigma(b)} = i_a$; that is, $a = \sigma(b)$. Therefore, the matrix for δ_J^I is exactly the permutation matrix of σ , so the determinant is $\text{sgn } \sigma$. $\square$

Lemma 3.2.13 (Properties of Elementary k -Covectors)

Let (E_i) be a basis for V , let (ε^i) be the dual basis for V^* , and let ε^I be as defined above.

- (i) If I has a repeated index, then $\varepsilon^I = 0$.
- (ii) If $J = I_\sigma$ for some $\sigma \in S_k$, then $\varepsilon^I = (\text{sgn } \sigma)\varepsilon^J$.
- (iii) The result of evaluating ε^I on a sequence of basis vectors is

$$\varepsilon^I(E_{j_1}, \dots, E_{j_k}) = \delta_J^I.$$

Proof. If I has a repeated index, then for any vectors $v_1, \dots, v_k$, the determinant in (3.1) has two identical rows and thus is equal to zero, which proves (a). On the other hand, if J is obtained from I by interchanging two indices, then the corresponding determinants have opposite signs; this implies (ii). Finally, (iii) follows immediately from the definition of ε^I . $\square$

Proposition 3.2.14

If $\omega_1, \dots, \omega_n$ is a basis of V^* then

$$\{\omega_{i_1} \wedge \dots \wedge \omega_{i_p} \mid i_1 < \dots < i_p\}$$

is a basis of $A^p V^*$ over $\mathbb{R}$. Hence $\dim(A^p V^*) = \frac{n!}{p!(n-p)!}$.

Proof. Let $\mathcal{E}$ denote the set in the proposition. We need to show that $\mathcal{E}$ spans $A^k(V^*)$ and is linearly independent. Let (E_I) be the basis for V dual to $\mathcal{E}$.

To show that $\mathcal{E}$ spans $A^k(V^*)$, let $\alpha \in A^k(V^*)$ be arbitrary. For each multi-index $I = (i_1, \dots, i_k)$ (not necessarily increasing), define a real number α_I by

$$\alpha_I = \alpha(E_{i_1}, \dots, E_{i_k}).$$

The fact that α is alternating implies that $\alpha_I = 0$ if I contains a repeated index, and $\alpha_J = (\text{sgn } \sigma)\alpha_I$ if $J = I_\sigma$ for $\sigma \in S_k$. For any multi-index J , therefore, Lemma 3.2.13 gives

$$\sum_I' \alpha_I \varepsilon^I(E_{j_1}, \dots, E_{j_k}) = \sum_I' \alpha_I \delta_J^I = \alpha_J = \alpha(E_{j_1}, \dots, E_{j_k}).$$

Thus $\sum_I' \alpha_I \varepsilon^I = \alpha$, so $\mathcal{E}$ spans $A^k(V^*)$.

To show that $\mathcal{E}$ is linearly independent, suppose the identity

$$\sum_I' \alpha_I \varepsilon^I = 0$$

holds for some coefficients α_I . Let J be any increasing multi-index. Applying both sides of the identity to the vectors $(E_{j_1}, \dots, E_{j_k})$ and using Lemma 3.2.13, we get

$$0 = \sum_I' \alpha_I \varepsilon^I(E_{j_1}, \dots, E_{j_k}) = \alpha_J.$$

Thus each coefficient α_J is zero. □

Corollary 3.2.15

For $\omega \in A^p V^*, \eta \in A^q V^*$, we have $\omega \wedge \eta = (-1)^{pq} \eta \wedge \omega$. Particularly, $\omega \wedge \omega = 0$ if $\omega \in A^1 V^*$.

Proof. We have

$$(\omega \wedge \eta)(v_1, \dots, v_{p+q}) = \frac{1}{p!q!} \sum_{\sigma} \text{sgn}(\sigma) \omega(v_{\sigma(1)}, \dots, v_{\sigma(p)}) \eta(v_{\sigma(p+1)}, \dots, v_{\sigma(p+q)})$$

and

$$(\eta \wedge \omega)(v_1, \dots, v_{p+q}) = \frac{1}{p!q!} \sum_{\rho} \text{sgn}(\rho) \eta(v_{\rho(1)}, \dots, v_{\rho(q)}) \omega(v_{\rho(q+1)}, \dots, v_{\rho(q+p)}).$$

Now, let $\tau \in S_{p+q}$ be a permutation which swaps the first p vectors with the last q vectors. Specifically,

$$\begin{aligned}\tau(1) &= p+1, & \dots, & \tau(q) = p+q \\ \tau(q+1) &= 1, & \dots, & \tau(q+p) = p.\end{aligned}$$

Then, τ is a composition of pq -transpositions, since for every number from $q+1$ to $q+p$, we need to shift it to the left q times. Therefore, $\text{sgn}(\tau) = (-1)^{pq}$. Furthermore, the composition with τ map is a bijection from S_{p+q} to S_{p+q} , since $\tau \in S_{p+q}$. Thus, we have as follows:

$$\begin{aligned}(\eta \wedge \omega)(v_1, \dots, v_{p+q}) &= \frac{1}{p!q!} \sum_{\rho} \text{sgn}(\rho\tau) \eta(v_{\rho\tau(1)}, \dots, v_{\rho\tau(q)}) \omega(v_{\rho\tau(q+1)}, \dots, v_{\rho\tau(q+p)}) \\ &= \frac{(-1)^{pq}}{p!q!} \sum_{\rho} \text{sgn}(\rho) \eta(v_{\rho(p+1)}, \dots, v_{\rho(p+q)}) \omega(v_{\rho(1)}, \dots, v_{\rho(p)}) \\ &= (-1)^{pq} (\omega \wedge \eta)(v_1, \dots, v_{p+q}).\end{aligned}$$

□

§3.2.i Wedge Product of Cotangent Bundles

Now we apply the above multi-linear algebra constructions to the cotangent bundle.

Theorem 3.2.16

Let X be a smooth manifold. For any nonnegative integer p , the set

$$A^p(T^*X) := \bigsqcup_{x \in X} A^p T_x^* X$$

admits a smooth manifold structure, and it is a smooth vector bundle over X .

Proof. The proof is almost identical to the proof that cotangent bundle is a smooth vector bundle. We take smooth charts on X and use the local basis given above. Then apply the vector bundle chart Lemma 3.1.7. □

Remark 3.2.17 — Note that when $p = 0$, we have $A^0(T^*X) = X \times \mathbb{R}$ as the product bundle.

Exercise 3.2.18. Let $X = \mathbb{R}^3$, and let $\omega = x^2 dy \wedge dz$. We see how this acts on $\left(\frac{\partial}{\partial x} + \frac{\partial}{\partial y}, \frac{\partial}{\partial z}\right)$.

Solution: Let $(v_1, v_2) = \left(\frac{\partial}{\partial x} + \frac{\partial}{\partial y}, \frac{\partial}{\partial z}\right)$. We just use the rule for wedge products.

$$\begin{aligned}\omega(v_1, v_2) &= x^2 dy(v_1) dz(v_2) - x^2 dy(v_2) dz(v_1) \\ &= x^2 \cdot 1 \cdot 1 - x^2 \cdot 0 \cdot 0 \\ &= x^2.\end{aligned}$$

□

Definition 3.2.19. Let X be a smooth manifold. A **differential p -form** is a smooth section of the vector bundle $A^p(T^*X)$. The space of differential p -forms is written as $\Omega^p(X)$.

Example 3.2.20

When $p = 0$, $\Omega^0(X)$ is the space of smooth functions on X , $\Omega^0(X) \cong C^\infty(X)$.

Example 3.2.21

When $p = 1$, $\Omega^1(X)$ is the set of sections of T^*X . This is precisely the definition of covector fields on X . That is, $\Omega^1(X) = \mathfrak{X}^*(X)$.

Next we discuss differential forms on $\mathbb{R}^n$ explicitly.

Example 3.2.22

Let x, y, z be the coordinate functions on $X = \mathbb{R}^3$. Note that dx, dy, dz is a basis for $T_x^*\mathbb{R}^3$ for any $q \in \mathbb{R}^3$. From Proposition 3.2.14 we know

1. A zero-form is a smooth function.
2. A one-form is $\omega = f_1(q)dx + f_2(q)dy + f_3(q)dz$ for three smooth functions f_1, f_2, f_3 .
3. A two-form is $\alpha = g_1dx \wedge dy + g_2dx \wedge dz + g_3dy \wedge dz$ for three smooth functions g_1, g_2, g_3 .
4. A three-form is $\eta = hdx \wedge dy \wedge dz$ for a smooth function h .

Generally, let $x_1, \dots, x_n$ be the coordinate functions on $\mathbb{R}^n$. Then a differential p -form on $\mathbb{R}^n$ can be written as

$$\omega = \sum_{i_1 < \dots < i_p} f_{i_1 \dots i_p} dx_{i_1} \wedge \dots \wedge dx_{i_p}$$

for smooth functions $f_{i_1 \dots i_p} \in C^\infty(\mathbb{R}^n)$.

Recall that if $F: X \rightarrow Y$ is a smooth map then we have defined a pullback map

$$F^*: \Omega^1(Y) \rightarrow \Omega^1(X).$$

Analogously, we can define the pullback of differential p -forms.

Definition 3.2.23. Let $F: X \rightarrow Y$ be a smooth map. We define the **pullback of a differential p -form** as follows. Let $p \geq 1$, for any $x \in X$ and $v_1, \dots, v_p \in T_x X$, define

$$(F^*\omega)_x(v_1, \dots, v_p) := \omega_{F(x)}(dF_x(v_1), \dots, dF_x(v_p)).$$

Remark 3.2.24 — If $p = 0$, define $F^*f := f \circ F$. The following properties of the pullback can be checked from the definitions.

Lemma 3.2.25

Let $F: X \rightarrow Y$ be smooth.

- (i) $F^*: \Omega^p(Y) \rightarrow \Omega^p(X)$ is a well-defined $\mathbb{R}$ -linear map for any p .
- (ii) Let $\omega \in \Omega^p(Y), \eta \in \Omega^q(Y)$. We have $F^*(\omega \wedge \eta) = (F^*\omega) \wedge (F^*\eta)$.
- (iii) Let $G: Y \rightarrow Z$ be smooth. Then, $F^* \circ G^* = (G \circ F)^*$. If F is a diffeomorphism, show $(F^*)^{-1} = (F^{-1})^*$.

Proof. (i). Let $\omega, \eta \in \Omega^p(Y)$ and $a, b \in \mathbb{R}$. Then,

$$\begin{aligned} F^*(a\omega + b\eta)_x(v_1, \dots, v_p) &= (a\omega + b\eta)_{F(x)}(dF_x(v_1), \dots, dF_x(v_p)) \\ &= (aF^*\omega + bF^*\eta)(v_1, \dots, v_p). \end{aligned}$$

Thus, it is $\mathbb{R}$ -linear. To show that this is well-defined, we just need to show that the image is indeed a differential p -form; that is, that $F^*\omega$ is a smooth section of the vector bundle $A^p(T^*X)$. Note that $F^*\omega: X \rightarrow A^p(T^*X)$ is defined as $x \mapsto (x, (F^*\omega)_x)$. Therefore,

$$(\pi \circ F^*\omega)(x) = \pi((x, (F^*\omega)_x)) = x,$$

so $F^*\omega$ is a section of $A^p(T^*X)$. To show that it is smooth, we just need to see that $x \mapsto F(x)$ and $x \mapsto dF_x$ are smooth, and ω is a smooth section (since it's in $\Omega^p(Y)$). Therefore, everything depends smoothly on x .

(ii). Explicitly, if we write

$$(\omega \wedge \eta)(v_1, \dots, v_{p+q}) := \sum_{\sigma} \text{sgn}(\sigma) \omega(v_{\sigma(1)}, \dots, v_{\sigma(p)}) \cdot \eta(v_{\sigma(p+1)}, \dots, v_{\sigma(p+q)})$$

and apply $F^*(-)_x$, we get

$$\sum_{\sigma} \text{sgn}(\sigma) \omega_{F(x)}(dF_x(v_{\sigma(1)}), \dots, dF_x(v_{\sigma(p)})) \cdot \eta_{F(x)}(dF_x(v_{\sigma(p+1)}), \dots, dF_x(v_{\sigma(p+q)})).$$

However, this is just $(F^*\omega)_x \wedge (F^*\eta)_x$.

(iii). Let $x \in X$. Then,

$$\begin{aligned} ((G \circ F)^*\omega)_x(v_1, \dots, v_p) &= \omega_{(G \circ F)(x)}(d(G \circ F)_x(v_1), \dots, d(G \circ F)_x(v_p)) \\ &= \omega_{(G \circ F)(x)}(dG_{F(x)} \circ dF_x(v_1), \dots, dG_{F(x)} \circ dF_x(v_p)) \\ &= (G^*\omega)_{F(x)}(dF_x(v_1), \dots, dF_x(v_p)) \\ &= ((F^* \circ G^*)\omega)_x(v_1, \dots, v_p). \end{aligned}$$

Thus, we are done. For the second part, just set $G = F^{-1}$, which exists because F is a diffeomorphism. Then, we see that $(F^{-1})^* = (F^*)^{-1}$. $\square$

Exercise 3.2.26. Given a smooth map

$$F: \mathbb{R}^n \rightarrow \mathbb{R}^m, \quad F(x_1, \dots, x_n) = (f_1(x_1, \dots, x_n), \dots, f_m(x_1, \dots, x_n)).$$

Check the explicit formula of the pullback map in coordinates:

$$F^*(g \, dy_1 \wedge \cdots \wedge dy_m) = (g \circ F)(df_1) \wedge \cdots \wedge (df_m).$$

Solution: Write $\omega := dy_1 \wedge \cdots \wedge dy_m$. We first claim that $F^*(g\omega) = (g \circ F)F^*\omega$. First, note that

$$g\omega_y(v_1, \dots, v_n) = g(y)\omega_y(v_1, \dots, v_n).$$

Then, we apply $F^*(-)_x$:

$$\begin{aligned} F^*(g\omega)_x(v_1, \dots, v_n) &= (g\omega)_{F(x)}(dF_x(v_1), \dots, dF_x(v_n)) \\ &= g(F(x)) \cdot \omega_{F(x)}(dF_x(v_1), \dots, dF_x(v_n)) \\ &= (g \circ F)_x(F^*\omega)_x \end{aligned}$$

Now, we want to show that $F^*(dy_1 \wedge \cdots \wedge dy_m) = (df_1) \wedge \cdots \wedge (df_m)$. In fact, by the previous lemma, the wedge product and F^* commute, so it suffices to show that $F^*(dy_i) = df_i$. Let $x \in \mathbb{R}^n$, $v \in T_x\mathbb{R}^n$.

$$\begin{aligned} (F^*dy_i)_x(v) &= (dy_i)_{F(x)}(dF_x(v)) \\ &= d(y_i \circ F)_x(v) \\ &= (df_i)_x(v) \end{aligned}$$

Here, we used the chain rule in the second equality and also that y_i is just the i -th coordinate of the map F . $\square$

Exercise 3.2.27. Let $F: M \rightarrow N$ be smooth. Then, in any smooth chart,

$$F^*\left(\sum_I \omega_I \, dy_{i_1} \wedge \cdots \wedge dy_{i_k}\right) = \sum_I (\omega_I \circ F) d(y_{i_1} \circ F) \wedge \cdots \wedge d(y_{i_k} \circ F),$$

where ω_I are continuous functions defined on the coordinate domain.

Solution: Let (U, φ) and (V, ψ) be some chart of M, N , respectively such that the local coordinate of U are given by x_i s and that of V are given by y_j s. Then, define $\tilde{F} := \psi \circ F \circ \varphi^{-1}: \mathbb{R}^m \rightarrow \mathbb{R}^n$. Basically running through the same steps as the above exercise, we get

$$\tilde{F}^*\left(\sum_I \omega_I dy_{i_1} \wedge \cdots \wedge dy_{i_k}\right) = \sum_I (\omega_I \circ \tilde{F}) (d(y_{i_1} \circ \tilde{F})) \wedge \cdots \wedge (d(y_{i_k} \circ \tilde{F}))$$

However, if we then apply φ^* , we then attain the desired form:

$$F^*\left(\sum_I \omega_I dy_{i_1} \wedge \cdots \wedge dy_{i_k}\right) = \sum_I (\omega_I \circ F) d(y_{i_1} \circ F) \wedge \cdots \wedge d(y_{i_k} \circ F).$$

Here we note there is slight abuse of notation with the $d(y_i \circ F)$, as in we really mean $d(y_i \circ \psi \circ F)$. $\square$

Example 3.2.28

Let $\iota: \mathbb{S}^1 \hookrightarrow \mathbb{R}^2$ be the canonical inclusion map, and $\alpha = xdy - ydx$. Let $U = \{(x, y) \in \mathbb{S}^1 : y > 0\}$, with the chart map $\pi: (x, y) \mapsto x$. Compute $\iota^*\alpha$ with respect to this chart.

Solution: In this chart, we note that $t \mapsto (t, \sqrt{1-t^2})$. Since we calculate $\iota^*\alpha$ using this coordinate chart, we calculate $(\varphi^{-1})^*\iota^*\alpha$. This is just $(\iota \circ \varphi^{-1})^*\alpha$. Using the exercise above, we have as follows.

$$\begin{aligned} (\iota \circ \varphi^{-1})^*\alpha &= (t)d(\sqrt{1-t^2}) - (\sqrt{1-t^2})dt \\ &= \frac{-t^2}{\sqrt{1-t^2}}dt - \sqrt{1-t^2}dt \\ &= \frac{-1}{\sqrt{1-t^2}}dt. \end{aligned}$$

□

Example 3.2.29

Suppose we have a smooth function $F: \mathbb{R}^n \rightarrow \mathbb{R}^n$, where the coordinates in the domain are $(x_1, \dots, x_n)$ and the target being $(y_1, \dots, y_n)$. For simplicity, we calculate here the case when $n = 3$.

Solution: Then,

$$F(x_1, x_2, x_3) = (f_1(x_1, x_2, x_3), f_2(x_1, x_2, x_3), f_3(x_1, x_2, x_3)).$$

We calculate the 3-form $\omega = dy_1 \wedge dy_2 \wedge dy_3$. Then we calculate the pullback of omega, $F^*\omega$.

$$\begin{aligned} F^*\omega &= F^*(dy_1 \wedge dy_2 \wedge dy_3) \\ &= (F^*dy_1) \wedge (F^*dy_2) \wedge (F^*dy_3). \end{aligned}$$

Then,

$$\begin{aligned} (F^*dy_1) \left(a \frac{\partial}{\partial x_1} + b \frac{\partial}{\partial x_2} + c \frac{\partial}{\partial x_3} \right) &= (a \frac{\partial}{\partial x_1} + b \frac{\partial}{\partial x_2} + c \frac{\partial}{\partial x_3})(y_1 \circ F) \\ &= a \frac{\partial f_1}{\partial x_1} + b \frac{\partial f_1}{\partial x_2} + c \frac{\partial f_1}{\partial x_3}. \end{aligned}$$

It follows that

$$F^*dy_1 = \frac{\partial f_1}{\partial x_1}dx_1 + \frac{\partial f_1}{\partial x_2}dx_2 + \frac{\partial f_1}{\partial x_3}dx_3.$$

Then, we can do the same thing for dy_2 and dy_3 .

$$\begin{aligned} F^*dy_2 &= \frac{\partial f_2}{\partial x_1}dx_1 + \frac{\partial f_2}{\partial x_2}dx_2 + \frac{\partial f_2}{\partial x_3}dx_3 \\ F^*dy_3 &= \frac{\partial f_3}{\partial x_1}dx_1 + \frac{\partial f_3}{\partial x_2}dx_2 + \frac{\partial f_3}{\partial x_3}dx_3 \end{aligned}$$

Now we need to wedge them together.

$$\begin{aligned} (F^*dy_1) \wedge (F^*dy_2) &= f_{11}f_{22}dx_1 \wedge dx_2 + f_{11}f_{23}dx_1 \wedge dx_3 \\ &\quad + f_{12}f_{21}dx_2 \wedge dx_1 + f_{12}f_{23}dx_2 \wedge dx_3 \\ &\quad + f_{13}f_{21}dx_3 \wedge dx_1 + f_{13}f_{22}dx_3 \wedge dx_2. \end{aligned}$$

Here, $f_{ij} = \frac{\partial f_i}{\partial x_j}$. We also have graded commutativity; that is, $\omega \wedge \eta = (-1)^{pq}\eta \wedge \omega$, where ω is a p -form and η is a q -form. Thus, we can further simplify this to get

$$(f_{11}f_{21} - f_{12}f_{21})dx_1 \wedge dx_2 + (f_{11}f_{23} - f_{13}f_{21})dx_1 \wedge dx_3 + (f_{12}f_{23} - f_{13}f_{22})dx_2 \wedge dx_3.$$

To finish off, we need to wedge with F^*dy_3 . Note that the last two terms vanish, so we only care about the first term wedged. We get

$$\begin{aligned} F^*\omega &= (f_{11}f_{22}f_{33} - f_{12}f_{21}f_{33})dx_1 \wedge dx_2 \wedge dx_3 \\ &\quad + (f_{11}f_{23}f_{32} - f_{13}f_{21}f_{32})dx_1 \wedge dx_3 \wedge dx_2 \\ &\quad + (f_{12}f_{23}f_{31} - f_{13}f_{22}f_{31})dx_2 \wedge dx_3 \wedge dx_1. \end{aligned}$$

To further simplify, we use graded commutativity again. The final answer is

$$\begin{aligned} F^*\omega &= (f_{11}f_{23}f_{32} - f_{13}f_{21}f_{32} - f_{11}f_{23}f_{32} + f_{13}f_{21}f_{32} \\ &\quad + f_{12}f_{23}f_{31} - f_{13}f_{22}f_{31})dx_1 \wedge dx_2 \wedge dx_3. \end{aligned}$$

However, this is just the determinant of the derivative matrix; that is, the Jacobian of F .

$$dF = \begin{pmatrix} f_{11} & f_{12} & f_{13} \\ f_{21} & f_{22} & f_{23} \\ f_{31} & f_{32} & f_{33} \end{pmatrix}$$

Therefore,

$$F^*\omega = \det(dF)dx_1 \wedge dx_2 \wedge dx_3.$$

□

The case for general $\mathbb{R}^n$ is as follows.

Proposition 3.2.30

Let $F: M \rightarrow N$ be a smooth map between n -manifolds. If (x^i) and (y^j) are smooth coordinates on open subsets $U \subseteq M$ and $V \subseteq N$, respectively, and u is a continuous real-valued function on V , then the following holds on $U \cap F^{-1}(V)$:

$$F^*(udy_1 \wedge \cdots \wedge dy_n) = (u \circ F)(\det DF)dx_1 \wedge \cdots \wedge dx_n.$$

where DF represents the Jacobian matrix of F in these coordinates.

Proof. Because the fiber of $A^n T^*M$ is spanned by $dx_1 \wedge \cdots \wedge dx_n$ at each point, it suffices to show that both sides of the above give the same result when evaluated on

$$\left(\frac{\partial}{\partial x_1}, \dots, \frac{\partial}{\partial x_n} \right).$$

The exercise above already told us that

$$F^*(udy_1 \wedge \cdots \wedge dy_n) = (u \circ F)dF_1 \wedge \cdots \wedge dF_n.$$

Further, Exercise 3.2.8 below shows that

$$dF_1 \wedge \cdots \wedge dF_n \left(\frac{\partial}{\partial x_1}, \dots, \frac{\partial}{\partial x_n} \right) = \det \left(dF_j \left(\frac{\partial}{\partial x_i} \right) \right) = \det \left(\frac{\partial F_j}{\partial x_i} \right).$$

Therefore, the LHS of the above gives $(u \circ F) \det DF$ when applied to

$$\left(\frac{\partial}{\partial x_1}, \dots, \frac{\partial}{\partial x_n} \right).$$

On the other hand, the right-hand side gives the same thing, because

$$dx_1 \wedge \cdots \wedge dx_n \left(\frac{\partial}{\partial x_1}, \dots, \frac{\partial}{\partial x_n} \right) = 1.$$

□

§3.2.ii Exterior Derivative

Recall that we identify $\Omega^0(M)$ with $C^\infty(M)$ and $\Omega^1(M)$ with covector fields. Note that if $f \in C^\infty(M)$, we defined $df \in \Omega^1(M)$ by $df(v) = v(f)$, where $d: \Omega^0(M) \rightarrow \Omega^1(M)$.

Now we define another important operator on differential forms, the *exterior derivative*. We will first define it on $\mathbb{R}^n$. Recall we already defined the operator for functions

$$d: \Omega^0(\mathbb{R}^n) \rightarrow \Omega^1(\mathbb{R}^n), \quad df := \sum_i \frac{\partial f}{\partial x_i} dx_i.$$

For a general p -form

$$\omega = \sum_{i_1 < \dots < i_p} f_{i_1 \dots i_p} dx_{i_1} \wedge \dots \wedge dx_{i_p},$$

we define $d: \Omega^p(\mathbb{R}^n) \rightarrow \Omega^{p+1}(\mathbb{R}^n)$ as

$$d\omega := \sum_{i_1 < \dots < i_p} (df_{i_1 \dots i_p}) \wedge dx_{i_1} \wedge \dots \wedge dx_{i_p}.$$

By definition, this operator is $\mathbb{R}$ -linear, which means that

$$d(a\omega + b\eta) = a \cdot d\omega + b \cdot d\eta, \quad a, b \in \mathbb{R}.$$

Several remarkable properties of d are the following.

Proposition 3.2.31

$$d \circ d = 0.$$

Proof. By linearity, it is sufficient to check $d \circ d(f dx_{i_1} \wedge \dots \wedge dx_{i_p}) = 0$. We do it by computation.

$$\begin{aligned} d \circ d(f dx_{i_1} \wedge \dots \wedge dx_{i_p}) &= d \left(\left(\sum_i \frac{\partial f}{\partial x_i} dx_i \right) \wedge dx_{i_1} \wedge \dots \wedge dx_{i_p} \right) \\ &= \sum_i \sum_j \frac{\partial^2 f}{\partial x_j \partial x_i} dx_j \wedge dx_i \wedge dx_{i_1} \wedge \dots \wedge dx_{i_p}. \end{aligned}$$

For any pair $i \neq j$, we have

$$\frac{\partial^2 f}{\partial x_j \partial x_i} dx_j \wedge dx_i = -\frac{\partial^2 f}{\partial x_i \partial x_j} dx_i \wedge dx_j$$

by the smoothness of f and Corollary 3.2.15. For any i , we also have

$$\frac{\partial^2 f}{\partial x_i^2} dx_i \wedge dx_i = 0$$

by Corollary 3.2.15. Therefore the above sum is zero. $\square$

Proposition 3.2.32 (Graded Leibniz Rule)

Let $\omega \in \Omega^p(\mathbb{R}^n)$, $\eta \in \Omega^r(\mathbb{R}^n)$, then

$$d(\omega \wedge \eta) = d\omega \wedge \eta + (-1)^p \omega \wedge d\eta.$$

Proof. By linearity, we only need to check $\omega = f dx_{i_1} \wedge \cdots \wedge dx_{i_p}$ and $\eta = g dx_{i_{p+1}} \wedge \cdots \wedge dx_{i_{p+r}}$ which follows from a direct computation:

$$\begin{aligned} d(\omega \wedge \eta) &= d(fg \wedge dx_{i_1} \wedge \cdots \wedge dx_{i_p} \wedge dx_{i_{p+1}} \wedge \cdots \wedge dx_{i_q}) \\ &= ((df)g + f(dg)) \wedge (dx_{i_1} \wedge \cdots \wedge dx_{i_p} \wedge dx_{i_{p+1}} \wedge \cdots \wedge dx_{i_q}) \\ &= ((df) \wedge dx_{i_1} \wedge \cdots \wedge dx_{i_p}) \wedge (g dx_{i_{p+1}} \wedge \cdots \wedge dx_{i_q}) \\ &\quad + (-1)^p (f dx_{i_1} \wedge \cdots \wedge dx_{i_p}) \wedge ((dg) \wedge dx_{i_{p+1}} \wedge \cdots \wedge dx_{i_q}) \\ &= d\omega \wedge \eta + (-1)^p \omega \wedge d\eta. \end{aligned}$$

□

Proposition 3.2.33

Let $F: \mathbb{R}^n \rightarrow \mathbb{R}^m$ be a smooth map. We have

$$F^*(d\omega) = d(F^*\omega)$$

for any differential form ω .

Proof. It suffices to verify the formula in coordinates, by using Exercise 3.2.26 and Lemma 3.1.42. On one hand we have

$$\begin{aligned} F^*(d(f dx_{i_1} \wedge \cdots \wedge dx_{i_p})) &= F^* df \wedge (dx_{i_1} \wedge \cdots \wedge dx_{i_p}) \\ &= d(f \circ F) \wedge d(x_{i_1} \circ F) \wedge \cdots \wedge d(x_{i_p} \circ F). \end{aligned}$$

On the other hand, we have

$$\begin{aligned} d(F^*(f dx_{i_1} \wedge \cdots \wedge dx_{i_p})) &= d((f \circ F) \cdot d(x_{i_1} \circ F) \wedge \cdots \wedge d(x_{i_p} \circ F)) \\ &= d(f \circ F) \wedge d(x_{i_1} \circ F) \wedge \cdots \wedge d(x_{i_p} \circ F). \end{aligned}$$

Here, we've used the graded Leibniz rule and that $d^2 = 0$. □

Exterior Derivative on Manifolds

Now we can define the operator d on general smooth manifolds.

Definition 3.2.34. Let X be a smooth manifold and ω be a differential form on X . For any $x \in X$, pick a smooth chart (U, ϕ) containing x , we define

$$(d\omega)_x := (\phi^*(d(\phi^{-1})^*\omega))_x.$$

That is, we first pullback ω from X to the Euclidean space, then take d as defined above, then pullback to X .

Remark 3.2.35 — From here, when we write Ω^* , we mean that this works for any p .

Proposition 3.2.36

The operator $d: \Omega^p(X) \rightarrow \Omega^{p+1}(X)$ is well-defined.

Proof. Since our definition involves a choice of chart, we need to show the independence of such a choice. Pick another chart (V, ψ) containing x , we will check

$$(\phi^*(d(\phi^{-1})^*\omega))_x = (\psi^*(d(\psi^{-1})^*\omega))_x.$$

Note that

$$\begin{aligned} (\phi^{-1})^*\psi^*d(\psi^{-1})^*\omega &= (\psi \circ \phi^{-1})^*d(\psi^{-1})^*\omega \\ &= d((\psi \circ \phi^{-1})^* \circ (\psi^{-1})^*\omega) \\ &= d((\phi^{-1})^* \circ \psi^* \circ (\psi^{-1})^*\omega) \\ &= d(\phi^{-1})^*\omega. \end{aligned}$$

The first and the third equality are the composition law for the pullback. The second equality is the commutativity law, Proposition 3.2.33. Now apply ϕ^* on both sides of the equation we get $(\psi^*(d(\psi^{-1})^*\omega))_x = (\phi^*(d(\phi^{-1})^*\omega))_x$. $\square$

Corollary 3.2.37

The operator $d: \Omega^p(X) \rightarrow \Omega^{p+1}(X)$ satisfies:

- (i) $d \circ d = 0$.
- (ii) For any smooth map $F: X \rightarrow Y$, $F^*d = dF^*$.
- (iii) $d(\omega \wedge \eta) = d\omega \wedge \eta + (-1)^p\omega \wedge d\eta$ for all $\omega \in \Omega^p(X)$, $\eta \in \Omega^q(X)$.

Proof. (i) and (iii) follow from computation in coordinates. We prove (ii) here. Pick a chart (U, ϕ) on X and a chart (V, ψ) on Y . Then $\psi \circ F \circ \phi^{-1}$ is a smooth map between Euclidean spaces. Pick $\omega \in \Omega^p(Y)$. Using Proposition 3.2.33 we get

$$(\psi \circ F \circ \phi^{-1})^*(d(\psi^{-1})^*\omega) = d(\psi \circ F \circ \phi^{-1})^*(\psi^{-1})^*\omega.$$

The composition law for the pullback shows that the right hand side can be simplified to $d(\phi^{-1})^*F^*\omega$. Therefore we get

$$\begin{aligned} d(F^*\omega) &= \phi^*d(\phi^{-1})^*F^*\omega \\ &= \phi^*(\psi \circ F \circ \phi^{-1})^*(d(\psi^{-1})^*\omega) \\ &= F^*\psi^*d(\psi^{-1})^*\omega \\ &= F^*(d\omega). \end{aligned}$$

The first and last equality are by definition. The second equality is given above. The third equality is the composition law for pull back. $\square$

Exercise 3.2.27 and the previous proposition now provide useful ways to calculate pullbacks of differential forms. Here is an example of this in action.

Example 3.2.38

Define $F: \mathbb{R}^2 \rightarrow \mathbb{R}^3$ by

$$F(u, v) = (u, v, u^2 - v^2),$$

and let ω be the 2-form $ydx \wedge dz + xdy \wedge dz$ on $\mathbb{R}^3$. The pullback $F^*\omega$ is computed as follows:

$$\begin{aligned} F^*(ydx \wedge dz + xdy \wedge dz) &= vdu \wedge d(u^2 - v^2) + udv \wedge d(u^2 - v^2) \\ &= vdu \wedge (2udu - 2vdv) + udv \wedge (2udu - 2vdv) \\ &= -2v^2 du \wedge dv + 2u^2 dv \wedge du. \end{aligned}$$

Here, we have used the fact that $du \wedge du = dv \wedge dv = 0$ by anti-commutativity. Because $dv \wedge du = -du \wedge dv$, this simplifies to

$$F^*\omega = -2(u^2 + v^2)du \wedge dv.$$

§3.2.iii de Rham Cohomology

Let X be a smooth manifold. Set

$$\Omega(X) := \bigoplus_{p \geq 0} \Omega^p(X).$$

Definition 3.2.39. $\Omega(X)$, as defined above, is an algebra over $\mathbb{R}$, with respect to the wedge product; $\wedge: \Omega^p(X) \otimes \Omega^q(X) \rightarrow \Omega^{p+q}(X)$. We call it the **de Rham algebra** of X . The operator $d: \Omega^p(X) \rightarrow \Omega^{p+1}(X)$ is called the **de Rham differential** on $\Omega(X)$.

Remark 3.2.40 — The identity/ unit of this algebra is the constant function identical to 1. Further, this is associative, by the associativity of the wedge product. However, it is not commutative.

Definition 3.2.41. A differential p -form is called **closed** if $d\omega = 0$, and it is called **exact** if there exists a differential $(p-1)$ -form η such that $d\eta = \omega$.

Remark 3.2.42 — Note that exact forms are also closed, since $d^2 = 0$.

In other words, we have a sequence of operators

$$\dots \rightarrow \Omega^{p-1}(X) \xrightarrow{d} \Omega^p(X) \xrightarrow{d} \Omega^{p+1}(X) \xrightarrow{d} \dots$$

By the $\mathbb{R}$ -linearity of d , the set of closed p -forms is a subspace of $\Omega^p(X)$, which we write as $Z^p(X)$. Similarly, the subspace of exact p -forms is written as $B^p(X)$. Since $d \circ d = 0$, we always have $B^p(X) \subset Z^p(X)$.

Definition 3.2.43. Let X be a smooth manifold. The **degree- p de Rham cohomology group** of X is defined as

$$H_{dR}^p(X) := Z^p(X)/B^p(X).$$

Since $Z^p(X), B^p(X)$ are real vector spaces, $H_{dR}^p(X)$ is also a real vector space. It is a very important invariant of manifolds. We now think upon some very simple computations. By definition, $H_{dR}^p(X) = 0$ if $p < 0$ or $p > \dim X$.

Example 3.2.44

Let X be a connected smooth manifold. Then, $H_{dR}^0(X) \cong \mathbb{R}$. By definition,

$$H_{dR}^0(X) = Z^0(X)/B^0(X) = \ker d / \text{Im} d.$$

Then,

$$0 = \Omega^{-1}(X) \xrightarrow{d} \Omega^0(X) \xrightarrow{d} \Omega^1(X) \rightarrow \dots$$

Then, $\text{Im} d = 0$, so $H_{dR}^0(X) \cong \ker d$. To find the kernel, note that $\Omega^0(X) \cong C^\infty(X)$, so if $f \in \Omega^0(X)$, then $df = 0$ implies f is constant. Thus, $\ker d$ is the space of constant functions. Then, by connectedness of X , we get that $\ker d \cong \mathbb{R}$.

Lemma 3.2.45

Let X be a smooth manifold. Then $H_{dR}^0(X)$ is isomorphic to $\mathbb{R}^k$, where k is the number of connected components of X .

Proof. Since $d: \Omega^0(X) \rightarrow \Omega^1(X)$ we have $B^0(X) = 0$. Hence we only need to compute $Z^0(X)$. A smooth function $f \in \Omega^0(X)$ with $df = 0$. Pick any point $x \in X$ consider the subset $g(t) = f(\gamma(t))$, where $\gamma: [0, 1] \rightarrow X$ is a smooth curve joining x to y . Then $g'(t) = d(f \circ \gamma)_t = df_{\gamma(t)}(\gamma'(t)) = 0$ since $df = 0$. Therefore, $f(x) = f(y)$. Now, if we let $U = \{y \in X: f(y) = f(x)\}$, then this is both open and closed. Particularly, this is open because $df = 0$ if and only if f is constant on every component of X . Thus, $Z^0(X)$ is isomorphic to the space of functions that are constant on each connected component of X . $\square$

Proposition 3.2.46

$$H_{dR}^1(\mathbb{R}) = 0$$

Proof. Note that $\Omega^2(\mathbb{R}) = 0$, since $\mathbb{R}$ is one dimensional. Thus, $\ker d = \Omega^1(\mathbb{R})$. Thus, $H_{dR}^1 = \Omega^1(\mathbb{R}) / \text{Im} d$. The claim is that every 1-form is exact. Pick a one-form (which is just covectors) $f(x)dx$. We want to find $g \in \Omega^0(\mathbb{R})$ such that $dg = fdx$. Consider $g(x) = \int_0^x f(t)dt$. We see $dg = fdx$. This implies that $\text{Im} d = \Omega^1(\mathbb{R})$, so we get that $H_{dR}^1 = 0$. $\square$

Exercise 3.2.47. Show that $H_{dR}^n(\mathbb{R}^n) = 0$.

Solution: We can mimic what is done above. Any n -form is closed, so we just need to show that any n -form is exact. For this, we can construct a primitive, or some $n - 1$ form which maps to the n -form by integration/ the fundamental theorem of calculus. $\square$

The following deep theorem builds a bridge between the analytical and the topological features of smooth manifolds.

Theorem 3.2.48 (de Rham Theorem)

For any smooth manifold and non-negative integer p , there is an isomorphism between the de Rham cohomology group $H_{dR}^p(X)$ and the singular cohomology group $H_{Sing}^p(X, \mathbb{R})$.

This theorem will not be proved or used in this course. We list its statement here to motivate the readers to appreciate the power of differential forms. Note that the singular cohomology of a topological space is almost impossible to compute from its definition. The de Rham theorem gives a possible and effective way to compute it when the space admits a smooth manifold structure. Interested readers are referred to chapter 18 in [1] and references therein for further discussions.

Proposition 3.2.49

The wedge product gives a well defined product on $H_{dR}^\bullet(X)$.

Proof. For $\omega \in \Omega^p(X)$, $\eta \in \Omega^q(X)$, both closed, they represent some equivalence class in the de Rham cohomology:

$$[\omega] \in H_{dR}^p(X), \quad [\eta] \in H_{dR}^q(X).$$

We define

$$[\omega] \wedge [\eta] = [\omega \wedge \eta].$$

The claim is that this is well-defined; that is, that it doesn't depend on the choice of representative. First, ω and η are closed, implies that $\omega \wedge \eta$ is closed, which is by the Leibniz rule. If we pick different representatives, we need

$$[\omega + d\alpha] \wedge [\eta + d\beta] = [\omega \wedge \eta].$$

The LHS is equal to

$$[\omega \wedge \eta + \omega \wedge d\beta + d\alpha \wedge \eta + d\alpha \wedge d\beta].$$

However, the last three terms can be written as $d\gamma$, for some γ . We use the generalized Leibniz rule/ $d^2 = 0$ as motivation, and find that $\gamma = (-1)^p \omega \wedge \beta + \alpha \wedge \eta + \alpha \wedge d\beta$ works. We compute these one by one to verify this is the case:

$$d((-1)^p \omega \wedge \beta) = (-1)^p d\omega \wedge \beta + (-1)^{2p} \omega \wedge d\beta.$$

Next,

$$d(\alpha \wedge \eta) = d\alpha \wedge \eta + (-1)^{\deg \alpha} \alpha \wedge d\eta.$$

Finally,

$$d(\alpha \wedge d\beta) = d\alpha \wedge d\beta + (-1)^{\deg \alpha} \alpha \wedge d^2 \beta.$$

Noting that $d\omega = d\eta = d^2 \beta = 0$ (the former two by being closed forms and the last because $d^2 = 0$), we get that

$$\omega \wedge d\beta + d\alpha \wedge \eta + d\alpha \wedge d\beta = d((-1)^p \omega \wedge \beta + \alpha \wedge \eta + \alpha \wedge d\beta).$$

Therefore, this part vanishes under the quotient. $\square$

Exercise 3.2.50. Let ω, η be two closed differential forms on a connected manifold M .

1. Show that the de Rham cohomology class of $\omega \wedge \eta$ depends only on the cohomology classes of ω and η .
2. Show the above wedge product gives the de Rham cohomology

$$R(M) := \bigoplus_p H_{dR}^p(M)$$

a ring structure. Is it commutative? What is the unit of this ring $R(M)$?

3. Let $F: M \rightarrow N$ be a smooth map. Show the pullback operation $F^*: \omega \mapsto F^*\omega$ induces a well-defined ring homomorphism from $R(N)$ to $R(M)$.

Solution: 1. The well-definedness of the wedge product follows from the above proposition.

2. This is also clear.

3. We just need to check that F^* is a ring homomorphism, and maps kernels to kernels, images to images. $\square$

4 Integration

The goal is as follows. If X is a compact n -dimensional *oriented* manifold and Ω is an n -form, how to define $\int_X \omega \in \mathbb{R}$. In other words, we want to get an operator from $\Omega^n(X) \xrightarrow{\int_X} \mathbb{R}$. We note that our focus on compactness makes sense, as when we define integral over $\mathbb{R}$. Now, recall that over $\mathbb{R}$, we have

$$\int_a^b f dx = - \int_b^a f dx.$$

This *flipping* of bounds is something unique to intervals, so we make this precise for the case of manifolds.

§4.1 Orientation

We begin with orientation on $\mathbb{R}^n$. Let $F: \mathbb{R}^n \rightarrow \mathbb{R}^n$ be a diffeomorphism. Let $x_1, \dots, x_n$ be the coordinates on the domain and $y_1, \dots, y_n$ be the coordinates on the target. Then we can check that

$$F^*(dy_1 \wedge \dots \wedge dy_n) = \det(JF) dx_1 \wedge \dots \wedge dx_n$$

where $\det(JF)$ is the determinate of the Jacobian matrix of F . We note that when F is a diffeomorphism, this determinant is nonzero for all $(x_1, \dots, x_n)$. It follows by the intermediate value theorem that $\det(JF) > 0$ or < 0 .

Definition 4.1.1. A diffeomorphism $F: (\mathbb{R}^n, x) \rightarrow (\mathbb{R}^n, y)$ is called **orientation preserving** if $\det(JF) > 0$, and **orientation reversing** if $\det(JF) < 0$. Similarly, one can define orientation preserving diffeomorphisms between open subsets of $\mathbb{R}^n$.

Example 4.1.2

The *flip* diffeomorphism

$$T: \mathbb{R}^n \rightarrow \mathbb{R}^n, \quad T(x_1, \dots, x_n) = (-x_1, x_2, \dots, x_n)$$

is orientation reversing. As we can calculate,

$$\det(JF) = \det(\text{diag}(-1, 1, \dots, 1)) = -1.$$

Definition 4.1.3. A smooth manifold is called **orientable** if it admits a smooth atlas $\mathcal{A} = \{(U_\alpha, \phi_\alpha)\}$ such that all transition maps $\phi \circ \psi^{-1}$ are orientation preserving. Such an atlas is called an **orientation** on X . Such a pair $(X, \mathcal{A})$ is called an **oriented** manifold.

Remark 4.1.4 — Here, we are not saying that $\mathcal{A}$ is maximal.

Example 4.1.5

The Euclidean space $\mathbb{R}^n$ is orientable, since it admits a global chart.

Proposition 4.1.6

Let X be a smooth manifold of dimension n . X is orientable if and only if it admits a nowhere-vanishing differential n -form.

Proof. ($\Leftarrow$): Let ω be a nowhere vanishing n -form on X . Start with an arbitrary atlas $\{(U_\alpha, \phi_\alpha)\}$ and we will modify it to be an atlas whose all transition maps are orientation preserving. Suppose that U_α is connected (if not just repeat this argument on its components). For any α , consider the n -form $\phi_\alpha^*(dx_1 \wedge \cdots \wedge dx_n)$. It is an n -form on U_α hence we can write

$$\phi_\alpha^*(dx_1 \wedge \cdots \wedge dx_n) = f_\alpha \omega,$$

where $\omega \in \Omega^n(X)$ and f_α is the determinant of the Jacobian. Since both $\phi_\alpha^*(dx_1 \wedge \cdots \wedge dx_n)$ and ω are nowhere zero, the function f_α is either everywhere positive or everywhere negative. If f_α is everywhere positive then we keep the chart (U_α, ϕ_α) . Otherwise we modify the chart map to be $T \circ \phi_\alpha$ where T is the flip map in Example 4.1.2. After modifying all charts (U_α, ϕ_α) , the functions f_α become always everywhere positive. A direct consequence is that all the transition maps are orientation preserving. Hence the modified atlas induces an orientation on X .

($\Rightarrow$): Suppose we have an atlas $\{(U_\alpha, \phi_\alpha)\}$ which gives an orientation on X . We will construct a nowhere vanishing n -form on X . For any α , set

$$\omega_\alpha := \phi_\alpha^*(dx_1 \wedge \cdots \wedge dx_n).$$

This is a nowhere vanishing n -form on U_α . Now take another chart (U_β, ϕ_β) , and we glue them together. For any α, β , we have

$$(\phi_\beta^{-1})^* \phi_\alpha^*(dx_1 \wedge \cdots \wedge dx_n) = \lambda_{\alpha\beta}(dx_1 \wedge \cdots \wedge dx_n).$$

Since the atlas gives an orientation on X , the functions $\lambda_{\alpha\beta}$ are always positive. Hence there holds

$$\phi_\alpha^*(dx_1 \wedge \cdots \wedge dx_n) = (\lambda_{\alpha\beta} \circ \phi_\beta) \cdot \phi_\beta^*(dx_1 \wedge \cdots \wedge dx_n).$$

This shows that any point, ω_α is always a positive multiple of ω_β . Now we pick a partition of unity ρ_α subordinate to the open cover $\{U_\alpha\}$ and define $\omega := \sum \rho_\alpha \omega_\alpha$. At any point p , ω is a positive multiple of some ω_α . Hence ω is globally nowhere non-vanishing. $\square$

Remark 4.1.7 — We note that this proof suggests that not every manifold is orientable.

Therefore an oriented manifold can be also thought of as a pair (X, ω) where ω is a nowhere vanishing n -form on an n -dimensional manifold X .

Definition 4.1.8. Let ω, ω' be two nowhere vanishing n -forms on X , then there is a smooth function f such that $\omega' = f\omega$ and f is either everywhere positive or everywhere negative, on each connected component of X . We say ω, ω' are **equivalent** if f is positive. In this case we also say they induce the **equivalent orientations** on X .

The above discussion shows that for any connected orientable smooth manifold, there are two equivalent classes of orientations.

Example 4.1.9

The two atlases $(\mathbb{R}^n, \text{Id})$ and $(\mathbb{R}^n, T)$ where T is the flip map give two different orientations on $\mathbb{R}^n$.

Remark 4.1.10 — A submanifold of an oriented manifold need not be orientable. For instance, consider the Möbius band inside $\mathbb{R}^3$. Note that the Möbius band is a manifold with boundary!

Exercise 4.1.11. Show that the product $X \times Y$ of two orientable manifolds is orientable.

Solution: Let $\{(U, \varphi)\}$ and $\{(V, \psi)\}$ be an oriented atlas for the manifolds X, Y . Then, we consider the atlas of $X \times Y$ formed by the product of these. That is, the atlas $\{(U \times V, \varphi \times \psi)\}$. Now, let $\varphi_1 \times \psi_1$, and $\varphi_2 \times \psi_2$ be two chart maps. Then we want to show that the transition map $(\varphi_2 \times \psi_2) \circ (\varphi_1 \times \psi_1)^{-1}$ is an orientation preserving diffeomorphism. Well, this composition is simply $(\varphi_2 \circ \varphi_1^{-1}) \times (\psi_2 \circ \psi_1^{-1})$. The Jacobian of this map is clearly positive, since the Jacobian of this total map is simply the Jacobian of the first transition with the second, which are both positive. $\square$

Example 4.1.12

Below, we will see that $\mathbb{S}^n$ is orientable. The exercise above implies then that the torus $\mathbb{T}^1 = \mathbb{S}^1 \times \mathbb{S}^1$ is orientable.

Exercise 4.1.13. Let $F: \mathbb{R}^n \rightarrow \mathbb{R}$ be a smooth function and c be a regular value of F . Show that $F^{-1}(c)$ is a orientable manifold.

Solution: c being regular value implies that $F^{-1}(c)$ is a submanifold of $\mathbb{R}^n$ of dimension $n-1$. To show that $F^{-1}(c)$ is orientable, we shall show that it admits a nowhere vanishing $(n-1)$ -form. Consider the vector field

$$\nabla F(p) = \sum_{i=1}^n \frac{\partial F}{\partial x_i}(p) \frac{\partial}{\partial x_i}.$$

Now let $\omega = dx_1 \wedge \cdots \wedge dx_n$. We define

$$\eta(\cdot, \dots, \cdot) = \omega(\nabla F, \cdot, \dots, \cdot).$$

We note this is a $(n-1)$ -form, since we are always specifying the first entry to be $\nabla F(p)$. For any p , since $\nabla F(p)$ is nowhere vanishing, we can extend it to a basis of $T_p \mathbb{R}^n$, say $\{\nabla F(p), v_1, \dots, v_{n-1}\}$. Then, η restricted to $F^{-1}(c)$ is a nowhere vanishing $(n-1)$ -form because we've chosen $v_1, \dots, v_{n-1}$ so that it forms a basis. $\square$

Exercise 4.1.14. Show that $\mathbb{S}^n$ is orientable.

Solution: This follows by the previous exercise. Particularly, we have that

$$\mathbb{S}^n = \{(x_1, \dots, x_{n+1}) \in \mathbb{R}^{n+1} : \sum_{i=1}^{n+1} x_i^2 = 1\},$$

so letting $F: \mathbb{R}^{n+1} \rightarrow \mathbb{R}$ be the map $(x_1, \dots, x_{n+1}) \rightarrow x_1^2 + \cdots + x_{n+1}^2$, it follows that $F^{-1}(1) = \mathbb{S}^n$. $\square$

Exercise 4.1.15. Let M be a connected, orientable, smooth manifold. Then M has exactly two orientations. Further, if two orientation forms of M agree at one point, they are equivalent.

Solution: M is orientable, so there exists a nowhere vanishing n -form ω on M . In particular, suppose there is another nonvanishing n -form ω' . Then, since M is connected, there exists some smooth function f on M such that $\omega = f\omega'$. Here, f is either everywhere positive on M or everywhere negative on M . This is because M is connected. If f is everywhere positive, they are equivalent, and if f is everywhere negative, they define two distinct orientations. $\square$

Example 4.1.16

The above exercise shows that $\mathbb{R}^n$ or S^n , have two equivalence classes of orientations.

Exercise 4.1.17. Suppose M is an oriented smooth manifold and $U \subseteq M$ is a smooth codimension-0 embedded submanifold. Then the orientation of M restricts to an orientation of U . If ω is an orientation form for M , then $\iota^*\omega$ is an orientation form for U .

Solution: If U is a smooth codimension 0 embedded submanifold of M , it is an open submanifold of M . Then, if ω is a nowhere vanishing top form for M , we get that $\iota^*\omega$ restricts to a nowhere vanishing top form on U . $\square$

Exercise 4.1.18. Let $F: M \rightarrow N$ be a local diffeomorphism and N be an oriented manifold. Show that M is orientable as well.

Solution: If N is oriented, then there exists a nowhere vanishing top form ω . Write $\omega = f dx_1 \wedge \cdots \wedge dx_n$. Then, $F^*\omega = (f \circ F)_x \det(JF)_x dy_1 \wedge \cdots \wedge dy_n$, which is a nowhere vanishing top form for N . $\square$

§4.2 Integration of Forms

Now we study how to integrate differential forms on manifolds. Still, we start with Riemann integrals on $\mathbb{R}^n$. Let f be a smooth function on $\mathbb{R}^n$ which has compact support. Such functions are called compactly supported. Then we can consider the Riemann integral of f on $\mathbb{R}^n$, written as

$$\int_{\mathbb{R}^n} f.$$

This compact support condition is only used for convergence. Let $F: (\mathbb{R}^n, x) \rightarrow (\mathbb{R}^n, y)$ be a diffeomorphism. The important change of variables formula is

$$\int_{(\mathbb{R}^n, y)} f = \int_{(\mathbb{R}^n, x)} (f \circ F) \cdot |\det(JF)|.$$

Now we define the integration of a compactly supported n -form in the following way. Pick $\omega \in \Omega^n(\mathbb{R}^n)$ with compact support. There is a unique smooth function f such that $\omega = f \cdot dx_1 \wedge \cdots \wedge dx_n$. We define

$$\int \omega := \int_{(\mathbb{R}^n, x)} f$$

where the right-hand side is the Riemann integral of f , $\int \int \cdots \int f(x_1, \dots, x_n) dx_1 dx_2 \dots dx_n$. Now, we note that (by Fubini's theorem), the order of integration does not matter. That is,

$$\int \int \cdots \int f(x_1, \dots, x_n) dx_1 dx_2 \dots dx_n = \int \int \cdots \int f(x_1, \dots, x_n) dx_2 dx_1 \dots dx_n.$$

This swaps the order of the n -form with respect to $x_2, x_1, \dots, x_n$ as $dx_2 \wedge dx_1 \wedge \cdots \wedge dx_n$. Let $\omega = g \cdot dx_2 \wedge dx_1 \wedge \cdots \wedge dx_n$. Then, what is g ? We have that $g = -f$. Thus,

$$\int_{\mathbb{R}^n} \omega = \int_{\mathbb{R}^n} g dx_1 dx_2 \dots dx_n = - \int_{\mathbb{R}^n} f dx_1 dx_2 \dots dx_n.$$

The point is that the change of coordinates $x_1, x_2, \dots, x_n$ to $x_2, x_1, \dots, x_n$ is orientation reversing. We note that we have the change of variable formula in the Riemann integral. Suppose $F: (\mathbb{R}^n, x_1, \dots, x_n) \rightarrow (\mathbb{R}^n, y_1, \dots, y_n)$. Then,

$$\int_{(\mathbb{R}^n, y)} f = \int_{(\mathbb{R}^n, x)} (f \circ F) |\det(JF)|.$$

Under a diffeomorphism, the integral of an n -form changes in the following way.

Proposition 4.2.1

Let $F: \mathbb{R}^n \rightarrow \mathbb{R}^n$ be a diffeomorphism. We have that $\int F^* \omega = \int \omega$ if F is orientation preserving and $\int F^* \omega = - \int \omega$ if F is orientation reversing.

Proof. Let y be the coordinates on the target $\mathbb{R}^n$ and x be the coordinates on the domain $\mathbb{R}^n$. Then by definition

$$\int \omega = \int_{(\mathbb{R}^n, y)} f,$$

where $\omega = f \cdot dy_1 \wedge \cdots \wedge dy_n$. Then we perform change of variables formula to differential forms and Riemann integrals, respectively.

$$\int_{(\mathbb{R}^n, y)} \omega = \int_{(\mathbb{R}^n, x)} f = (f \circ F) \cdot |\det(JF)|,$$

$$\int F^* \omega = \int (f \circ F) \det(JF) dx_1 \wedge \cdots \wedge dx_n = \int_{(\mathbb{R}^n, x)} (f \circ F) \cdot \det(JF).$$

Hence we can see the sign change directly. $\square$

As discussed above, the main difference between two integrals under change of variables is that one ignores the sign, putting $\det(JF)$ inside the absolute value, while the other remembers the sign. With this understood, we can define integrals of differential forms.

Let X be a smooth manifold of dimension n and let $\omega \in \Omega^n(X)$ be a differential n -form with compact support. Suppose X admits an atlas $\mathcal{A} = \{(U_\alpha, \varphi_\alpha)\}$ which gives an orientation on X . We will define $\int_{X, \mathcal{A}} \omega$. First, assume the support of ω is contained in some U_α , we define

$$\int_{U_\alpha} \omega := \int_{\mathbb{R}^n} (\varphi_\alpha^{-1})^* \omega.$$

We need to show that this definition is well defined.

Proposition 4.2.2

Let $(U_\beta, \varphi_\beta) \in \mathcal{A}$ be another chart containing the support of ω , then

$$\int_{\mathbb{R}^n} (\varphi_\alpha^{-1})^* \omega = \int_{\mathbb{R}^n} (\varphi_\beta^{-1})^* \omega.$$

Hence the definition of $\int_X \omega$ does not depend on the choice of chart containing the support of ω .

Proof. Consider the transition map $\varphi_\alpha \circ \varphi_\beta^{-1}: \mathbb{R}^n \rightarrow \mathbb{R}^n$. Then, by Proposition 4.2.1,

$$\int_{\mathbb{R}^n} (\varphi_\beta^{-1})^* \omega = \int_{\mathbb{R}^n} (\varphi_\beta \circ \varphi_\alpha^{-1})^* (\varphi_\beta^{-1})^* \omega = \int_{\mathbb{R}^n} (\varphi_\alpha^{-1})^* \circ \varphi_\beta^* \circ (\varphi_\beta^{-1})^* \omega = \int_{\mathbb{R}^n} (\varphi_\alpha^{-1})^* \omega.$$

The assumption that the atlas $\mathcal{A}$ is an orientation (so orientation preserving on overlaps) is crucial here. $\square$

To integrate over general forms, we combine this with the notion of partition of unities. Let ω be an n -form with compact support, we pick a partition of unity $\{\rho_\alpha\}$ subordinate to the open cover $\{U_\alpha\}$. We define

$$\int_{X, \mathcal{A}} \omega := \sum_\alpha \int_{\mathbb{R}^n} (\varphi_\alpha^{-1})^* (\rho_\alpha \cdot \omega).$$

First note that the sum on the RHS is locally finite, since $\{\rho_\alpha\}$ is a partition of unity. Further, $\text{supp}(\rho_\alpha) \subseteq U_i$. Therefore, it makes sense to write

$$\int_{X, \mathcal{A}} \omega = \sum_X (\rho_\alpha \cdot \omega).$$

To make this well-defined, we need to see if this is invariant under changing our partition of unity and atlas.

Proposition 4.2.3

Let $\mathcal{B} = \{(V_\beta, \psi_\beta)\}$ be another atlas for X giving the same orientation as the atlas $\mathcal{A}$. Let $\{\tau_\beta\}$ be a partition of unity subordinate to $\{V_\beta\}$. Then,

$$\int_{X, \mathcal{A}, \rho} \omega = \int_{X, \mathcal{B}, \tau} \omega.$$

Hence we can just write them as $\int_X \omega$.

Proof. For a fixed α , we have

$$\int_X \rho_\alpha \omega = \int_X \underbrace{\left(\sum_\beta \tau_\beta \right)}_{=1} \rho_\alpha \omega = \sum_\beta \int_X \tau_\beta \rho_\alpha \omega,$$

since partition of unity is a locally finite sum. Therefore we get

$$\begin{aligned}
 \int_{X, \mathcal{A}, \rho} \omega &= \sum_{\alpha} \int_{X, \mathcal{A}, \rho} \rho_{\alpha} \omega \\
 &= \sum_{\alpha} \sum_{\beta} \int_{X, \mathcal{A}, \rho} \tau_{\beta} \rho_{\alpha} \omega \\
 &= \sum_{\beta} \sum_{\alpha} \int_{X, \mathcal{B}, \tau} \rho_{\alpha} \tau_{\beta} \omega \\
 &= \sum_{\alpha} \int_{X, \mathcal{B}, \tau} \sum_{\alpha} \rho_{\alpha} \tau_{\beta} \omega \\
 &= \sum_{\beta} \int_{X, \mathcal{B}, \tau} \rho_{\beta} \omega \\
 &= \int_{X, \mathcal{B}, \tau} \omega.
 \end{aligned}$$

The middle equality holds because that $\rho_{\alpha} \rho_{\beta} \omega$ is supported in both U_{α} and V_{β} . The integral does not depend on the choice of a single chart, by Proposition 4.2.2. $\square$

Remark 4.2.4 — We note that this integral is really useful in abstract arguments more than concrete calculations, since it may not even be possible.

Example 4.2.5

Consider the unit circle $S = \{x^2 + y^2 = 1\} \subseteq \mathbb{R}^2$. Let $\lambda = xdy - ydx \in \Omega^1(\mathbb{R}^2)$. Compute

$$\int_S \iota^* \lambda,$$

where $\iota: S \rightarrow \mathbb{R}^2$ is the inclusion map.

Solution: Consider the chart (U, φ) , where U is the set of points with positive x -coordinate and φ is the projection to y -axis. The projection map is $\varphi(x, y) = y$. Then,

$$\int_U \iota^* \lambda = \int_{\mathbb{R}, y} (\varphi^{-1})^* \iota^* \lambda.$$

We calculate the pullback. We have that

$$\lambda = xdy - ydx.$$

Noting that we can write $x = \sqrt{1 - y^2}$, we get that

$$(\varphi^{-1})^* \iota^* \lambda = \sqrt{1 - y^2} dy - y \frac{(-2y)}{2\sqrt{1 - y^2}} dy = \frac{1}{\sqrt{1 - y^2}} dy.$$

Thus, the integral is

$$\int_{-1}^1 \frac{1}{\sqrt{1 - y^2}} dy = \pi.$$

We must repeat the calculation the other side. $\square$

Remark 4.2.6 — Note that in the above, we didn't consider the north pole, but this is a measure zero set, so we were able to neglect it.

Remark 4.2.7 — We didn't follow the partition of unity method in the example above. If one wants to do this, one should construct a bump function that is 1 from $(-1 + \epsilon, 1 - \epsilon)$, for some small ϵ , and the integral should be taken as this ϵ shrinks.

Exercise 4.2.8. Compute the above in polar coordinates.

Solution: Consider the chart $\theta: \mathbb{S}^1 \setminus \{(1, 0)\}$ defined by $\theta: (x, y) \mapsto \arctan(\frac{y}{x})$. Then, θ^{-1} is just the map $\theta \mapsto (\cos \theta, \sin \theta)$. We evaluate $\int_S \iota^* \lambda$ using this coordinate chart. Note that we are missing out on the point $(1, 0)$, but this is a measure zero set so doesn't affect the value of the integral. We have as follows:

$$\begin{aligned} \int_S \iota^* \lambda &= \int_0^{2\pi} (\theta^{-1})^* \iota^* \lambda \\ &= \int_0^{2\pi} (\theta^{-1})^* \iota^* (x dy - y dx) \\ &= \int_0^{2\pi} \cos^2 \theta d\theta + \sin^2 \theta d\theta \\ &= \int_0^{2\pi} d\theta \\ &= 2\pi. \end{aligned}$$

□

Exercise 4.2.9. Generalize the calculation above to $\mathbb{S}^{n-1} \subseteq \mathbb{R}^n$.

Solution: The correct $n - 1$ -form to consider is

$$\omega = \sum_{i=1}^{n-1} (-1)^{i+1} x_i dx_1 \wedge \dots \wedge \hat{dx}_i \wedge \dots \wedge dx_{n-1}.$$

Then, all we need to do is find a good atlas. Omitting the north pole, we can actually find that $\varphi: (0, 2\pi) \times (0, \pi)^{n-1} \rightarrow \mathbb{S}^n$ defined by spherical coordinates is good. What will happen is that we will get the surface area of $\mathbb{S}^{n-1}$. We shall do this for $n = 2$: The spherical coordinates are:

$$\begin{aligned} x_1 &= \cos \theta \sin \phi \\ x_2 &= \sin \theta \sin \phi \\ x_3 &= \cos \phi \end{aligned}$$

This is a map from $\varphi: (0, 2\pi) \times (0, \pi) \rightarrow \mathbb{S}^2 \setminus \{(0, 0, 1)\}$; here θ is the first coordinate and ϕ is the second. Again, $\{(0, 0, 1)\}$ is a measure zero set so it is fine. We have as follows:

$$\begin{aligned} \int_{\mathbb{S}^2} \iota^* \lambda &= \int_{(0, 2\pi)} \int_{(0, \pi)} \varphi^* \iota^* \lambda \\ &= \int_{(0, 2\pi)} \int_{(0, \pi)} \varphi^* \iota^* (x_1 dx_2 \wedge dx_3 - x_2 dx_1 \wedge dx_3 + x_3 dx_1 \wedge dx_2). \end{aligned}$$

After computing these pullbacks, this integral simplifies to

$$\int_{(0,2\pi)} \int_{(0,\pi)} \sin \phi d\theta \wedge d\phi.$$

Since our basis is (θ, ϕ) , we find that $d\theta \wedge d\phi$ is positive. This as follows:

$$\int_{(0,2\pi)} \int_{(0,\pi)} \sin \phi d\theta d\phi = 2\pi \cdot (-1) \cdot (-1 - 1) = 4\pi.$$

This is what we expected. □

Exercise 4.2.10. Calculate the volume for $\mathbb{S}^n$.

Therefore when we fix an orientation on X , we can define the integral of a compactly supported n -form.

Definition 4.2.11. Let X be an oriented n -dimensional smooth manifold and let ω be a compactly supported n -form on X . We define $\int_X \omega$ as $\int_{X,\mathcal{A}} \omega$ for some atlas $\mathcal{A}$ inducing the same orientation.

Remark 4.2.12 — This definition is independent of the choices of atlases giving the same orientation and the partition of unity.

Exercise 4.2.13. If $\mathcal{B}$ is an atlas for X giving the opposite orientation as the atlas $\mathcal{A}$, show

$$\int_{X,\mathcal{A}} \omega = - \int_{X,\mathcal{B}} \omega.$$

Solution: By linearity of the integral (proven below), it suffices to simply show this for one chart such that $\rho_\alpha \omega$ doesn't vanish. Let $(U, \varphi) \in \mathcal{A}$ and $(U, \psi) \in \mathcal{B}$. Then, $\phi \circ \psi^{-1}$ is in particular, a diffeomorphism which is orientation reversing. Therefore,

$$\begin{aligned} \int_{U,\mathcal{A}} \rho_\alpha \omega &= \int_{\mathbb{R}^n} (\varphi^{-1})^* \rho_\alpha \omega \\ &= - \int_{\mathbb{R}^n} (\varphi \circ \psi^{-1})^* (\varphi^{-1})^* \rho_\alpha \omega \\ &= - \int_{\mathbb{R}^n} (\psi^{-1})^* \rho_\alpha \omega \\ &= - \int_{U,\mathcal{B}} \rho_\alpha \omega. \end{aligned}$$

Summing over the partition of unity then provides the desired result. □

Exercise 4.2.14. For an oriented manifold X and any number $a, b \in \mathbb{R}$, there holds

$$\int_X a\omega + b\eta = a \int_X \omega + b \int_X \eta$$

for any compactly supported n -forms ω, η .

Solution: This follows from the local calculation that the integral is a linear operator. Indeed, let (U, φ) be a chart for X such that ω, η doesn't vanish on U . Then,

$$\begin{aligned} \int_X a\omega + b\eta &= \int_{\mathbb{R}^n} (\varphi^{-1})^*(a\omega + b\eta) \\ &= a \cdot \int_{\mathbb{R}^n} (\varphi^{-1})^*\omega + b \cdot \int_{\mathbb{R}^n} (\varphi^{-1})^*\eta \\ &= a \int_X \omega + b \int_X \eta. \end{aligned}$$

□

Exercise 4.2.15. For an oriented n -dimensional manifold X , if ω is a nowhere vanishing n -form on X determining the same orientation, then $\int_{X, \mathcal{A}} \omega > 0$.

Solution: Let ω' be an orientation induced by the orientation on X . Then, $\int_{X, \mathcal{A}} \omega' > 0$ by definition. Since ω and X determine the same orientation, this means $\omega = f\omega'$, where $f > 0$. Therefore,

$$\int_{X, \mathcal{A}} \omega = \int_{X, \mathcal{A}} f \cdot \omega' > 0,$$

because f is positively valued.

□

§4.3 The Stokes' Theorem

Now we discuss the Stokes' theorem, which is a far-reaching generalization of the fundamental theorem of calculus. Recall if we have a smooth function f on $\mathbb{R}$, the fundamental theorem of calculus says that

$$\int_a^b f'(x)dx = f(b) - f(a).$$

In particular, if f has compact support, then

$$\int_{\mathbb{R}} f'(x)dx = f(b) - f(a) = 0 - 0 = 0,$$

where $b \gg 0$, $a \ll 0$. In the language of manifolds, we can view the closed interval $[a, b]$ as a *manifold with boundary* and $f'(x)dx$ as a one-form. Moreover, the one-form $f'(x)dx$ is exact since $f'(x)dx = df$. Then the theorem means that the integral of df over $[a, b]$ is determined by the values of f on the boundary of $[a, b]$, in this case, two points $\{a, b\}$. The Stokes' theorem is a natural generalization of this phenomenon to the integral of differential forms on manifolds with boundary. Since we did not talk about the theory of manifolds with boundary in this course, we will focus on a special case.

Example 4.3.1

Let $X = \mathbb{R}$ and f be a smooth function on $\mathbb{R}$ with compact support. Pick a small enough and b large enough such that $[a, b]$ contains the support of f . Then we have

$$\int_{\mathbb{R}} f'(x) dx = \int_a^b f'(x) dx = f(b) - f(a) = 0 - 0 = 0.$$

This example generalizes to the following theorem:

Theorem 4.3.2

Let X be an oriented smooth manifold of dimension n . Let η be a differential $(n-1)$ -form on X with compact support, $\eta \in \Omega_{cpt}^{n-1}(X)$. We have

$$\int_X d\eta = 0.$$

The proof of this theorem follows from a local computation and partition of unity. For completeness, we shall state the following theorem, which we will not cover.

Theorem 4.3.3

Let X be an oriented smooth manifold of dimension n . Let η be a differential $(n-1)$ -form on X with compact support, $\eta \in \Omega_{cpt}^{n-1}(X)$. Then,

$$\int_X d\eta = \int_{\partial X} \eta$$

Now to go back to Theorem 4.3.2. We first prove a lemma in the local case, which is a direct corollary of the fundamental theorem of calculus.

Lemma 4.3.4

Let η be a differential $(n-1)$ -form on $\mathbb{R}^n$ with compact support. There holds

$$\int_{\mathbb{R}^n} d\eta = 0.$$

Proof. We note that since η is an $(n-1)$ -form, we have

$$\eta = f_1 dx_2 \wedge \cdots \wedge dx_n + f_2 dx_1 \wedge \cdots \wedge dx_n + \cdots + f_n dx_1 \wedge \cdots \wedge dx_{n-1},$$

where each f_i has compact support. By the linearity of integral, we only need to check for $\eta = f_1 dx_2 \wedge \cdots \wedge dx_n$, where f is a smooth function with compact support. We have

$$\begin{aligned} \int_{\mathbb{R}^n} d\eta &= \int_{\mathbb{R}^n} d(f_1 dx_2 \wedge \cdots \wedge dx_n) \\ &= \int_{\mathbb{R}^n} \frac{\partial f_1}{\partial x_1} dx_1 \wedge dx_2 \wedge \cdots \wedge dx_n \\ &= \int \cdots \int \frac{\partial f_1}{\partial x_1} dx_1 dx_2 \cdots dx_n. \end{aligned}$$

Now, pick $r > 0$ large enough such that $[-r, r]^n$ contains the support of f . Then, the last integral is

$$\int_{[-r, r]^{n-1}} \int_{-r}^r \frac{\partial f}{\partial x_1} dx_1 dx_2 \cdots dx_n,$$

and this first integral $\int_{-r}^r \frac{\partial f_1}{\partial x_1} dx_1 = f_1(r) - f_1(-r) = 0 - 0 = 0$ by the fundamental theorem of calculus. Therefore, the entire integral is 0. $\square$

Remark 4.3.5 — In the above, we mean with respect to the standard coordinates (and whenever we don't mention it).

Proof of Theorem 4.3.2. Let $\{(U_\alpha, \varphi_\alpha)\}$ be an atlas giving the same orientation as X . Since the support of η is compact, there exists a finite subcover $U_1, \dots, U_\ell$ which covers its support. Pick a partition of unity $\{\rho_\alpha\}$ subordinate to this finite subcover. So we have

$$\eta = (\rho_1 + \dots + \rho_\ell)\eta, \quad d\eta = d(\rho_1\eta) + \dots + d(\rho_\ell\eta).$$

Each of $d(\rho_i\eta)$ is supported in U_i . Hence its integral is zero by the above lemma:

$$\int_{U_i} d(\rho_i\eta) = \int_{\mathbb{R}^n} (\varphi_i^{-1})^* d(\rho_i\eta) = 0.$$

Since the integral is linear, so we get $\int_X d\eta = 0$. $\square$

Corollary 4.3.6

Let X be a compact oriented manifold of dimension n . Then $\int_X d\eta = 0$ for any $\eta \in \Omega^{n-1}(X)$.

Proof. When X itself is compact, the compactly supported condition is automatically satisfied in the definition above. $\square$

From here, assume that X is compact and oriented.

$$\begin{aligned} \int_X : \Omega^n(X) &\rightarrow \mathbb{R} \\ \omega &\mapsto \int_X \omega. \end{aligned}$$

There are two claims.

Claim 4.3.7. $\int$ is surjective.

Proof. By linearity, we just need to find one form which is nonzero. However, by the orientedness of the manifold, there exists a nowhere vanishing $\omega \in \Omega^n(X)$. Then, $\int_X \omega \neq 0$, so $\int_X c \cdot \omega \in \mathbb{R}$. $\square$

The second is related to the de Rham cohomology.

Claim 4.3.8. $\int$ gives a well-defined group homomorphism $\int_X : H_{\text{dR}}^n(X) \rightarrow \mathbb{R}$.

Proof. For a compact oriented manifold X of dimension n , recall that the degree- n de Rham cohomology group $H_{\text{dR}}^n(X)$ is defined as the space of closed n -forms modulo exact n -forms. Therefore $\int_X$ is a well-defined since the integral operation gives a well-defined group morphism. $\square$

Remark 4.3.9 — We note that this induced map on the de Rham cohomology is surjective from the previous claim; the nowhere vanishing n -form defines a class in the $H_{\text{dR}}^n(X)$, so the induced map on cohomology is not zero. Further, this induced operator on the de Rham cohomology is injective, which is a nontrivial fact.

This provides a first glimpse of an extremely deep and important fact about manifolds called **Poincaré duality**.

Here is an example which shows that when X is non-compact, then there is some $(n - 1)$ -form that is nonzero.

Exercise 4.3.10. Let $X = \mathbb{R}$, and define some bump function $f(x)$ on $\mathbb{R}$ such that $\int f(x)dx = 1$.

Bibliography

- [1] J. Lee. *Introduction to Smooth Manifolds*. Graduate Texts in Mathematics. Springer New York, 2012. ISBN: 9781441999825. URL: <https://books.google.co.jp/books?id=xygVcKGPsNwC>.